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Rasashastra & Bhaishajya Kalpana | Semester II

ABQ QUESTION BANK

Applied Basics of Rasashastra and Bhaishajya Kalpana

COMPANION TO VOLUME 0 OF THE SERIES

AYPG – AB – RB

Case-based Application Questions and
Analytical Structured LAQs

with model answer frameworks — as per the NCISM Curriculum

53 Case ABQs • 10 Analytical LAQs • 1 Complete University Paper Solved

Every ABQ-eligible unit of Table 6F • Bloom's and Miller's tags

Ready Reckoner of verified limits and rules

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Parul Institute of Ayurved and Research

Parul University, Vadodara, Gujarat, Bharat

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For MD / MS (Ayurveda) Scholars, Semester II — Course code AYPG-AB-RB, National Commission for Indian System of Medicine

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EXAMINATION QUESTIONS REPRODUCED

The complete question paper of the MS/MD (Ayurved) Preliminary Examination, Semester 2, October 2026, Faculty of Ayurved, Parul University (Questions 1, 2.1 to 2.4 and 3.1 to 3.8), is reproduced for teaching, with its original wording. All other cases were constructed for this book.

NOTE ON PREPARATION

This question bank was prepared by the author with the assistance of **Claude Opus**, a large language model developed by Anthropic. The choice of cases, the mapping to the NCISM curriculum, the verification of facts and references, and the review of all content were directed and carried out by the author, who accepts sole and full responsibility for its contents.

DISCLAIMER

This book is intended solely for the academic use of postgraduate scholars and teachers of Ayurveda. The cases describe hazardous substances and procedures in order to teach their safe control. **No procedure mentioned in this book should be attempted except in a licensed pharmacy, a recognised teaching Rasashala or an approved laboratory, under the direct supervision of a qualified teacher, and in accordance with the prescribed safety measures and statutory requirements.** Model answer frameworks indicate the points an examiner is likely to look for; they are not official answer keys of any university. Regulatory provisions and pharmacopoeial limits are stated as verified in October 2026 and are revised from time to time; readers must confirm the current position from the primary source. Nothing in this book is clinical advice. The author and the publisher accept no liability for any loss or injury arising from its use.

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HOW TO USE THIS QUESTION BANK

This book is for practice. It assumes you have studied the unit from the textbook (Volume 0), and it trains one skill: answering a long case in the way the university paper now asks.

How the book is arranged

- **Introduction** — how a case ABQ is built, a seven-step method, and how to read the Domain and Level tags.
- **The October 2026 Preliminary Paper — Solved** — all 14 questions (one ABQ, four LAQs, eight SAQs), in the order set, with a check of the paper against Tables 6F and 6G.
- **Part A** — 53 case-based ABQs (20 marks each), from the 17 units that Table 6F marks "Yes" for ABQ. Each case is numbered by module, unit and serial number: ABQ 7.1.2 is the second case on Module 7, Unit 1.
- **Part B** — 10 case-based analytical structured LAQs (10 marks each), from LAQ units that have no ABQ. Each is numbered by module and unit: LAQ B6.4 is on Module 6, Unit 4.
- **Ready Reckoner** — verified limits, rules, numbers and an instrument chooser. **Topic Finder** — every case on a topic, across modules.

How each case is laid out

- A heading line gives the *anchor* unit (the Table 6F unit the case is set from), any units it also draws on, and a suggested time.
- **The case** is in a shaded box. Its findings are numbered F1, F2… so that answers can refer to them.
- **The question table** follows the format of the university paper: sub-part, question, marks, Domain (Bloom's) and Level (Miller's).
- **The model answer framework** begins on a new page, so you can attempt the case first. It gives the points, mark by mark, usually in tables, followed by *Where marks are lost* and a page reference to the textbook.

How to practise

- Attempt one case in 35 minutes, on paper, with the book closed after you have read the question page.
- Then mark yourself against the framework. Count how many findings you used. A finding left unused is a mark left unclaimed.
- Rewrite one sub-part you did badly. This is where most improvement happens.
- Before the examination, do at least one case from every module you are revising — and remember that the module that carries the ABQ will carry no other question.

A word on language and terms

The language is kept plain on purpose. Sanskrit terms are written in ordinary Roman letters without diacritical marks. Where the university paper spells a term differently, the paper's wording is kept in the question and the standard term is used in the answer.

TABLE 6F — THE UNIT MAP

Table 6F of the curriculum decides which unit can carry which type of question. This map shows every unit, its eligibility, the questions in this book, and where the unit begins in the textbook.

Unit	ABQ	SAQ	LAQ	In this book	Vol 0 page
Module 1 — Pharmaceutics and Pharmaceutical Technology (5 to 20 marks)					
U1 Introduction to Pharmaceutics	No	Yes	Yes	LAQ B1.1	9
U2 Pharmaceutical Dosage Forms	No	Yes	No	—	32
U3 Pharmaceutical Formulation	Yes	Yes	Yes	ABQ 1.3.1–1.3.4	44
U4 Pharmaceutical Manufacturing and Quality Control	No	Yes	No	—	56
U5 Biopharmaceutics	No	Yes	Yes	LAQ B1.5	66
Module 2 — Databases and Information Systems (5 to 15 marks)					
U1 Introduction to Databases and Information Systems	No	Yes	No	—	93
U2 Metal and Mineral Databases	No	No	Yes	Paper Q2.1; LAQ B2.2	102
U3 Plant Databases	No	Yes	No	Paper Q3.6	109
U4 Database Application	No	Yes	No	—	117
U5 Challenges and Limitation of Databases	No	Yes	No	—	125
Module 3 — Essential Chemistry (5 to 20 marks)					
U1 Atomic, Molecular Structure and Periodic Table Elements	Yes	Yes	No	ABQ 3.1.1–3.1.3; Paper Q3.1	145
U2 Specialised Chemistry Fields	Yes	Yes	No	ABQ 3.2.1–3.2.3	153
U3 Organic Chemistry	Yes	Yes	No	ABQ 3.3.1–3.3.3	163
U4 Phytochemistry	No	Yes	Yes	LAQ B3.4; Paper Q2.4	171
U5 Metallurgy	No	Yes	Yes	LAQ B3.5	183
Module 4 — Shodhana and Marana (5 to 20 marks)					
U1 Shodhana and Marana Procedures	Yes	Yes	No	ABQ 4.1.1–4.1.3; Paper Q2.3	207
U2 SOP of Shodhana Processes	No	Yes	Yes	LAQ B4.2	215
U3 SOP of Marana Process	No	Yes	Yes	LAQ B4.3	224
U4 Chemical Changes in Shodhana and Marana	Yes	Yes	No	ABQ 4.4.1–4.4.3	232
U5 Case Studies, Troubleshooting and Emerging Trends	No	Yes	No	—	240
Module 5 — Classical and Advanced Instruments (5 to 20 marks)					
U1 Puta and its Modern Aspects	No	Yes	Yes	LAQ B5.1	257

Unit	ABQ	SAQ	LAQ	In this book	Vol 0 page
U2 Musha and its Modern Aspects	No	Yes	No	Paper Q3.2	265
U3 Chromatography	Yes	Yes	Yes	ABQ 5.3.1–5.3.3	272
U4 Spectroscopy	Yes	Yes	Yes	ABQ 5.4.1–5.4.3	283
U5 Microscopy	Yes	Yes	No	ABQ 5.5.1–5.5.3; Paper Q3.3	292
U6 Calibration Techniques	Yes	Yes	No	ABQ 5.6.1–5.6.3; Paper Q3.8	299
U7 Overview of Characterisation Instruments	Yes	Yes	Yes	ABQ 5.7.1–5.7.3	305
Module 6 — Animal House and Animal Handling (5 to 15 marks)					
U1 Introduction to Animal House Management	No	Yes	No	Paper Q3.4	325
U2 Study of Animal Behaviour	No	Yes	No	—	332
U3 Species-specific Handling Techniques	No	Yes	No	—	338
U4 Animal Health and Disease Prevention	No	Yes	Yes	Paper Q2.2; LAQ B6.4	343
U5 Animal Welfare and Ethics	No	Yes	No	—	350
Module 7 — Quality Control Techniques (5 to 20 marks)					
U1 Quality Control and Quality Assurance	Yes	No	Yes	ABQ 7.1.1–7.1.3; Paper Q1	368
U2 Classical Aspects of QC Parameters	No	Yes	No	(as in-process tests within cases)	377
U3 QC Parameters for Solid Dosage Forms	Yes	Yes	No	ABQ 7.3.1–7.3.3	385
U4 QC Parameters for Semi-solid Dosage Forms	Yes	Yes	No	ABQ 7.4.1–7.4.3; Paper Q1	394
U5 QC Parameters for Liquid and Gaseous Dosage Forms	Yes	Yes	No	ABQ 7.5.1–7.5.4; Paper Q3.5	399
Module 8 — Nano-science and Nanotechnology (5 to 20 marks)					
U1 Introduction to Nanotechnology	No	Yes	Yes	LAQ B8.1	421
U2 Ayurvedic Formulations and Nanotechnology	Yes	Yes	No	ABQ 8.2.1–8.2.3	429
U3 Nanocarriers for Ayurvedic Formulations	No	Yes	No	Paper Q3.7	439
U4 Regulatory Aspects and Safety Concerns	Yes	Yes	No	ABQ 8.4.1–8.4.3	446
U5 Recent Advances and Future Directions	No	Yes	No	—	454

All modules belong to Semester II. LAQ units that already carry ABQs (M1U3, M5U3, M5U4, M5U7, M7U1) are covered by Part A; their cases can also be practised as LAQs by attempting any two sub-parts.

CODES AND ABBREVIATIONS

Domain (Bloom's taxonomy) and Level (Miller's pyramid) — as in the curriculum

CK	Cognitive / Knowledge	**K**	Knows
CC	Cognitive / Comprehension	**KH**	Knows how
CAP	Cognitive / Application	**SH**	Shows how
CAN	Cognitive / Analysis	**D**	Does
CS	Cognitive / Synthesis		
CE	Cognitive / Evaluation		
PSY-SET, -GUD, -MEC, -ADT, -ORG	Psychomotor: Set, Guided response, Mechanism, Adaptation, Origination		
AFT-REC, -RES, -VAL, -SET, -CHR	Affective: Receiving, Responding, Valuing, Organisation, Characterisation		

Examination and curriculum

ABQ	Application-Based Question (20 marks)
LAQ	Long Answer Question — analytical, structured (10 marks)
SAQ	Short Answer Question (5 marks)
M7U1	Module 7, Unit 1
F1, F2…	Findings of a case, numbered in this book
P1, P2.1…P3.8	Questions of the October 2026 preliminary paper, solved (used in the Topic Finder)

Regulatory and institutional

API	Ayurvedic Pharmacopoeia of India
AFI	Ayurvedic Formulary of India
ASU	Ayurveda, Siddha and Unani
BMR	Batch Manufacturing Record
CAPA	Corrective and Preventive Action
CCSEA	Committee for Control and Supervision of Experiments on Animals (formerly CPCSEA)
CDSCO	Central Drugs Standard Control Organisation
CoA	Certificate of Analysis
CoPP	Certificate of Pharmaceutical Product
CRM	Certified Reference Material
GMP	Good Manufacturing Practices
G.S.R.	General Statutory Rules (Gazette notification number)
IAEC	Institutional Animal Ethics Committee
ICH	International Council for Harmonisation
IP	Indian Pharmacopoeia
NABL	National Accreditation Board for Testing and Calibration Laboratories
NCISM	National Commission for Indian System of Medicine

OECD	Organisation for Economic Co-operation and Development
PCIM&H	Pharmacopoeia Commission for Indian Medicine and Homoeopathy
QA / QC	Quality Assurance / Quality Control
SOP	Standard Operating Procedure

Analytical and instrumental

AAS	Atomic Absorption Spectroscopy
AUC, C_{max}, T_{max}	Area under the curve; peak concentration; time to peak
D10, D50, D90	Particle sizes below which 10, 50 and 90 percent of the sample lies
DLS	Dynamic Light Scattering
DTA / TGA	Differential Thermal Analysis / Thermogravimetric Analysis
FTIR	Fourier Transform Infrared Spectroscopy
GC, GC-MS	Gas Chromatography; with Mass Spectrometry
HPLC, HPTLC	High-Performance Liquid / Thin-Layer Chromatography
ICP-OES, ICP-MS	Inductively Coupled Plasma Optical Emission Spectroscopy / Mass Spectrometry
LC-MS	Liquid Chromatography–Mass Spectrometry
LOD / LOQ	Limit of Detection / Limit of Quantification (in Part A also: Loss on Drying, written in full)
NMR	Nuclear Magnetic Resonance
PDA	Photodiode Array detector
PDI	Polydispersity Index
PSA	Particle Size Analyser
R_f	Retardation factor
SEM-EDAX	Scanning Electron Microscopy with Energy-Dispersive X-ray Analysis
TEM	Transmission Electron Microscopy
TLC	Thin-Layer Chromatography
VSM	Vibrating Sample Magnetometer
XRD / XRF	X-ray Diffraction / X-ray Fluorescence

Introduction — Reading and Answering a Case

Why this book

The NCISM curriculum gives the Application-Based Question (ABQ) one-fifth of the theory paper: one question, 20 marks. In the exam corners of Volume 0, ABQs were set as short stems of two or three lines. University papers now set them differently. The October 2026 preliminary examination of Parul University gave a long case: a pharmacy, an inspection, four concrete defects, and four sub-parts, each tagged with a Bloom's domain and a Miller's level. The analytical LAQs in the same paper were also built on short cases.

Scholars asked for practice in this format. This book gives it: 53 long case ABQs from every unit that Table 6F allows, 10 case-based analytical LAQs, the October 2026 preliminary paper solved in full — all 14 questions — and a model answer framework for each.

What the curriculum says about the theory paper

Rule (Table 6E and 6G)	What it means for you
One ABQ of 20 marks; 8 SAQs of 5 marks; 4 LAQs of 10 marks	The ABQ is the single biggest question. Plan 35 minutes for it
Questions are drawn only from units marked "Yes" in Table 6F	ABQs can come from only 17 units. All 17 are covered in Part A
No other question is asked from the module on which the ABQ is framed	If the ABQ is from Module 7, the rest of the paper skips Module 7
Except for the ABQ module, at least one SAQ comes from each module	Every module needs revision for SAQs
LAQs are "analytical based structured questions assessing the higher cognitive ability"	LAQs are no longer "write an essay on…". Expect a short case or a critical question. See Part B

How a case ABQ is built

Every case in this book, like the university paper, has three layers.

1. **The setting** — a pharmacy, a laboratory, a research project, an inspection.
2. **The findings** — four to seven concrete observations, numbered here as F1, F2 and so on. Each finding is there for a reason. Most sub-parts can be answered only by using them.
3. **The sub-parts** — usually (a) to (d). Each carries a domain tag and a level tag. The tag tells you what kind of thinking the examiner wants.

A finding that you do not use in your answer is a mark you did not claim.

The seven-step method

Step	Time	What to do
1	2 min	Read the *sub-parts* first. You then read the case knowing what to look for
2	3 min	Read the case. Number the findings F1, F2… in the margin of the question paper
3	2 min	Against each finding, write one word: what does it show? (contamination, wrong endpoint, no record, wrong phase…)
4	1 min	Map findings to sub-parts. Most findings feed more than one sub-part
5	2 min	Plan each sub-part as a heading and a table or a short list
6	23 min	Write. Give each sub-part space in proportion to its marks — about one page for 5 marks
7	2 min	Close each sub-part with a one-line judgement. Check that every finding has been used

The whole paper is 100 marks in 180 minutes — a little under two minutes a mark. The ABQ therefore deserves about 35 minutes.

Reading the domain tag

The domain tag uses the codes of the curriculum (Bloom’s taxonomy, cognitive domain). Each code asks for a different kind of answer.

Tag	Meaning	Verbs you will see	What a good answer does
CK	Knowledge	List, define, name, state	Gives correct facts, completely, in an organised list or table
CC	Comprehension	Explain, describe, discuss	Explains *why* and *how*, in your own words, with examples from the case
CAP	Application	Apply, calculate, select, use	Uses a principle or formula on the data given; shows working
CAN	Analysis	Analyse, interpret, compare, identify causes	Breaks the case into parts; links each finding to its cause or meaning
CS	Synthesis	Design, plan, propose, develop, write an SOP	Builds something new and complete: a protocol, a formula, an SOP, a plan
CE	Evaluation	Evaluate, judge, justify, critique	Weighs a claim against evidence; says what is supported and what is not; gives a verdict

Reading the level tag

The level tag uses Miller's pyramid: Knows (K), Knows how (KH), Shows how (SH), Does (D). A written paper can test only the first two. Shows how and Does are tested in the practical examination, the viva and the logbook. So in this book, written sub-parts are tagged K or KH.

- **K (Knows):** recall of facts.
- **KH (Knows how):** using knowledge to explain, interpret, decide or plan. Almost every ABQ sub-part is KH.

An answer template for one sub-part

> **(b) Six QC parameters for a finished Avaleha — 5 marks**
>
> *One opening line* that answers the question directly.
>
> *A table* — parameter, method, why it matters, which finding it addresses (F1, F3…).
>
> *One closing line* — the judgement, or the boundary between what is established and what is not.

Examiners award marks for structure. A table that links findings to answers shows the examiner, at a glance, that you have read the case.

Six ways to lose marks

1. Writing a general essay on the topic and never using the findings.
2. Giving lists without reasons ("add a preservative" — which, where, why?).
3. Ignoring numbers in the case. If the case gives 22 percent moisture or 14 ppm lead, compare it with a limit.
4. Over-claiming: "proves", "completely", "non-toxic", "100 percent nano". A careful answer scores higher than an over-claimed one.
5. Leaving out safety where the case involves mercury, arsenic, lead or other hazards.
6. Spending 50 minutes on the ABQ and running short for the LAQs.

Marking the boundary

This subject attracts over-claiming. Many cases in this book end with a claim — by a scholar, a supplier or a sponsor — that goes beyond the data. Learn to write in three columns:

What the data show	What is reasonably proposed	What is not established
Measured results, stated plainly	Mechanisms that fit the data and published work	Claims that need more evidence

Where published data support a classical claim, say so. Where they do not yet, say that too. A tradition defended by overstatement is not defended.

How to use the model answer frameworks

Each framework gives the points an examiner looks for, in the order of the sub-parts, with a mark split. It is a skeleton, not an essay. Your written answer should put these points into full sentences, add a diagram where it helps, and keep the tables.

The frameworks start on a fresh page. Attempt the case first, in 35 minutes, on paper. Then turn the page.

The October 2026 Preliminary Paper — Solved

About this paper

Parul University, Faculty of Ayurved · MS/MD (Ayurved) Preliminary Examination, Semester 2, October 2026 · Subject: Applied Basics of Rasashastra and Bhaishajya Kalpana · Subject code AYPG-AB-RB (printed on the paper as "AYPG-AB-RG") · Time: 3 hours · Maximum marks: 100 · Instructions: all questions compulsory · Reproduced for teaching.

This chapter solves every question of the paper, in the order set: one ABQ, four analytical LAQs and eight SAQs — the pattern of Table 6E. Each question is reproduced with its original wording and its printed Domain and Level tags. Finding numbers (F1, F2…) and a suggested split of marks have been added to help you plan; the paper itself did not split the marks.

The paper against the blueprint (Tables 6F and 6G)

Q	Topic	Unit	Table 6F for this type	Note
1 (ABQ)	QC and QA of Avaleha	M7U1, M7U4	Yes	—
2.1 (LAQ)	Metal and mineral databases	M2U2	Yes	—
2.2 (LAQ)	Disease in laboratory animals	M6U4	Yes	—
2.3 (LAQ)	Importance of Shodhana and Marana	M4U1	No	M4U1 is eligible for ABQ and SAQ, not LAQ
2.4 (LAQ)	Phytochemistry and extraction	M3U4	Yes	—
3.1 (SAQ)	Chemistry principles in Rasashastra	M3U1	Yes	—
3.2 (SAQ)	Musha and crucibles	M5U2	Yes	—
3.3 (SAQ)	Microscopy	M5U5	Yes	—
3.4 (SAQ)	Animal house management	M6U1	Yes	—
3.5 (SAQ)	QC of liquid dosage forms	M7U5	Yes	Same module as the ABQ (Table 6G, instruction 5)
3.6 (SAQ)	Plant databases	M2U3	Yes	—

Q	Topic	Unit	Table 6F for this type	Note
3.7 (SAQ)	Types of nanocarriers	M8U3	Yes	—
3.8 (SAQ)	Principles and types of calibration	M5U6	Yes	—

Table 6G (instruction 6) also asks for at least one SAQ from every module except the one carrying the ABQ. This paper has no SAQ from Module 1 or Module 4; Module 1 does not appear at all. These notes are given so that scholars understand the curriculum’s rules. In any examination, answer the question that is set.

Q1 — Application-Based Question (20 marks)

Q 1 of the paper · Anchor: M7U1 (1.1 to 1.5) and M7U4 (4.1) · Also draws on: M7U2 · Domain and level as printed: CC, KH · Suggested time: 35 minutes

THE CASE

A GMP certified pharmacy manufactures Chyavanprasha Avaleha. During a routine inspection, the Drug Inspector found the following:

F1 The Amalaki pulp used was not tested for heavy metals or pesticide residues before use.

F2 The Paka endpoint was decided by the operator's guess; no Tantuvata or Ghanavata Pariksha was recorded in the BMR.

F3 The finished product showed fungal growth within 20 days, and the moisture content was 22%.

F4 The Batch Manufacturing Record had no signature of the QA officer, and the jaggery lot number was missing.

Q	Question	Marks	Domain	Level
(a)	Explain why QC and QA are critically important in Avaleha Kalpana with reference to the defects observed above (fungal growth, untested raw material).	5	CC	KH
(b)	List six QC parameters in detail that must be tested for a finished Avaleha to prevent the defects seen in this case.	5	CC	KH
(c)	Which regulatory provisions (Schedule T / API / AFI) were violated?	5	CC	KH
(d)	How would proper documentation have prevented these failures? Explain in detail.	5	CC	KH

The paper gave one Domain/Level tag (CC, KH) for the whole question and did not split the marks. A split of 5 marks per sub-part is assumed here. The finding numbers F1 to F4 have been added. This ABQ draws on Module 7, Units 1 and 4 (both ABQ-eligible), and uses the Avaleha Siddhi Lakshana of Unit 2 as in-process tests.

Step 3 of the method, done for you

Finding	What it shows	Feeds sub-part
F1	Raw material control failed — a safety hazard entered the batch and cannot be removed later	(a), (b), (c), (d)
F2	In-process control failed — endpoint not tested, not recorded	(a), (c), (d)
F3	Consequence: too much water → high water activity → mould. A product failure	(a), (b), (c)
F4	Documentation and release failed — no QA approval; no traceability of jaggery	(a), (c), (d)

Model answer — written in full

(a) Why QC and QA are critically important in Avaleha Kalpana — 5 marks

QC tests the product and its materials; QA is the system that prevents defects from arising. Avaleha needs both more than most dosage forms, for four reasons.

Feature of Avaleha	Why it makes QC and QA critical	Defect in this case
Many raw materials — Chyavanaprasha has several dozen herbs, Amalaki pulp, jaggery or sugar, ghee, honey	Each is a route for contamination: heavy metals and pesticides from fruit and soil, microbes from pulp, adulterated jaggery	F1, F4
Raw material defects cannot be removed by processing	Lead or pesticide in the pulp goes straight into the finished product. Only testing before use can stop it	F1
The quality depends on the Paka endpoint	Under-cooking leaves too much water; over-cooking burns and hardens. The endpoint must be tested and recorded every time	F2
High sugar and fruit pulp	Safe only when water content is low. Above that, water activity rises and yeasts and moulds grow	F3

QC would have detected the untested pulp (F1) and the 22% moisture (F3) before release. *QA* would have prevented them: by requiring raw material release before use, a written endpoint procedure, a complete BMR, and release only after QA review (F2, F4). Fungal growth within 20 days shows that neither worked. The result is a product unsafe to use, a recall, and loss of trust.

(b) Six QC parameters for a finished Avaleha — 5 marks

No.	Parameter and method	Why it prevents the defects
1	*Loss on drying (moisture)* — API general method	The single most important test here. A properly made Avaleha has low moisture; API monographs set the limit for each Avaleha (for example, the Vasavaleha monograph allows not more than about 12%). 22% (F3) is far above any such limit and explains the mould
2	*Total sugar, reducing and non-reducing sugar*	Shows the degree of inversion during Paka. Too little inversion → crystallisation; too much water and low sugar → fermentation
3	*pH (of a 10% aqueous solution)*	A falling pH on storage is an early sign of microbial activity; also a batch-consistency check
4	*Fat content; acid value and peroxide value of the extracted fat*	Detects rancidity of the ghee and oil
5	*Microbial limits* (API Appendix 2.4) — total bacterial count not more than 10^5/g; total yeast and mould not more than 10^3/g; *E. coli*, *Salmonella*, *S. aureus*, *P. aeruginosa* absent	Directly tests for the fungal contamination seen in F3; must pass before release

No.	Parameter and method	Why it prevents the defects
6	*Heavy metals* (API Appendix 2.3: lead 10, arsenic 3, cadmium 0.3, mercury 1 ppm) *and pesticide residues* (API Appendix 2.5); *aflatoxins* (Appendix 2.7)	Safety tests that address F1 — on the raw pulp before use, and on the finished product

Also tested: identity (organoleptic, TLC/HPTLC fingerprint), total ash and acid-insoluble ash. And as *in-process* tests at the Paka endpoint, the Avaleha Siddhi Lakshana: Tantumatva (the mass draws into threads between fingers — the "Tantuvata" of the question), Apsu majjanam (a drop sinks in water without dispersing), Peedite mudra (a finger impression remains), and firmness on cooling — each recorded in the BMR.

(c) Regulatory provisions violated — 5 marks

Provision	Requirement	Violated by
Schedule T of the Drugs Rules, 1945 (GMP for ASU drugs, under Rule 157)	Raw materials tested and approved before use; in-process controls; complete batch manufacturing records with lot numbers of all materials; QC testing of finished product; release only after QC/QA approval; proper storage	F1, F2, F4 (and release of F3)
Ayurvedic Pharmacopoeia of India (API)	Raw drug monograph for Amalaki, including "other requirements" — heavy metals, pesticide residues, microbial limits, aflatoxins; general methods in the appendices; finished product standards where a monograph exists	F1, F3
Ayurvedic Formulary of India (AFI)	Chyavanaprasha must be made as per the formula and method of the authoritative text, including the Paka described	F2 (method not followed as described), F3
Drugs and Cosmetics Act, 1940, Chapter IVA	A drug that contains any filthy, putrid or decomposed substance may be deemed adulterated (Section 33EE)	F3

Note the 2026 update: Schedule T was substantially revised by the Ministry of Ayush (Drugs (Eleventh Amendment) Rules, 2026, G.S.R. 657(E), dated 24 July 2026), with a risk-based quality management system. Existing units have until 31 July 2029 to comply. The defects in this case violate both the old and the revised Schedule T.

(d) How proper documentation would have prevented these failures — 5 marks

Documentation is not paperwork after the event. It is the control that forces each check to happen, and proves that it happened. "If it is not written, it was not done."

Document	How it would have prevented the failure	Finding
Raw material specification and test record; approved supplier list; status labels (Quarantine / Approved / Rejected)	Pulp could not be issued to production without an “Approved” label backed by a test report for heavy metals and pesticides	F1
SOP for Paka with a defined endpoint; BMR with spaces for endpoint tests, time and temperature, with operator and supervisor signatures	The operator could not complete the BMR without recording Tantumatva and the other tests; a guess would show as a blank	F2
In-process and finished-product test records (moisture, microbial limits)	22% moisture would have been caught before filling; the batch would have been held	F3
BMR listing lot numbers of every material (jaggery included)	Traceability: if jaggery is the source of contamination, all batches using that lot can be found and recalled	F4
QA review and release record	No batch leaves without a QA signature after all results are reviewed	F4, F3
Deviation, complaint and recall procedures; retained samples	Allow investigation and quick withdrawal when a failure is found	All

Good records follow ALCOA+: attributable, legible, contemporaneous, original, accurate — and complete, consistent, enduring and available. A BMR without a QA signature and with a missing lot number fails the very first test of a record: it cannot show who approved the batch, or what went into it.

Revise: Volume 0 — Module 7, Unit 1 (p. 368), Unit 2 (p. 377) and Unit 4 (p. 394).

Q2 — Analytical Structured LAQs (10 marks each)

Each LAQ starts on a new page. The paper gives each LAQ as a single question; a suggested structure and split of marks is added so that the answer covers every part of what is asked.

Q2.1 — Databases and the Heavy Metal Debate

Q 2.1 of the paper · Anchor: M2U2 (2.1 to 2.5) · Domain and level as printed: CC, KH · Suggested time: 18 minutes

THE CASE

Critically analyze the role of mineral and metal databases in addressing the global safety concerns regarding heavy metal toxicity in Ayurvedic medicines. How can these databases help regulators like AYUSH differentiate between prepared therapeutic bhasma (e.g. Abhraka Bhasma) and accidental toxic contaminants (e.g., lead or arsenic contamination)?

Q	Question (structure suggested for answering)	Marks	Domain	Level
(a)	The global concern, and the databases available	3	CC	KH
(b)	How the databases help to tell an intended Bhasma phase from a contaminant	4	CC	KH
(c)	Critical analysis: limitations, and what is needed	3	CC	KH

Model answer framework

(a) The concern and the databases — 3 marks

- *Concern:* published surveys abroad have found lead, mercury and arsenic in some Ayurvedic products, and case reports of poisoning exist. The debate mixes two different things: metals that are *intended* ingredients of Rasaushadhi, and metals that are *contaminants*.
- *Databases:*

Database	What it holds
USGS mineral resources databases (for example the Mineral Resources Data System)	Locations, deposits and commodity data for mineral resources
Mindat (“Mineral Database”)	Mineral species, chemistry, associated minerals, localities
Webmineral	Mineral data: formula, crystal system, physical properties, and diffraction data
Crystallographic and diffraction databases (RRUFF; American Mineralogist Crystal Structure Database; Crystallography Open Database; ICDD Powder Diffraction File)	Reference X-ray diffraction patterns and Raman spectra for phase identification
Metal and materials databases (for example NIST Chemistry WebBook; Materials Project)	Properties, thermodynamic data, structures of metals and compounds

(b) How they help tell a Bhasma from a contaminant — 4 marks

1. *Phase identification.* XRD of a Bhasma is matched against reference patterns. Abhraka Bhasma shows expected phases (iron oxide such as hematite, and silicate phases). A contaminant appears as a phase that should not be there — for example lead oxide, metallic lead, or arsenic trioxide (arsenolite).
2. *Expected elements.* Mineral databases give the formula and usual companion elements of the source mineral (biotite: K, Mg, Fe, Al, Si). Lead or arsenic is not part of biotite. Their presence points to contamination from the source ore, the process or the equipment.
3. *Speciation, not totals.* Databases link each compound to its solubility and properties, supporting the principle that risk depends on the chemical form.
4. *Building a reference.* Regulators and the Pharmacopoeia Commission could compile reference patterns and elemental profiles of authentic Bhasmas — a “fingerprint library” — against which market samples are compared.

(c) Critical analysis — 3 marks

- Databases describe *minerals and pure compounds*, not processed formulations. A Bhasma is a mixture of phases, amorphous matter and organic residue; database matching shows only the crystalline part.
- Detection limits of XRD (about 1 to 5 percent) are far above toxic contamination levels (ppm). Elemental analysis (ICP-MS, AAS) and speciation methods are still essential.
- No database decides safety. Safety needs dose, solubility, absorption and toxicity data.
- What is needed: an Ayurveda-specific, curated database of authenticated Bhasma reference data (XRD, elemental profiles, particle size), linked to pharmacopoeial standards. At present this is a proposal, not an established regulatory tool.

Revise: Volume 0 — Module 2, Unit 2 (p. 102) and Unit 4 (p. 117).

Q2.2 — Sneezing Rats and Fur Mites

Q 2.2 of the paper · Anchor: M6U4 (4.1, 4.2, 4.3) · Domain and level as printed: CC, KH · Suggested time: 18 minutes

THE CASE

During an Ayurvedic drug safety study, 10 out of 50 Wistar rats develop sneezing and weight loss. Simultaneously, fur mites are found in the mouse colony. No vaccination records exist. Identify the most probable disease in the rats. Is it bacterial, viral, or parasitic? How would this infection compromise the toxicity study data? List five preventive measures you would implement to avoid such outbreaks in the future.

Q	Question (structure suggested for answering)	Marks	Domain	Level
(a)	Most probable disease; its nature; the meaning of the mites and of "no vaccination records"	4	CC	KH
(b)	How the infection compromises the toxicity data	3	CC	KH
(c)	Five preventive measures	3	CC	KH

Model answer framework

(a) The disease — 4 marks

- *Most probable:* murine respiratory mycoplasmosis, caused by *Mycoplasma pulmonis* — a **bacterial** infection (mycoplasmas are bacteria without a cell wall). It causes sneezing, snuffling, laboured breathing, weight loss and chronic lung disease in rats.
- Viral infections (for example sialodacryoadenitis virus or Sendai virus) can also cause sneezing and often occur together with mycoplasma, worsening it. Confirm by PCR, serology, culture and lung histopathology.
- *Fur mites* in the mouse colony (for example *Myobia*, *Radfordia* or *Myocoptes* species) are a separate, **parasitic** infestation. Their presence shows a breakdown of colony health control and biosecurity.
- *"No vaccination records":* routine vaccination is not the usual method of disease control in laboratory rats and mice. Their health is protected by sourcing from health-monitored suppliers, quarantine, barrier housing and routine health monitoring. The absence of *health monitoring* records is the real failure.

(b) How the data are compromised — 3 marks

- Weight loss confounds body weight and food intake, which are key endpoints.
- Lung lesions and inflammatory changes in histopathology may be wrongly attributed to the test drug.
- Infection alters immune function, liver enzymes, blood counts and drug metabolism — so clinical chemistry and haematology become unreliable.
- Increased variability and possible deaths reduce statistical power. The study fails quality standards and may need to be repeated, wasting animals — against the 3Rs principle.

(c) Five preventive measures — 3 marks

1. Procure animals only from accredited breeders with health certificates; specific pathogen-free stock where possible.
2. Quarantine and health screening of new arrivals before they join the colony.
3. A health monitoring programme with sentinel animals and periodic testing (for example following FELASA recommendations); keep health records.
4. Barrier husbandry: separate rooms for species and studies; individually ventilated cages; good ventilation (ammonia worsens respiratory mycoplasmosis); regular sanitation; pest control.
5. Parasite control: periodic examination of fur and faeces; treat infested colonies (for example with an acaricide under veterinary advice); decontaminate rooms; staff PPE and change of gowns between rooms.

Also: institutional animal ethics approval and facility standards as per CCSEA guidelines.

Revise: Volume 0 — Module 6, Unit 4 (p. 343) and Unit 1 (p. 325).

Q2.3 — Importance of Shodhana and Marana in Clinical Practice

Q 2.3 of the paper · Anchor: M4U1 (1.1 to 1.4) · Also draws on: M4U4, M7U2 · Domain and level as printed: CAN, KH · Suggested time: 18 minutes

> **THE CASE**
>
> Describe the importance of ‘Sodhana’ and ‘Marana’ in Rasashastra with special reference to clinical practice.

Q	Question (structure suggested for answering)	Marks	Domain	Level
(a)	Meaning and aims of Shodhana and Marana	2	CAN	KH
(b)	Importance in pharmaceutical terms — what the processes change, with examples	3	CAN	KH
(c)	Importance in clinical practice	3	CAN	KH
(d)	What the evidence shows, and the practitioner’s duties	2	CAN	KH

Note on the blueprint. This topic belongs to Module 4, Unit 1. Table 6F marks that unit “Yes” for ABQ and SAQ but “No” for LAQ. Under the curriculum, the same content would come as an LAQ only through Unit 2 or Unit 3 (the SOPs of Shodhana and Marana). The paper spells the term “Sodhana”; the standard spelling, Shodhana, is used in the answer.

Model answer framework

(a) Meaning and aims — 2 marks

- *Shodhana* is the set of processes that remove the Doshas of a drug — physical impurities and harmful properties — and impart desired qualities. It is general (Samanya) or specific to a drug (Vishesha).
- *Marana* converts a Shodhita metal or mineral into a fine, light, lustreless powder fit for internal use (Bhasma), through Bhavana and repeated Puta.
- Together they aim to make the drug *safe, assimilable, effective in a small dose, and stable*.

(b) Pharmaceutical importance — 3 marks

Drug	What the process does	Chemical basis (where known)
Tamra	Removes free copper that causes vomiting and giddiness	Copper and its oxide converted to sulphide; Amla Pariksha turns negative
Lauha	Makes the metal brittle, then converts it to a fine Bhasma	Metallic iron oxidised step by step to hematite (Fe_2O_3); no free metal by XRD
Vatsanabha (Shodhana in Gomutra)	Reduces toxicity	Diester alkaloids hydrolysed to far less toxic monoester and non-ester forms
Kupilu (Gomutra, then milk)	Reduces toxicity	Published studies report lower strychnine and brucine

Drug	What the process does	Chemical basis (where known)
Bhallataka (rubbed with brick powder)	Removes the irritant oil	The oil causes contact dermatitis; removing it protects skin and mucosa
Shankha, Godanti	Converts to a fine, soft powder	Calcination ($CaCO_3$), dehydration of gypsum ($CaSO_4 \cdot 2H_2O \rightarrow CaSO_4$)

General points: size reduction, change of form, removal of free metal, formation of stable compounds, coating with organic matter from the media (Samskara — "Samskaro hi gunantaradhanam"), and defined endpoints (Rekhapurnata, Varitara, Nishchandra, Apunarbhava, Niruttha).

(c) Importance in clinical practice — 3 marks

- *Safety.* Classical texts describe adverse effects of Ashuddha or Apakva (unprocessed or under-processed) drugs. Several match modern chemistry — for example, soluble copper is a gastric irritant and emetic; free mercury vapour is toxic.
- *Small dose.* Bhasmas act in milligram doses, which makes long-term administration practical.
- *Tolerance and palatability.* A properly made Bhasma is free of metallic taste and is tolerated without gastric irritation.
- *Action and assimilation.* Classical texts describe processed drugs as Sukshma and Yogavahi — able to carry and enhance the action of co-administered drugs. Bhavana adds properties of the herbs used.
- *Choice of drug.* Shodhita Vatsanabha, Kupilu and Bhallataka are used clinically only in the processed form.

(d) Evidence, and the practitioner's duties — 2 marks

Established	Proposed	Not established
Chemical changes documented by XRD, HPLC and elemental analysis; reduced acute toxicity of several processed drugs in animal studies	Better absorption; a protective role of organic coating; relevance of small particle size	Long-term human safety of metal-containing medicines through large studies; safety at any dose

Duties of the practitioner: prescribe only properly processed, tested medicines from licensed sources; use the right dose, Anupana and duration; advise Pathya; monitor patients on long-term metal-containing medicines (liver and kidney function, and blood metal levels where indicated); report adverse reactions through the Ayush pharmacovigilance network.

Revise: Volume 0 — Module 4, Unit 1 (p. 207) and Unit 4 (p. 232); see also ABQ 4.1.1, 4.4.1 and 3.3.1 in this book.

Q2.4 — Phytochemistry and Extraction Techniques

Q 2.4 of the paper · Anchor: M3U4 (4.1, 4.2, 4.4) · Domain and level as printed: CC, KH · Suggested time: 18 minutes

THE CASE

What is phytochemistry and its significance in explaining Ayurvedic pharmacology. Describe in detail different extraction techniques.

Q	Question (structure suggested for answering)	Marks	Domain	Level
(a)	Definition of phytochemistry and its significance in explaining Ayurvedic pharmacology	4	CC	KH
(b)	Extraction techniques in detail	6	CC	KH

Model answer framework

(a) Phytochemistry and its significance — 4 marks

Definition. Phytochemistry is the study of the chemical constituents of plants — their structure, biosynthesis, extraction, isolation, identification and biological role.

Significance in explaining Ayurvedic pharmacology:

Ayurvedic concept	Phytochemical correlate (proposed)	Example
Tikta Rasa	Alkaloids, bitter terpenoids and glycosides	Berberine in Daruharidra; andrographolide in Kalmegha
Kashaya Rasa	Tannins (protein-precipitating, astringent)	Haritaki, Amalaki
Madhura Rasa	Sugars; sweet saponins	Glycyrrhizin in Yashtimadhu
Katu Rasa	Pungent compounds	Piperine in Pippali and Maricha; gingerols in Shunthi
Amla Rasa	Organic acids	Ascorbic acid in Amalaki; citric acid in Nimbu
Yogavahi	Bioavailability enhancers	Piperine raises blood levels of curcumin
Specific actions	Active constituents	Vasicine (Vasa) — bronchodilator; withanolides (Ashwagandha); guggulsterones (Guggulu)

Other uses: identity and standardisation by marker compounds; detection of adulteration; stability studies; safety (toxic constituents such as aconitine).

Boundary: these correlations are partial. Rasa, Guna, Virya and Vipaka describe the whole drug, and a single compound rarely explains a whole formulation. Synergy between constituents is a reasonable proposal, still being studied.

(b) Extraction techniques — 6 marks

Technique	Principle	Suited to	Merits	Limits	Ayurvedic parallel
Maceration	Drug soaked in solvent at room temperature, with shaking, for days	Heat-sensitive constituents	Simple; no heat	Slow; incomplete	Hima (cold infusion)
Infusion	Short soaking in hot water	Readily soluble, delicate constituents	Quick	Weak extract	Phanta
Decoction	Boiling in water and reducing	Hard, heat-stable, water-soluble constituents	Simple; traditional	Heat damages labile constituents	Kwatha
Percolation	Fresh solvent passes slowly through packed drug	Exhaustive extraction	More complete than maceration	Slow; more solvent	—
Soxhlet (continuous hot extraction)	Condensed solvent repeatedly cycles through the drug	Exhaustive extraction with little solvent	Efficient	Long heating degrades labile compounds	—
Ultrasound-assisted	Cavitation breaks cell walls	Most plant materials	Fast; low temperature	Equipment; scale-up	—
Microwave-assisted	Microwaves heat solvent and cell water rapidly	Most plant materials	Very fast; less solvent	Uneven heating; not for volatile or heat-labile compounds without control	—
Supercritical fluid extraction	CO_2 above its critical point (31 °C, 73.8 bar) acts as a tunable solvent	Lipophilic constituents, essential oils	No solvent residue	Costly; CO_2 is non-polar (co-solvent needed for polar compounds)	—
Steam / hydro-distillation	Volatile oils carried over with steam and condensed	Essential oils	Simple	Heat; only volatiles	Arka
Expression	Pressing fresh material	Juices	No solvent	Perishable	Svarasa
Fermentation-based	Self-generated alcohol extracts constituents over time	Mixed constituents	Self-preserving	Slow; needs control	Asava-Arishta

Choice of solvent: "like dissolves like" — hexane for fats and waxes; chloroform or ethyl acetate for aglycones and alkaloid bases; methanol, ethanol and water for glycosides, phenolics and sugars. Successive extraction from non-polar to polar separates classes. Report the yield and the conditions.

Revise: Volume 0 — Module 3, Unit 4 (p. 171); see also LAQ B3.4 in this book.

Q3 — Short Answer Questions (5 marks each)

An SAQ of 5 marks deserves about eight to nine minutes and one page. Open with a one-line definition, give one well-organised table or list, and close with one line of application or judgement.

Q3.1 — Basic Chemistry Principles in Rasashastra

Q	Question	Marks	Domain	Level
3.1	Describe the role of basic chemistry principles and their application in Rasashastra.	5	CC	KH

Model answer

Rasashastra transforms metals, minerals and poisonous drugs by heat, liquids and trituration. Chemistry explains *how* these changes happen and gives tools to check them. (Anchor: M3U1.)

Principle	Application in Rasashastra
Atomic structure and the periodic table	Reactivity of metals: zinc and lead oxidise easily (Jarana); gold and silver are noble and need special media (Parada, Gandhaka) for Marana
Chemical bonding	HgS (Kajjali, Rasasindura) is a stable, mainly covalent compound; Lavana and Kshara are ionic salts
Oxidation and reduction	Marana as oxidation (iron → Fe_2O_3); Hingulottha Parada (HgS → Hg); Satvapatana as reduction of ore to metal
Variable oxidation states	Fe(II)/Fe(III) and Cu(I)/Cu(II) in Lauha and Tamra Bhasma; the basis of Amla Pariksha
Acids, bases and salts	Acidic media (Nimbu, Kanji) and alkaline media (Gomutra, Kshara) in Shodhana
Complex formation	Binding of metal ions by plant constituents in Bhavana and Amritikarana (proposed)
Thermal decomposition	Shankha ($CaCO_3$ → $CaO + CO_2$); dehydration of Godanti
Crystal structure and polymorphism	Black HgS in Kajjali, red HgS in Rasasindura; phase identification by XRD
Stoichiometry	Proportions of ingredients; HgS contains 86 percent mercury

Closing line: chemistry explains the processes and allows their standardisation; it works alongside the classical endpoints, not in place of them.

Q3.2 — Modern Aspects of Musha: Crucibles

Q	Question	Marks	Domain	Level
3.2	Analyse modern aspect of Musha with special reference to crucibles.	5	CAN	KH

Model answer

Musha is the classical crucible of Rasashastra. It is made of clay mixed with husk (Tusha), charcoal, iron rust (Lauha Kitta) and fibre — in modern terms, a refractory clay body with grog, organic burn-out material and fibre for strength. Types include Vrintaka, Gostani (for Satvapatana), Vajra (for severe heat), Malla and Andha Musha. (Anchor: M5U2.)

Modern crucible	Strengths	Limits	Use in Rasashastra and QC
Porcelain	Cheap; smooth; inert to most samples	Poor thermal shock resistance	Ash value; loss on ignition
Fireclay / clay	Cheap; refractory	Porous; can react with the charge	Melting low-melting metals
Alumina	Very refractory; chemically stable	Costly; poorer thermal shock resistance	High-temperature Bhasma trials
Graphite and clay-graphite	Excellent for melting metals; conducts heat	Oxidises in air — not for oxidising conditions	Melting and Satvapatana under reducing conditions
Silicon carbide	Strong; good thermal conductivity and shock resistance	Costly	Metal melting
Fused silica	Resists thermal shock and acids	Attacked by alkalis	Acidic charges
Platinum	Chemically inert; analytical use	Very costly; damaged by easily reducible metals such as lead and arsenic	Precise analytical ignition

Analysis. Modern crucibles offer known composition, standard size, high temperature use in electric and induction furnaces, reusability and record-keeping (each crucible logged by use). Choose by temperature, atmosphere (oxidising or reducing), chemical nature of the charge and thermal shock. But the classical Musha is not inert — its iron rust and organic matter can take part in the process. Whether an inert alumina crucible gives the same product must be answered by comparative study, not assumed.

Q3.3 — Microscopy: Principle, Types and Specifications

Q	Question	Marks	Domain	Level
3.3	Describe the principle, types and specifications of microscopy instruments.	5	CC	KH

Model answer

Principle. A microscope produces a magnified image of a small object. Three things matter: *magnification* (objective × eyepiece), *resolution* — the smallest distance between two points seen as separate, d = λ ÷ (2 NA), about 0.2 µm for the best light microscope — and *contrast*. Electron microscopes use electrons of much shorter wavelength, so they resolve far smaller detail. (Anchor: M5U5.)

Type	Principle	Use
Bright-field (compound)	Transmitted visible light	Powder microscopy of crude drugs
Stereo	Low-magnification three-dimensional view	Macroscopy; sorting
Polarising (petrological)	Polarised light; anisotropic materials show interference colours	Minerals; crystals; starch
Phase contrast and dark field	Convert phase differences or scattered light into contrast	Unstained cells; fine particles
Fluorescence	Specimen emits light when excited by UV or blue light	Specific constituents
SEM (with EDAX)	Scanned electron beam; secondary electrons form a surface image	Particle shape and size; elements
TEM	Electrons pass through a thin specimen	Nanoparticles; internal structure

Specifications to check: objectives (4×, 10×, 40×, 100× oil) with their numerical aperture; eyepieces (10×) with a reticle; Köhler illumination; condenser; mechanical stage; trinocular head with camera and measuring software; eyepiece and stage micrometers; polarising attachment. For electron microscopes: resolution, accelerating voltage, magnification range, detectors and EDAX.

Q3.4 — Animal House Management

Q	Question	Marks	Domain	Level
3.4	Write a short note on animal house management.	5	CC	KH

Model answer

Animal house management is the organised care of laboratory animals so that they stay healthy, their welfare is protected, and experimental results are reliable. (Anchor: M6U1.)

Area	Key points
Regulation	Establishment registered with CCSEA (formerly CPCSEA); Institutional Animal Ethics Committee (IAEC) approval for every study; Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998
Design and layout	Separate areas for quarantine, breeding, holding, procedures, cage washing, feed and bedding store, and waste; clean and dirty corridors; species kept apart; restricted entry
Environment (rats and mice, as in the BIS draft code of 2024, aligned with the CPCSEA guidelines of 2015)	Temperature about 20 to 26 °C; relative humidity 40 to 70 percent; 10 to 15 air changes per hour with fresh air; 12-hour light–dark cycle on a timer; low noise; ammonia kept low
Husbandry	Suitable cages and bedding; clean feed and water; regular cage change; environmental enrichment
Health	Quarantine of new animals; health monitoring with sentinels; veterinary care; pest control
Records and documentation	Animal procurement and stock register; breeding records; daily environment log; health and treatment records; IAEC approvals; SOPs
People	Trained staff; protective clothing; occupational health; waste and carcass disposal as per rules

Closing line: good management is both an ethical duty and a scientific need — stressed or sick animals give unreliable data.

Q3.5 — Quality Control Parameters of Liquid Dosage Forms

Q	Question	Marks	Domain	Level
3.5	Write about the quality control parameters of liquid dosage forms.	5	CC	KH

Note on the blueprint. This topic is Module 7, Unit 5 — eligible for SAQ in Table 6F. But the ABQ of this paper (Q1) is also from Module 7, and Table 6G (instruction 5) says no other question should be asked from the module on which the ABQ is framed. Under the curriculum's rules, a university paper with a Module 7 ABQ would not carry this SAQ.

Model answer

Liquid dosage forms are prone to microbial growth, sedimentation and chemical change, so their QC covers identity, physical stability and safety. (Anchor: M7U5.)

Common parameters: organoleptic (colour, odour, taste, clarity); pH; specific gravity; total solids; viscosity; TLC or HPTLC fingerprint and marker assay; microbial limits; heavy metals; fill volume and container integrity.

Dosage form	Specific parameters
Kashaya (bottled)	Total solids; sediment; preservative content; microbial limits; in-use period after opening

Dosage form	Specific parameters
Arka	Characteristic odour; volatile oil content; GC fingerprint and marker (for example thymol in Yavani Arka); clarity
Sharkara	Brix or total solids; total and reducing sugar; viscosity; absence of crystallisation and fermentation
Asava-Arishta	Alcohol content (self-generated); methanol at no more than traces; pH and total acidity; reducing and non-reducing sugar
Netra Bindu / Karna Bindu	Sterility (eye drops); clarity and particulate matter; pH and tonicity; preservative efficacy; for oil-based Karna Bindu — acid and peroxide value
Aerosols and sprays	Delivered dose uniformity; spray pattern; droplet size distribution; leak test; number of actuations

Closing line: the parameters are chosen by the risk of each form — sterility for the eye, alcohol for Arishta, sugar concentration for Sharkara.

Q3.6 — Plant Databases

Q	Question	Marks	Domain	Level
3.6	Write a shot note on Plant databases.	5	CC	KH

The paper prints “shot note”; read it as “short note”.

Model answer

Plant databases are organised online collections of plant names, classification, distribution, specimens, chemistry and genes. They help an Ayurveda scholar use the correct, current botanical name for a Dravya and find reliable data on it. (Anchor: M2U3.)

Database	Content	Note
The Plant List (TPL)	Working list of plant names; Kew and Missouri Botanical Garden	Not updated since 2013 (version 1.1). Use World Flora Online for current work
World Flora Online (WFO) and WFO Plant List	Consortium-curated taxonomic backbone; accepted names and synonyms	The WFO Plant List was launched in 2021 as the replacement for TPL
International Plant Names Index (IPNI)	Names of seed plants, ferns and lycophytes with their place of publication	A collaboration of Kew, Harvard University Herbaria and the Australian National Herbarium
Royal Botanic Gardens, Kew databases	Plants of the World Online (POWO) — accepted names, distribution, descriptions; herbarium catalogue	Useful for checking a name with its distribution
Other botanical resources	Tropicos (Missouri Botanical Garden); GBIF (occurrence records)	Specimens and distribution

Database	Content	Note
Plant genetic databases	PlantGDB (plant genome sequences and annotation); NCBI; Phytozome	PlantGDB has not been updated since 2015; current genomic work uses NCBI and Phytozome
Phytochemical databases (India)	IMPPAT — phytochemicals of Indian medicinal plants	Links plants to their constituents

Use in Ayurveda: verifying names and synonyms of Dravya, avoiding outdated names in theses and labels, checking distribution for authentication, and linking a plant to its chemistry. Always record the database and the date of access.

Q3.7 — Types of Nanocarriers

Q	Question	Marks	Domain	Level
3.7	Discuss the types of Nano carriers.	5	CC	KH

Model answer

Nanocarriers are nanometre-sized systems that carry a drug, protect it and improve its solubility, absorption or targeting. (Anchor: M8U3.)

Type	Structure	Advantages	Example of use with herbal drugs
Liposomes	Phospholipid bilayer vesicles with a watery core	Carry both water- and fat-soluble drugs; biocompatible	Curcumin liposomes (research)
Niosomes	Vesicles of non-ionic surfactants	More stable and cheaper than liposomes	Topical herbal preparations
Phytosomes	Complexes of a plant constituent with phospholipid	Better absorption of polyphenols	Silybin and curcumin phytosomes
Solid lipid nanoparticles (SLN) and nanostructured lipid carriers (NLC)	Solid (or solid–liquid) lipid core	Controlled release; stability	Lipophilic plant actives
Polymeric nanoparticles	Biodegradable polymers (for example PLGA, chitosan)	Controlled release; targeting	Curcumin nanoparticles
Polymeric micelles	Self-assembled amphiphilic polymers	Solubilise poorly soluble drugs	Poorly soluble constituents
Nanoemulsions	Oil droplets of nanometre size in water (or the reverse)	Easy to make; improve spreading and absorption	Conversion of Taila into an oil-in-water nanoemulsion

Type	Structure	Advantages	Example of use with herbal drugs
Dendrimers	Branched, tree-like polymers	Precise size; many surface groups	Mainly research
Inorganic and metallic nanoparticles	Gold, silver, silica	Imaging, antimicrobial uses	Green-synthesised silver nanoparticles
Ethosomes and transfersomes	Deformable, ethanol- or surfactant-rich vesicles	Skin penetration	Topical herbal products

Closing line: most nanocarrier work with Ayurvedic drugs is still pre-clinical; each carrier adds new excipients and needs its own safety data.

Q3.8 — Principles and Types of Calibration

Q	Question	Marks	Domain	Level
3.8	Discuss the principles and types of calibration.	5	CC	KH

Model answer

Calibration is the comparison of an instrument's reading with a reference standard of known value, under stated conditions, to find and record its error — and, where needed, to adjust it. (Anchor: M5U6.)

Principles:

- Comparison with a reference standard of higher accuracy.
- *Traceability* through an unbroken chain to national standards — in India, the CSIR–National Physical Laboratory — usually through a NABL-accredited calibration laboratory.
- Checks at several points across the working range, not one.
- Stated acceptance limits and measurement uncertainty.
- A defined frequency based on risk and use; documented results; a status label on the instrument; and an out-of-calibration procedure.

Type	Meaning	Example
Instrument calibration	The instrument itself checked against standards	Balance with certified weights; furnace thermocouple against a reference thermocouple
Method calibration	Response of an analytical method related to concentration	HPLC calibration curve; AAS standards
Internal (in-house) and external (accredited)	Routine checks by the laboratory; periodic calibration by an accredited agency	Daily balance check; annual certificate

Type	Meaning	Example
Single-point and multi-point	One reference value, or several across the range	pH meter with two or three buffers
By quantity measured	Mass, temperature, volume, pH, wavelength and absorbance, time, pressure	Pipettes and volumetric glassware; UV wavelength check with a holmium oxide filter

Closing line: in a Rasashala, an uncalibrated balance or Puta thermocouple affects the dose of Visha dravya and the completeness of Marana — calibration is a safety measure, not paperwork.

Revise: Volume 0 — Module 3, Unit 1 (p. 145); Module 5, Units 2, 5 and 6 (pp. 265, 292, 299); Module 6, Unit 1 (p. 325); Module 7, Unit 5 (p. 399); Module 2, Unit 3 (p. 109); Module 8, Unit 3 (p. 439).

PART A — Case-based Application-Based Questions

Fifty-three cases, drawn only from the seventeen units that Table 6F of the curriculum marks “Yes” for ABQ. Each case carries 20 marks and is set in the format of the university paper.

Attempt each case first, on paper, in 35 minutes. Then turn the page to the model answer framework.

Module 1 — Pharmaceutics and Pharmaceutical Technology

SEMESTER 2 · AYPG-AB-RB

In the university paper (Table 6F)	
Marks range for this module	5 to 20
Unit eligible for ABQ	Unit 3 — Pharmaceutical Formulation
Units eligible for LAQ	Units 1, 3 and 5 (see Part B for Units 1 and 5)
Cases in this module	ABQ 1.3.1 to ABQ 1.3.4

Remember Table 6G, instruction 5. If the ABQ of your paper comes from Module 1, no other question in that paper will come from Module 1. Use this when you plan your revision.

Unit 3 — Pharmaceutical Formulation

Syllabus points. 3.1 Principles of pharmaceutical formulation · 3.2 Excipients and their functions · 3.3 Solid dosage forms · 3.4 Liquid and semi-solid dosage forms · 3.5 Parenteral dosage forms

Table 6F. ABQ Yes · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 1, Unit 3, page 44.

ABQ 1.3.1 — The Sticky Dashamula Tablet

Anchor: M1U3 (3.1, 3.2, 3.3) · Also draws on: M1U4 · Suggested time: 35 minutes

THE CASE

A GMP-certified unit wants to convert Dashamula Kwatha into a tablet. Working patients find it hard to prepare a decoction twice a day. The development chemist prepares the decoction, concentrates it, dries it in a tray drier and powders the dried extract. He mixes the extract with lactose and talc and compresses 500 mg tablets directly on a single-punch machine. The pilot batch shows the following.

F1 The dried extract turns sticky and lumpy within 30 minutes when left open in the room, where relative humidity is 70 to 75 percent.

F2 The blend does not flow evenly into the die. Tablet weights range from 440 mg to 560 mg.

F3 Many tablets split at the top or into layers when they are ejected. Friability is 2.8 percent.

F4 Powder sticks to the punch faces. Tablet surfaces are pitted.

F5 Disintegration time in water is 48 minutes.

F6 To match the daily dose of the classical Kwatha, a patient would need 8 tablets a day.

F7 After one month in a clear PVC bottle, the tablets have softened and darkened.

Q	Question	Marks	Domain	Level
(a)	Explain the principles of pharmaceutical formulation that were not applied in this pilot. Link each principle to a finding.	5	CAN	KH
(b)	Select excipients to correct the problems. For each, state its function and justify the choice for this extract.	6	CAP	KH
(c)	Explain the causes of the defects in F3, F4 and F5 and suggest process remedies.	5	CAN	KH
(d)	How would you solve the dose burden (F6) and the stability problem (F7)? What must be shown before the tablet can be called equivalent to the Kwatha?	4	CE	KH

Model answer framework

(a) Principles that were not applied — 5 marks

Principle of formulation	What it requires	Finding
Pre-formulation study of the active material	Test the extract first: moisture, hygroscopicity, flow (angle of repose, Carr's index, Hausner ratio), compressibility, stability	F1, F2
Choice of process to suit the material	Direct compression suits only free-flowing, compressible powders. A sticky extract needs granulation (dry granulation, or wet granulation with a non-aqueous binder)	F2, F3
Function-led selection of excipients	Each problem needs an excipient with a defined role. Lactose and talc alone give no binder, disintegrant or lubricant	F3 to F5
Control of environment	Hygroscopic material is handled and compressed in a dehumidified area	F1, F7
Patient acceptability of dose	The unit dose must give a practical number of tablets a day	F6
Packaging as part of design	Container and closure are chosen on stability data, not after the product fails	F7

(b) Excipients, function and justification — 6 marks

Problem	Excipient (examples)	Function	Why it fits here
Moisture pick-up	Colloidal silicon dioxide; microcrystalline cellulose or dibasic calcium phosphate as diluent	Adsorbent and moisture scavenger; diluent	Silica holds moisture on its large surface. These diluents are less sensitive to moisture than a sticky extract alone
Poor flow	Colloidal silicon dioxide, talc	Glidant	Reduces friction between particles so the die fills evenly
Weak bonding	Polyvinylpyrrolidone (PVP) in isopropyl alcohol; or microcrystalline cellulose as dry binder	Binder	A non-aqueous binder avoids adding water to a hygroscopic extract
Sticking	Magnesium stearate (blended last, for a short time); talc	Lubricant; anti-adherent	Stops adhesion to punches and eases ejection
Slow disintegration	Croscarmellose sodium, sodium starch glycolate or crospovidone, split between inside and outside the granule	Superdisintegrant	Swells or wicks water quickly, so the tablet breaks before the sticky extract forms a gel layer
Moisture and light	Film coat based on HPMC, with an opacifier	Barrier coat	Slows moisture entry and protects colour

Write one line saying that only excipients of pharmacopoeial quality that are permitted for ASU drugs will be used.

(c) Causes and remedies of F3, F4 and F5 — 5 marks

Defect	Likely causes in this batch	Remedies
Capping and lamination (F3)	Air trapped in fine powder; too many fines; no binder; high machine speed; worn punches or dies	Granulate; add binder; keep a set granule moisture; slower speed or pre-compression; inspect tooling
Sticking and picking (F4)	Moist, hygroscopic granules; too little lubricant; rough punch faces	Dry granules to a set loss on drying; add lubricant and anti-adherent; polish punches; compress at low humidity
Slow disintegration (F5)	No disintegrant; extract forms a gel barrier; tablets too hard; over-mixing with hydrophobic magnesium stearate	Add superdisintegrant; set hardness limits; control lubricant mixing time

(d) Dose, stability and equivalence — 4 marks

- *Dose (F6).* Use a stronger, standardised extract and state its drug-to-extract ratio. Or make a larger caplet, or change to granules in sachets. State the crude-drug equivalent on the label.
- *Stability (F7).* Pack in Alu-Alu blisters, or in HDPE bottles with a desiccant. Run real-time and accelerated stability studies. The expiry date must rest on stability data. Under the 2026 amendment to the Drugs Rules, 1945 (G.S.R. 657(E)), shelf life of ASU medicines must be supported by scientific data, and a change of dosage form needs its own stability data.
- *Equivalence.* Compare the HPTLC or HPLC fingerprint of the tablet with a freshly made Kwatha. Compare the marker content per daily dose. Check disintegration and, where a method exists, dissolution.
- *Mark the boundary.* A matching fingerprint shows chemical similarity. It does not prove equal clinical effect. Volatile and heat-sensitive constituents may be lost on drying. The licensing category of a classical formula in a new dosage form is decided by the State Licensing Authority under Rule 158-B.

WHERE MARKS ARE LOST

- Writing a general essay on tablets with no reference to F1 to F7.
- Naming excipients without stating the function and the reason.
- Calling the tablet “bioequivalent” without any data.

Revise: Volume 0 — Module 1, Unit 3 (p. 44) and Unit 4 (p. 56).

ABQ 1.3.2 — A Children’s Syrup that Fermented

Anchor: M1U3 (3.1, 3.2, 3.4) · Suggested time: 35 minutes

THE CASE

A paediatric clinic asks the college pharmacy for a syrup of Vasa (Adhatoda vasica) for children with cough, because children refuse Vasavaleha. The pharmacist dissolves sugar in a hot aqueous extract of Vasa leaves, filters it, and fills it in 100 ml amber bottles with ordinary screw caps. No preservative is added. The following problems are reported within six weeks.

F1 Several bottles show froth at the neck and a sour, alcoholic smell. Some caps have bulged.

F2 Sugar crystals form around the cap, and the cap becomes hard to open.

F3 A brown sediment settles at the bottom. It does not mix back evenly on shaking.

F4 Mothers say the children spit the syrup out because it is very bitter.

F5 The pH of a fresh bottle is 5.4. After four weeks it is 4.1.

F6 Sugar content of the batch was not measured. The label says “Shake well. Use within 6 months.”

Q	Question	Marks	Domain	Level
(a)	Using the principles of liquid formulation, explain why each of the defects F1 to F5 occurred.	6	CAN	KH
(b)	Design a corrected formula. Name each excipient, its function and a typical role in this syrup.	6	CS	KH
(c)	List the quality control tests you would apply to the finished syrup, with the purpose of each.	4	CAP	KH
(d)	Comment on the label in F6. What should the label and pack include?	4	CE	KH

Model answer framework

(a) Why each defect occurred — 6 marks

Finding	Explanation
F1 Fermentation	Sugar solution with plant extract is a good medium for yeast. If sugar is below the self-preserving level, and no preservative is used, yeast ferments sugar to alcohol and carbon dioxide. Gas lifts the cap.
F2 Cap-locking	Syrup dries at the neck. Water evaporates and sugar crystallises. Inversion and crystallisation are not controlled because sugar content was not set.
F3 Sediment	The extract has insoluble and colloidal matter. Without clarification or a suspending agent, it settles and forms a firm deposit.
F4 Bitterness	Vasa alkaloids (vasicine and related) are bitter. No sweetener balance or flavour was planned for children.
F5 Falling pH	Microbial growth produces organic acids. A falling pH is a sign of spoilage, and it can also affect stability of constituents.

Close with one line: the formula was designed without pre-formulation and without a preservation strategy.

(b) A corrected formula — 6 marks

Component	Function	Note for this syrup
Standardised Vasa extract	Active	Fix the quantity per 5 ml, standardised to a marker such as vasicine by HPTLC or HPLC
Sucrose at a high, defined concentration	Sweetener, viscosity, self-preservation	A high sugar level lowers water activity. Set and test it for each batch
Sorbitol or glycerin (part)	Co-solvent, humectant, anti-cap-locking	Reduces crystallisation at the neck
Preservative (for example sodium benzoate, or a paraben combination)	Antimicrobial	Choice depends on pH. Benzoate works better in acid pH. Use only preservatives permitted for ASU drugs, within limits
Buffer (for example citrate)	Holds pH steady	Protects both preservative action and constituents
Suspending agent or clarification step	Prevents hard sediment	Filter after cold storage, or add a small amount of a suspending agent
Flavour and taste-masking agent	Palatability	Choose child-acceptable flavours; avoid alcohol for children
Purified water	Vehicle	Microbial quality of water is part of GMP

(c) Quality control tests — 4 marks

Test	Purpose
Appearance, odour, taste	First signs of spoilage or change
pH	Stability and preservative efficacy
Specific gravity and total solids	Consistency between batches; indirect check of sugar
Total sugar and reducing sugar	Confirms self-preserving level and inversion
Viscosity	Pourability and dose accuracy
TLC or HPTLC fingerprint; marker assay	Identity and strength
Microbial limits (API Appendix 2.4)	Safety
Fill volume, leak test of closure	Pack integrity

(d) Label and pack — 4 marks

- "Use within 6 months" has no basis. It must come from a stability study, including an in-use study after first opening.
- The label must follow Rule 161 of the Drugs Rules, 1945: name, quantity of ingredients, dose, batch number, manufacturing licence number, date of manufacture and date of expiry. Long ingredient lists may now be given through a QR code (2026 amendment).
- Add a measuring cup or a marked dose; give a dose by age; add storage directions ("store in a cool place; keep the cap clean and tightly closed").
- Use a child-resistant closure where possible. Amber glass protects from light.

WHERE MARKS ARE LOST

- Treating F1 as "contamination" without explaining sugar concentration and water activity.
- Listing preservatives without linking them to pH.
- Ignoring the label question or writing only "add expiry date".

Revise: Volume 0 — Module 1, Unit 3 (p. 44); Module 7, Unit 5 for QC of Sharkara (p. 399).

ABQ 1.3.3 — Jatyadi Taila into a Cream

Anchor: M1U3 (3.1, 3.2, 3.4) · Suggested time: 35 minutes

THE CASE

A surgeon in the hospital wants a wound cream made from Jatyadi Taila, because the oil drips and stains dressings. The pharmacy makes the cream by warming Jatyadi Taila with beeswax and adding warm water while stirring by hand. The batch is filled in 50 g wide-mouth plastic jars. Observations over two months are as follows.

F1 On the third day a layer of oil appears on the surface of the cream.

F2 The cream feels gritty in some jars.

F3 Black spots of mould appear on the surface of opened jars after three weeks.

F4 The cream develops a rancid smell. The acid value of the oil phase rises.

F5 Patients complain that the cream is hard to spread on a wound in winter and too runny in summer.

F6 No emulsifier, preservative or antioxidant was used.

Q	Question	Marks	Domain	Level
(a)	What type of semi-solid has the pharmacy tried to make? Explain the formulation principles of such a preparation, and why the product failed in F1 and F2.	5	CAN	KH
(b)	Explain F3 and F4. Propose excipients that would prevent them, with function.	5	CAN	KH
(c)	Propose a complete formula and process for a stable cream, giving the function of each component.	6	CS	KH
(d)	List the quality control tests for the finished cream.	4	CAP	K

Model answer framework

(a) Type of preparation and why it failed — 5 marks

- The pharmacy tried to make an emulsion-based cream. Oil and water were mixed with beeswax only. Beeswax thickens the oil phase but is a weak emulsifier on its own.
- Principles: choose the type of emulsion (oil-in-water for a washable wound cream); choose an emulsifier system by its HLB value; heat both phases to the same temperature; add the internal phase slowly with high shear; cool with stirring; homogenise.
- *F1 Oil separation (creaming, then cracking).* No proper emulsifier, poor shear by hand, unequal phase temperatures.
- *F2 Grittiness.* Wax or high-melting components solidified as lumps because the phases were at different temperatures, or because of slow, uneven cooling. Undissolved particles from the oil may also be present.

(b) Mould and rancidity — 5 marks

Finding	Cause	Prevention
F3 Mould	Water phase supports microbes; open wide-mouth jar; repeated finger contact	Preservative in the water phase (for example parabens or another permitted preservative); tube pack instead of jar; GMP hygiene
F4 Rancidity	Oxidation of unsaturated fatty acids in the oil, speeded by air, light, heat and trace metals	Antioxidant in the oil phase (for example tocopherol or BHT where permitted); well-filled, light-protected container; cool storage

(c) Formula and process — 6 marks

Component	Function
Jatyadi Taila (fixed proportion)	Active phase
Cetostearyl alcohol or stearic acid	Stiffening agent; co-emulsifier
Emulsifier pair (for example a non-ionic surfactant pair chosen to the required HLB)	Forms and stabilises an oil-in-water emulsion
Glycerin or propylene glycol	Humectant; prevents drying; aids spreading
Preservative	Microbial protection
Antioxidant	Prevents rancidity
Purified water	External phase

Process. Melt the oil phase (oil, wax, stiffening agent, oil-soluble emulsifier, antioxidant) and heat the water phase (water, humectant, preservative, water-soluble emulsifier) to the same temperature. Add one phase to the other with high-shear mixing. Cool with slow stirring. Homogenise. Fill into collapsible tubes. A bench-scale trial with a range of HLB values decides the final emulsifier ratio.

(d) Quality control tests — 4 marks

Appearance, colour and odour; pH; spreadability; viscosity; type of emulsion (dilution test or dye test); centrifugation or freeze–thaw test for phase separation; globule size by microscopy; acid value and peroxide value of the extracted oil phase; microbial limits; fill weight; TLC fingerprint of the oil phase for identity.

WHERE MARKS ARE LOST

- Calling the product an “ointment”. An ointment has no water phase. This is a cream.
- Writing “add preservative” without saying which phase it goes into.
- Not proposing a pack change from jar to tube.

Revise: Volume 0 — Module 1, Unit 3 (p. 44).

ABQ 1.3.4 — “Can We Inject This Bhasma?”

Anchor: M1U3 (3.1, 3.5) · Also draws on: M1U1 (regulatory framework) · Suggested time: 35 minutes

THE CASE

A research scholar proposes to give Yashada Bhasma by injection to rats in a study of wound healing, “to avoid the uncertainty of oral absorption”. The plan is to suspend 10 mg of the Bhasma in normal saline, shake it, and inject it intramuscularly. A private company is also watching the project. It hopes to sell a ready-to-use injection of a herbal extract under an Ayurvedic licence. The following points come up at the department research committee.

F1 Laser diffraction shows that the Bhasma has a median particle size of about 4 µm. Some particles are larger than 20 µm.

F2 The Bhasma is not soluble in water or saline.

F3 The suspension was prepared on an open bench. There is no plan for sterilisation or a pyrogen test.

F4 The pH and tonicity of the suspension have not been measured.

F5 The company’s regulatory officer believes that an injection made from classical ingredients can be licensed as an Ayurvedic patent or proprietary medicine.

Q	Question	Marks	Domain	Level
(a)	List the essential requirements of a parenteral dosage form. Using F1 to F4, show which requirements this plan fails.	6	CAN	KH
(b)	Explain the dangers of injecting an insoluble particulate material.	4	CC	KH
(c)	Is the company’s belief in F5 correct? Explain with reference to the Drugs and Cosmetics Act, 1940.	4	CE	KH
(d)	Suggest a scientifically sound alternative plan for the scholar’s study.	6	CS	KH

Model answer framework

(a) Requirements of a parenteral and how the plan fails — 6 marks

Requirement	What it means	Failed by
Sterility	Free of living micro-organisms; made by aseptic process or terminal sterilisation	F3
Freedom from pyrogens and endotoxins	Bacterial endotoxin test or pyrogen test	F3
Freedom from visible and sub-visible particles (for solutions); controlled particle size (for suspensions)	Particles block capillaries and cause reactions	F1, F2
Isotonicity	Close to body fluids to avoid pain and tissue damage	F4
pH close to physiological	Avoids irritation; supports stability	F4
Stability, compatible container, validated process	Glass ampoules or vials, clean-room manufacture	F3
Defined dose and dose uniformity	Each injection gives the same dose; suspension must re-disperse evenly	F2

(b) Dangers of injected insoluble particles — 4 marks

Local inflammation and granuloma at the site; abscess if the material is not sterile; particles taken up by macrophages and carried to liver, spleen and lymph nodes; persistence of metal in tissue; embolism and blockage of small vessels if given by a vascular route; unknown long-term toxicity of zinc deposits. Oral Bhasma is used after Marana and classical processing for oral route only. No classical text supports injection of Bhasma.

(c) The company’s belief — 4 marks

The belief is incorrect. Under Section 3(h) of the Drugs and Cosmetics Act, 1940, a patent or proprietary medicine in the ASU systems excludes a medicine given by the parenteral route. An injection made from herbal ingredients would be regulated as a new drug through CDSCO, under the New Drugs and Clinical Trials Rules, 2019. It cannot be licensed as an Ayurvedic proprietary medicine.

(d) A sound alternative plan — 6 marks

- Keep the oral route, as in classical use. Give the Bhasma in a suitable Anupana or vehicle by oral gavage, at doses set from the human dose with a standard conversion.
- If a topical effect on the wound is the aim, prepare a topical form (an ointment or gel containing the Bhasma) and study it on the wound.
- Characterise the test material first: XRD, particle size, zinc content, heavy metal profile.
- Measure absorption directly if that is the question: zinc levels in blood and tissue by ICP-MS or AAS at set times.

- Take approval from the Institutional Animal Ethics Committee under CCSEA guidelines. Follow OECD test guidelines for toxicity.
- Mark the boundary in the protocol: “oral absorption is uncertain” is a research question, not a reason to change route.

WHERE MARKS ARE LOST

- Listing parenteral requirements without matching them to the findings.
- Missing the Section 3(h) exclusion of the parenteral route.
- Proposing a “sterile Bhasma injection” as the answer to (d).

Revise: Volume 0 — Module 1, Unit 1 (p. 9) and Unit 3 (p. 44); Module 6, Unit 5 (p. 350).

Module 3 — Essential Chemistry: Unlocking the Rasashastra Techniques

SEMESTER 2 · AYPG-AB-RB

In the university paper (Table 6F)	
Marks range for this module	5 to 20
Units eligible for ABQ	Unit 1 — Atomic, Molecular Structure and Periodic Table; Unit 2 — Specialised Chemistry Fields; Unit 3 — Organic Chemistry
Units eligible for LAQ	Units 4 and 5 (see Part B)
Cases in this module	ABQ 3.1.1 to ABQ 3.3.3

How chemistry is tested in an ABQ. The examiner gives you a Rasashastra procedure and its observations. You must explain them with chemistry: write the reaction, name the bond or the crystal form, and say what is proved and what is not. A chemistry answer without a Rasashastra link scores poorly. So does a Rasashastra answer without chemistry.

Unit 1 — Atomic, Molecular Structure and Periodic Table Elements

Syllabus points. 1.1 Importance of knowledge of chemistry for understanding various processing methods of Rasashastra and Oushadha nirmana · 1.2 Atomic structure, Periodic table, Chemical bonding, Isomerism — application of these principles in Shodhana, Marana, Amruteekarana and Chaturvidha Rasayana etc. · 1.3 Periodic Table, Main group elements, Transition metals: understanding of chemical properties of non-metals, metals, variable oxidation states, thermal stability, and the ability to form complex compounds

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 3, Unit 1, page 145.

ABQ 3.1.1 — Five Metals in One Week

Anchor: M3U1 (1.1, 1.3) · Also draws on: M4U4 · Suggested time: 35 minutes

THE CASE

A first-year scholar works with five metals in the same week. She records the following in her practical diary.

F1 Naga (lead) melts in an iron ladle on a coal stove within minutes. Tamra (copper) and Lauha (iron) do not melt at this heat.

F2 When molten Naga is stirred with plant powder during Jarana, a yellow-to-reddish powder forms on its surface.

F3 Yashada (zinc) heated strongly in an open crucible burns with a bright flame and gives white fluffy fumes. The residue is yellow while hot and white when cold.

F4 Copper sheets heated red-hot in air develop a black scale.

F5 A gold leaf kept in the same Puta as the copper comes out with no change in colour or weight.

F6 Iron filings after the first Puta are strongly attracted to a magnet. After many Putas, the product is only weakly attracted.

Q	Question	Marks	Domain	Level
(a)	Using the position of these metals in the periodic table and the reactivity series, explain why they behave so differently (F1 to F5).	6	CAN	KH
(b)	Explain the role of variable oxidation states and thermal stability of oxides in F4 and F6.	5	CC	KH
(c)	Why do noble metals such as Swarna need special Marana media, such as Parada, Hingula or Gandhaka? Explain with chemistry.	5	CAN	KH
(d)	Write balanced equations for the changes in F2, F3, F4 and F6.	4	CAP	KH

Model answer framework

(a) Why the metals behave differently — 6 marks

Metal	Position and nature	Melting point (approx.)	Behaviour seen
Lead (Pb)	Group 14, post-transition metal; soft; weak metallic bonding	327 °C	Melts easily (F1); oxidises on stirring in air (F2)
Zinc (Zn)	Group 12; reactive; low boiling point (about 907 °C)	420 °C	Vaporises and burns to ZnO (F3)
Copper (Cu)	Group 11, transition metal	1085 °C	Does not melt on a stove; forms black CuO scale (F4)
Iron (Fe)	Group 8, transition metal	1538 °C	Does not melt; oxidises stepwise (F6)
Gold (Au)	Group 11, noble metal; very high reduction potential	1064 °C	No oxidation at any furnace temperature (F5)

Then write two lines: reactivity in air falls in the order Zn > Fe > Pb > Cu > Au. Melting point depends on the strength of metallic bonding. Transition metals have strong bonding because d-electrons take part.

(b) Variable oxidation states and thermal stability — 5 marks

- Transition metals show more than one oxidation state because the 3d and 4s electrons are close in energy.
- *Copper (F4):* Cu(I) and Cu(II). In plenty of air, black CuO forms. In limited air or at higher temperature, red Cu_2O can form.
- *Iron (F6):* Fe(II) and Fe(III). Iron first forms magnetite, Fe_3O_4 (a mixed Fe(II)/Fe(III) oxide, strongly magnetic). On repeated heating in air it is oxidised to hematite, α-Fe_2O_3 (Fe(III), only weakly magnetic). Hematite is the most stable iron oxide in air. This is why magnetic attraction falls with more Putas.
- Thermal stability: oxides of reactive metals (ZnO) are very stable. Oxides of less reactive metals are less stable. HgO decomposes on heating — which is why mercury can be recovered from its compounds by heat.

(c) Why noble metals need special media — 5 marks

- Gold has a very high standard reduction potential. It does not react with oxygen, and it is not attacked by ordinary acids. Heat alone cannot convert it.
- Classical Marana of Swarna uses Parada (or Kajjali, Hingula) and Gandhaka with plant juices. Mercury dissolves gold to form an amalgam. This breaks the solid metal into a fine state.
- On heating, mercury volatilises, leaving gold in a very finely divided form. Published characterisation of Swarna Bhasma reports elemental gold as the main phase, with small particle size.
- Mark the boundary: the Bhasma is mainly metallic gold, not a gold oxide. The role of organic matter from plant media is proposed, and it is still being studied.

(d) Equations — 4 marks

- F2: $2Pb + O_2 \rightarrow 2PbO$ (litharge, red; massicot, yellow)
- F3: $2Zn + O_2 \rightarrow 2ZnO$ (yellow when hot, white when cold)
- F4: $2Cu + O_2 \rightarrow 2CuO$
- F6: $3Fe + 2O_2 \rightarrow Fe_3O_4$; then $4Fe_3O_4 + O_2 \rightarrow 6Fe_2O_3$

WHERE MARKS ARE LOST

- Writing only melting points without linking them to the periodic table.
- Saying gold “oxidises slowly”. It does not.
- Forgetting to explain why magnetism falls in F6.

Revise: Volume 0 — Module 3, Unit 1 (p. 145); Module 3, Unit 5 (p. 183); Module 4, Unit 4 (p. 232).

ABQ 3.1.2 — Black Kajjali, Red Rasasindura

Anchor: M3U1 (1.2) · Also draws on: M3U2, M4U4 · Suggested time: 35 minutes

THE CASE

In a teaching Rasashala the following are recorded.

F1 Shuddha Parada and Shuddha Gandhaka in equal parts are triturated in a Khalva for many hours. A black, smooth powder (Kajjali) forms. By the end, no mercury globules are seen under a lens.

F2 The same Kajjali is heated in a glass bottle on a sand bath (Kupipakwa). A red crystalline mass collects at the neck of the bottle (Rasasindura).

F3 XRD of the Kajjali shows metacinnabar (β-HgS) and some free sulphur. XRD of the Rasasindura shows cinnabar (α-HgS).

F4 A trainee asks: “Both contain mercury. Why is mercury sulphide said to be safer than mercury metal?”

F5 Last month a thermometer broke in the laboratory. The attendant swept up the mercury with a broom and threw it in the dustbin.

Q	Question	Marks	Domain	Level
(a)	Explain the type of chemical bonding in mercury sulphide. How does trituration alone form a compound in F1?	5	CC	KH
(b)	Explain F2 and F3 using the concept of polymorphism. How is this different from isomerism?	5	CAN	KH
(c)	Answer the trainee’s question in F4. Mark clearly what is established and what is not.	5	CE	KH
(d)	What was wrong in F5? Write the correct procedure for a mercury spill and five rules of mercury safety for a Rasashala.	5	CAP	KH

Model answer framework

(a) Bonding and mechanochemical formation — 5 marks

- Mercury (Group 12) and sulphur (Group 16) form HgS. The bond is mainly covalent, with some ionic character, because the electronegativity difference is small.
- Mercury has a very high affinity for sulphur. This is why HgS is very insoluble.
- Trituration supplies mechanical energy. It breaks mercury into tiny droplets, renews fresh surfaces and brings them into close contact with sulphur. The reaction Hg + S → HgS proceeds at room temperature. This is a mechanochemical reaction.
- Nishchandra (no visible globules) shows that free mercury is no longer visible. It does not prove that no free mercury remains. XRD and analysis for free mercury are needed for that.

(b) Polymorphism, not isomerism — 5 marks

Point	Black HgS (metacinnabar)	Red HgS (cinnabar)
Found in	Kajjali (F1, F3)	Rasasindura (F2, F3)
Crystal form	Cubic (zinc blende type)	Trigonal (chain-like)
Colour	Black	Red
How formed here	Trituration at room temperature	HgS vaporises in the hot bottle and condenses at the cooler neck as the red form

- Same chemical formula, different crystal structure: this is *polymorphism*.
- *Isomerism* is a term for molecules (mostly organic) with the same molecular formula but different arrangement of atoms within the molecule, for example structural or optical isomers. HgS is not a molecular compound. So “polymorphism” is the correct term here.
- Cinnabar is the stable form at ordinary temperature. Free sulphur in Kajjali (F3) shows that excess sulphur was used, as the classical ratio intends.

(c) Is HgS safer than mercury? — 5 marks

Established	Not established
Toxicity depends on the chemical form (speciation), not only on total mercury	That HgS is free of risk on long-term use
Mercury vapour from the metal is readily absorbed through the lungs. Liquid mercury swallowed is poorly absorbed	That traditional processing removes all free mercury in every batch
HgS is extremely insoluble, and animal studies show it is far less toxic than soluble mercury salts or methylmercury	That safety in animals at test doses proves safety in humans over months
Some mercury from HgS can still be absorbed, and it accumulates mainly in the kidney	Large human studies on long-term safety

Conclude: “Lower toxicity is not zero toxicity. Dose, duration, quality of preparation and monitoring decide the risk.”

(d) The spill — 5 marks

What was wrong. Sweeping breaks mercury into fine droplets and spreads vapour. Throwing it in the dustbin contaminates general waste.

Correct procedure. Keep people away and ventilate the room. Wear gloves. Do not sweep or vacuum. Collect droplets with a card and a syringe, or use a mercury spill kit with sulphur or zinc powder to bind fine droplets. Seal everything in a labelled, airtight container. Hand it over as hazardous waste. Never wash mercury down the drain.

Five rules for a Rasashala. (1) Store mercury under water in a closed, labelled, unbreakable container in a locked cupboard. (2) Work with mercury in a well-ventilated area with exhaust, on trays that catch spills. (3) Wear gloves, apron and mask; no eating or drinking in the area. (4) Keep a spill kit and a written spill SOP. (5) Record all mercury issued and returned; monitor staff health.

WHERE MARKS ARE LOST

- Calling metacinnabar and cinnabar “isomers”.
- Writing “HgS is non-toxic” without qualification.
- Leaving out the spill procedure, which carries a quarter of the marks.

Revise: Volume 0 — Module 3, Unit 1 (p. 145) and Unit 2 (p. 153); Module 7, Unit 3 for Kupipakwa QC (p. 385).

ABQ 3.1.3 — The Arsenic Shelf

Anchor: M3U1 (1.2, 1.3) · Also draws on: M4U4 · Suggested time: 35 minutes

THE CASE

During a stock audit of a pharmacy raw material store, the following are noted.

F1 Three arsenic drugs sit side by side: yellow Haratala (orpiment, As_2S_3), orange-red Manahshila (realgar, As_4S_4) and a white powder labelled “Malla” (arsenic trioxide, As_2O_3). The white powder has no warning label. The attendant handles all three with bare hands.

F2 In a teaching demonstration, a small amount of Haratala heated in an open crucible gives white fumes with a garlic-like smell.

F3 A jar of Manahshila left on a sunny window for months has turned into a yellow powder on the surface.

F4 ICP-OES shows that all three samples contain arsenic as the main element. The storekeeper says, “Arsenic is arsenic. If one is allowed, all are the same.”

F5 The pharmacy makes formulations that contain these drugs, which are poisonous substances listed under Schedule E(1).

Q	Question	Marks	Domain	Level
(a)	Describe the position of arsenic in the periodic table, its nature and oxidation states. Compare the three compounds.	5	CC	KH
(b)	Explain F2 and F3 chemically. Write the equation for F2.	5	CAN	KH
(c)	Evaluate the storekeeper’s statement in F4. What analysis would answer the real question?	5	CE	KH
(d)	Prepare a safety plan for storage, handling and labelling of arsenic drugs in this pharmacy.	5	CS	KH

Model answer framework

(a) Arsenic and its compounds — 5 marks

- Arsenic: Group 15, period 4. It is a *metalloid* — it has properties between metals and non-metals.
- Common oxidation states: −3, +3 and +5. In all three drugs here arsenic is in the +3 state.

Drug	Compound	Colour	Bonding and structure	Solubility in water
Haratala	As_2S_3 (orpiment)	Yellow	Covalent layered sulphide	Very low
Manahshila	As_4S_4 (realgar)	Orange-red	Covalent cage molecules	Very low
Malla	As_2O_3 (arsenic trioxide)	White	Oxide	Appreciably soluble; highly toxic

(b) F2 and F3 — 5 marks

- *F2.* Heating a sulphide in air oxidises it. Arsenic trioxide sublimes as white fumes, and sulphur dioxide forms: $2As_2S_3 + 9O_2 \rightarrow 2As_2O_3 + 6SO_2$ The garlic-like smell is typical of arsenic vapour. This is a serious hazard: it converts a less soluble sulphide into the most toxic common arsenic compound, in the air the worker breathes.
- *F3.* Realgar is sensitive to light. Light changes it into a yellow form (pararealgar) and partly oxidises it. So the jar no longer contains the material it was labelled as. Store arsenic sulphides in the dark.

(c) "Arsenic is arsenic" — 5 marks

- Wrong. Toxicity depends on the chemical form (speciation), solubility and dose, not only on the element.
- As_2O_3 dissolves and is absorbed easily. The sulphides are much less soluble. But sulphides can contain oxide as an impurity, and heating converts sulphide to oxide (F2).
- ICP-OES gives only total arsenic. It cannot tell oxide from sulphide.
- The real questions are: what compound is present, and how much soluble arsenic is there? Use XRD for the compound; Raman spectroscopy for the sulphide forms; and speciation analysis (for example HPLC coupled with ICP-MS) for soluble As(III) and As(V). An in-vitro solubility test in simulated gastric fluid adds useful data.

(d) Safety plan — 5 marks

Area	Measures
Storage	Separate locked cupboard for poisonous drugs; each container labelled with name, "POISON" warning and hazard symbol; arsenic sulphides kept away from light; stock register
Handling	Nitrile gloves, mask and apron; weigh in a fume hood or ventilated enclosure; no eating, drinking or touching the face; wash hands after work
Heating	Only in a fume hood with exhaust; never roast arsenic sulphides in an open room

Area	Measures
Labelling of products	Products containing Schedule E(1) substances must carry the warning “Caution: to be taken under medical supervision” (Rule 161 of the Drugs Rules, 1945)
Waste and spills	Collect arsenic waste separately as hazardous waste; never into drains
People	Training, written SOPs, and health surveillance of exposed workers

WHERE MARKS ARE LOST

- Treating all arsenic compounds as equally toxic, or as equally safe.
- Writing the equation for F2 without the sulphur dioxide.
- A safety plan without labelling and Schedule E(1).

Revise: Volume 0 — Module 3, Unit 1 (p. 145); Module 4, Unit 4 (p. 232).

Unit 2 — Specialised Chemistry Fields: An Outline

Syllabus points. 2.1 Crystallography — detoxification, transformation of compound · 2.2 Organometallic chemistry — Chaturvidha Rasayana · 2.3 Bioinorganic chemistry — Amruteekarana (Metal-Carbon bond — stable compound formation) · 2.4 Application in Shodhana, Marana, Amruteekarana, Chaturvidha Rasayana

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 3, Unit 2, page 153.

ABQ 3.2.1 — What the X-rays Saw in Lauha Bhasma

Anchor: M3U2 (2.1, 2.4) · Also draws on: M5U7 · Suggested time: 35 minutes

THE CASE

A PhD scholar prepares Lauha Bhasma by the classical method. She takes samples after Puta 1, Puta 7 and Puta 21 and sends them for X-ray diffraction (XRD). The results are as follows.

F1 After Puta 1: sharp peaks of metallic iron (α-Fe) and peaks of magnetite (Fe_3O_4).

F2 After Puta 7: magnetite is the main phase. Peaks of hematite (α-Fe_2O_3) appear.

F3 After Puta 21: hematite is the main phase. No peak of metallic iron is seen.

F4 Peaks become broader with each stage. Crystallite size by the Scherrer equation is 85 nm, then 42 nm, then 28 nm.

F5 The Puta 21 sample passes Rekhapurnata and Varitara. It is weakly attracted by a magnet.

F6 In her thesis she writes: “XRD proves that Lauha Bhasma is completely detoxified and is a nanomedicine.”

Q	Question	Marks	Domain	Level
(a)	Explain the principle of X-ray diffraction and how a crystalline phase is identified.	5	CC	KH
(b)	Interpret F1 to F3 and F5 as a record of transformation during Marana. Write the sequence of reactions.	6	CAN	KH
(c)	What does peak broadening in F4 mean? How is crystallite size calculated, and how is it different from particle size?	4	CAP	KH
(d)	Evaluate the statement in F6.	5	CE	KH

Model answer framework

(a) Principle of XRD — 5 marks

- A crystal is a regular, repeating arrangement of atoms in planes. X-rays have wavelengths similar to the spacing between these planes.
- X-rays scattered from parallel planes reinforce one another only at certain angles. This is *Bragg's law*: $n\lambda = 2d \sin\theta$, where λ is the wavelength, d is the spacing between planes and θ is the angle of incidence.
- A powder diffractometer records intensity against 2θ. Each crystalline compound gives its own set of peaks — a "fingerprint".
- The phase is identified by matching the peak positions and intensities with reference patterns in databases, such as the ICDD Powder Diffraction File or the Crystallography Open Database.
- Amorphous material gives no sharp peaks, only a broad hump.

(b) The transformation record — 6 marks

Stage	Phases	Meaning
Puta 1 (F1)	α-Fe + Fe_3O_4	Oxidation has begun at the surface; much metal remains
Puta 7 (F2)	Fe_3O_4 + α-Fe_2O_3	Metal is used up; magnetite is being oxidised to hematite
Puta 21 (F3)	α-Fe_2O_3	Conversion to the most stable oxide is nearly complete; no free metal detected

Reactions: $3Fe + 2O_2 \rightarrow Fe_3O_4$; $4Fe_3O_4 + O_2 \rightarrow 6Fe_2O_3$.

F5: a weak magnetic response fits mainly hematite with a small amount of magnetite remaining, or with very small magnetic particles. The classical tests and the XRD agree that no free metal remains.

(c) Peak broadening — 4 marks

- Smaller crystallites give broader peaks. Strain in the crystal also broadens peaks.
- Scherrer equation: $D = K\lambda / (\beta \cos\theta)$, where D is crystallite size, K is a shape factor (about 0.9), λ is the X-ray wavelength, β is the peak width at half maximum (in radians, after correcting for the instrument) and θ is the Bragg angle.
- *Crystallite size* is the size of a single coherent crystal domain. *Particle size* is the size of the grain one sees under SEM or measures by a particle size analyser. One particle may contain many crystallites, and particles stick together into agglomerates. So a 28 nm crystallite does not mean 28 nm particles.

(d) Evaluating F6 — 5 marks

Part of the claim	Judgement
“XRD proves detoxification”	XRD shows that metallic iron is no longer detected and that a stable oxide has formed. That supports transformation. It does not measure toxic trace elements (lead, arsenic), amorphous content, or biological safety. Safety needs elemental analysis and toxicity studies
“completely”	XRD cannot detect phases below about 1 to 5 percent of the sample. “Not detected” is not “absent”
“is a nanomedicine”	Crystallite size is not particle size. A nano claim needs particle size distribution (for example by TEM and a particle size analyser) and evidence that nano-size matters for the effect

Suggested wording for the thesis: “XRD shows progressive conversion of metallic iron to hematite, with decreasing crystallite size. Metallic iron was not detected after Puta 21.”

WHERE MARKS ARE LOST

- Writing Bragg’s law without explaining what d and θ are.
- Equating crystallite size with particle size.
- Accepting “proves” and “completely” without comment.

Revise: Volume 0 — Module 3, Unit 2 (p. 153); Module 5, Unit 7 (p. 305); Module 8, Unit 2 (p. 429).

ABQ 3.2.2 — Is Rasasindura “Organometallic”?

Anchor: M3U2 (2.2, 2.3, 2.4) · Suggested time: 35 minutes

THE CASE

A student group submits a manuscript to the department journal. You are asked to review it. The manuscript reports the following.

F1 The FTIR spectrum of Rasasindura shows weak bands near 2920 and 2850 cm^{-1} and a band near 1630 cm^{-1}.

F2 The authors conclude: “Mercury is bound to carbon. Rasasindura is an organometallic compound, and this explains its safety.”

F3 XRD of the same sample shows only cinnabar (α-HgS).

F4 Elemental (CHNS) analysis shows carbon below 1 percent.

F5 The discussion states that Amritikarana creates “metal–carbon bonds” in all Bhasmas.

Q	Question	Marks	Domain	Level
(a)	Define organometallic chemistry and bioinorganic chemistry. Differentiate them with examples.	4	CC	KH
(b)	Evaluate the evidence in F1, F3 and F4. Does it support the claim in F2? What do the FTIR bands most likely represent?	6	CE	KH
(c)	What analytical evidence would be needed to prove a metal–carbon bond in such a product?	5	CAP	KH
(d)	Explain how bioinorganic chemistry offers a more defensible model for Amritikarana and Chaturvidha Rasayana. Mark what is proposed and what is established.	5	CAN	KH

Model answer framework

(a) Definitions — 4 marks

Point	Organometallic chemistry	Bioinorganic chemistry
Definition	Study of compounds with at least one direct metal–carbon bond	Study of metals and metal compounds in living systems, and their interaction with biological molecules
Bond type	Metal–carbon (M–C)	Usually coordination bonds from O, N or S donor atoms to the metal
Examples	Ferrocene; Grignard reagents; methylmercury; tetraethyl lead	Iron in haemoglobin; zinc in many enzymes; copper in ceruloplasmin; metal–polyphenol complexes

(b) Evaluating the evidence — 6 marks

- Bands at about 2920 and 2850 cm^{-1} are C–H stretching of CH_2 groups. They show traces of organic matter — residue of plant media, or contamination. They say nothing about a bond to mercury.
- The band near 1630 cm^{-1} is commonly due to adsorbed water (H–O–H bending), or to C=O / C=C of organic residue.
- XRD shows only α-HgS (F3). Carbon is below 1 percent (F4). So the bulk of the product is mercury sulphide.
- Conclusion: the evidence shows a small amount of organic matter *associated* with HgS. It does not show a mercury–carbon bond. The claim in F2 is not supported.
- Note the irony: well-known organomercury compounds such as methylmercury are among the *most* toxic mercury compounds. So an M–C bond, if present, would not explain safety.

(c) Evidence needed for an M–C bond — 5 marks

- X-ray absorption spectroscopy (EXAFS) showing mercury–carbon distances in the coordination shell.
- Single-crystal X-ray structure, if a defined compound can be isolated.
- NMR evidence where applicable (for example ^{199}Hg or ^{13}C NMR showing coupling between Hg and C).
- Mass spectrometry showing fragments that contain both Hg and C.
- A quantitative balance: the amount of carbon must be enough to bind a meaningful share of mercury.

(d) A more defensible model — 5 marks

Proposed (supported by some data, still under study)	Established
Plant constituents (polyphenols, organic acids) coordinate to metal ions through O and N donor atoms and coat particle surfaces	Bhasmas and Kupipakwa products contain small amounts of organic matter
This coating may limit dissolution, reduce free metal ions and stabilise particles — similar to capping in green synthesis of nanoparticles	Plant extracts reduce metal salts and cap nanoparticles in the laboratory (green synthesis)
Amritikarana may lower reactivity of residual free metal	The main inorganic phase is identified by XRD (for example HgS, Fe_2O_3, ZnO)

Conclude: the curriculum uses the phrase “metal–carbon bond — stable compound formation” for Amruteekarana. A scholar should present it as a hypothesis, and should state what evidence would confirm it.

WHERE MARKS ARE LOST

- Accepting C–H bands as proof of a metal–carbon bond.
- Not noticing that the XRD and CHNS data contradict the claim.
- Writing “organometallic compounds are safe”.

Revise: Volume 0 — Module 3, Unit 2 (p. 153); Module 5, Unit 4 for FTIR (p. 283).

ABQ 3.2.3 — Where Did the Shine of Abhraka Go?

Anchor: M3U2 (2.1, 2.4) · Also draws on: M4U1, M7U2 · Suggested time: 35 minutes

THE CASE

A scholar studies Abhraka Bhasma made from Krishna Abhraka (black mica, identified as biotite). Samples are tested at each stage.

F1 Raw Abhraka splits easily into thin, shining, flexible sheets.

F2 XRD of raw Abhraka shows strong, sharp peaks typical of mica, especially the basal reflections.

F3 After Shodhana and Dhanyabhraka, the XRD pattern is almost unchanged, but the sheets are thinner and cleaner.

F4 After 20 Putas, the mica peaks are weaker. New peaks of hematite (α-Fe_2O_3) appear. The powder still shows shining particles in sunlight.

F5 After 60 Putas, the mica peaks are largely absent. Hematite and silicate phases are present. The powder no longer shines in sunlight (Nishchandra).

F6 SEM shows flat, layered plates in the raw material and rounded, fused grains in the 60-Puta Bhasma.

Q	Question	Marks	Domain	Level
(a)	Describe the crystal structure of mica. How does it explain F1 and the “Chandrika” (shine)?	5	CC	KH
(b)	Interpret F3 to F6 as a transformation of the compound.	6	CAN	KH
(c)	Relate these findings to the classical aim of Nishchandra and to the idea of “detoxification” in the syllabus.	5	CAN	KH
(d)	What are the limits of XRD in this study? Name two techniques that would complete the picture.	4	CE	KH

Model answer framework

(a) Structure of mica — 5 marks

- Mica is a *sheet silicate* (phyllosilicate). Biotite is a potassium-magnesium-iron aluminosilicate with hydroxyl groups.
- Each layer is a sandwich of silica–alumina tetrahedral sheets around an octahedral sheet. Layers are held to each other only by weak bonds through potassium ions.
- This gives *perfect basal cleavage*: the mineral splits into thin, flexible sheets (F1).
- Flat, smooth cleavage surfaces reflect light like tiny mirrors. This is the Chandrika (shine) of Abhraka.
- In XRD, the regular stacking of layers gives strong basal reflections (F2).

(b) The transformation — 6 marks

Stage	Change	Interpretation
Shodhana and Dhanyabhraka (F3)	Pattern unchanged; thinner, cleaner sheets	Mainly physical: cleaning, delamination, size reduction
20 Putas (F4)	Mica peaks weaker; hematite appears; still shines	Heat drives off hydroxyl water (dehydroxylation). Iron(II) in the lattice is oxidised to iron(III), and hematite separates. Some mica sheets survive
60 Putas (F5, F6)	Mica pattern largely gone; hematite and silicates; no shine; rounded, fused grains	The layered structure has broken down. New, stable oxide and silicate phases have formed. Partial sintering rounds the grains

(c) Nishchandra and “detoxification” — 5 marks

- Nishchandra means “free of shine”. It is a classical endpoint for Abhraka Bhasma. These results show *why* it works: shine comes from intact mica sheets. When the sheets are destroyed, the shine goes.
- So Nishchandra is a simple visual test for complete structural transformation.
- Classical texts say that improperly processed Abhraka causes disease. A modern reading: intact, sharp mica flakes are hard, insoluble and abrasive. Converting them into stable, fine, rounded grains changes how the material behaves in the body.
- Mark the boundary: “detoxification” here means structural transformation of the compound. It does not by itself prove safety. Safety still needs elemental analysis and toxicity data.

(d) Limits of XRD — 4 marks

- Detects only crystalline phases. Glassy or amorphous material formed on heating is missed.
- Minor phases below about 1 to 5 percent may not be detected.

- Gives no information on particle size distribution, organic matter or trace toxic elements.
- Complementary techniques: SEM-EDAX (morphology and elemental composition), XRF or ICP (elemental and trace analysis), particle size analysis, and TGA (mass loss and stability).

WHERE MARKS ARE LOST

- Describing XRD in general without discussing the mica structure.
- Missing the link between shine and the cleavage of mica sheets.
- Writing “Abhraka becomes non-toxic” as a proven fact.

Revise: Volume 0 — Module 3, Unit 2 (p. 153); Module 4, Unit 1 (p. 207); Module 7, Unit 2 for Bhasma Pariksha (p. 377).

Unit 3 — Organic Chemistry

Syllabus points. 3.1 Structure of organic molecules: understanding of Herbal components · 3.2 Interaction of herbs and animal products with metals for detoxification, compound formation, and compound stability · 3.3 Functional groups: Reactivity and properties of organic molecules influencing compound action

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 3, Unit 3, page 163.

ABQ 3.3.1 — Vatsanabha in Gomutra: An Ester Story

Anchor: M3U3 (3.1, 3.3) · Also draws on: M4U2 · Suggested time: 35 minutes

THE CASE

A Rasashala does Shodhana of Vatsanabha (Aconitum ferox) roots by soaking them in Gomutra for three days, changing the Gomutra daily, and then drying them. A research group analyses the roots by HPLC.

F1 The raw root has a high content of diester-type diterpenoid alkaloids (aconitine-type).

F2 After Shodhana, the diester alkaloids are much lower. Monoester and non-ester alkaloids are higher.

F3 The Gomutra used is alkaline.

F4 In a parallel batch soaked in plain water, the fall in diester alkaloids is much smaller.

F5 A worker who cut raw roots without gloves reports tingling and numbness of the fingers.

F6 In a published animal study, the processed root still showed analgesic activity.

Q	Question	Marks	Domain	Level
(a)	Describe the structural features and functional groups of aconitine-type alkaloids that decide their toxicity.	5	CC	KH
(b)	Explain the chemistry behind F2 to F4. Write the general reaction.	6	CAN	KH
(c)	Why might useful activity remain (F6) while toxicity falls? Mark what is established.	4	CE	KH
(d)	Explain F5. List the precautions for handling Vatsanabha in a Rasashala and the first-aid steps.	5	CAP	KH

Model answer framework

(a) Structure and functional groups — 5 marks

- Aconitine-type alkaloids are C19-diterpenoid alkaloids: a large, rigid ring skeleton with a tertiary nitrogen.
- Key functional groups: two *ester* groups (an acetyl ester and an aromatic ester such as benzoyl), several methoxy ($–OCH_3$) and hydroxyl (–OH) groups, and the tertiary amine nitrogen.
- The two ester groups are the main reason for the high toxicity. These alkaloids keep voltage-gated sodium channels open in nerve and heart muscle. This causes numbness, tingling and dangerous heart rhythm disturbances.
- When the esters are removed, toxicity falls sharply.

(b) The chemistry of Shodhana here — 6 marks

- *Ester hydrolysis.* An ester reacts with water to give an acid and an alcohol. The reaction is speeded up by alkali (base-catalysed hydrolysis) and by time.
- General reaction: $R–CO–O–R' + H_2O \rightarrow R–COOH + R'–OH$ (faster in alkaline medium).
- Diester alkaloid → monoester alkaloid (loss of the acetyl group) → non-ester alkaloid (loss of the aromatic ester as well). Each step gives a far less toxic product (F2).
- Alkaline Gomutra (F3) promotes hydrolysis. Water alone, close to neutral, does it much more slowly (F4). Some alkaloid also leaches into the medium, which is why the medium is changed daily.

(c) Activity retained, toxicity reduced — 4 marks

- Established: published HPLC studies show that Shodhana in Gomutra lowers diester alkaloids and raises their hydrolysed products. Animal studies report lower toxicity after Shodhana.
- Proposed: the monoester alkaloids and other constituents may carry part of the analgesic and anti-inflammatory activity at a much wider safety margin.
- Not established: the exact contribution of each product to the clinical effect. Remember also that Shodhana lowers but does not remove the hazard. The processed drug is still a Schedule E(1) poison.

(d) Handling and first aid — 5 marks

- *F5.* Aconite alkaloids are absorbed through skin, especially through cuts. Local sodium-channel action causes tingling and numbness.
- *Precautions.* Gloves, mask and apron while cutting or powdering; powder only in a closed system or fume hood; no eating, drinking or touching the face; wash hands with soap; separate, labelled, locked storage; record of quantities issued; label finished products with the Schedule E(1) caution.

- *First aid.* Remove contaminated gloves and clothing; wash skin with plenty of soap and water; stop work and observe the worker. If numbness spreads, or palpitations, giddiness, vomiting or breathlessness appear, take the worker to hospital at once. Aconite poisoning can cause fatal arrhythmia. Inform the doctor of the exposure.

WHERE MARKS ARE LOST

- Writing “Gomutra removes the poison” without naming ester hydrolysis.
- Forgetting the reaction.
- Leaving out first aid, or writing it vaguely.

Revise: Volume 0 — Module 3, Unit 3 (p. 163); Module 4, Unit 2 (p. 215).

ABQ 3.3.2 — Triphala and Iron: Phenols at Work

Anchor: M3U3 (3.2, 3.3) · Also draws on: M3U2 · Suggested time: 35 minutes

THE CASE

Lauha Bhasma is given Bhavana with Triphala Kwatha before the final Puta, as described in the classical text followed by the department.

F1 After Bhavana and drying, the reddish-brown Bhasma becomes dark brown to black.

F2 FTIR of the treated sample shows a broad O–H band near 3400 cm^{-1} and bands near 1700, 1610 and 1200 cm^{-1}. These are absent in the untreated sample.

F3 In the scholar's in-vitro test, the treated sample releases more soluble iron into simulated gastric fluid (pH 1.2) than the untreated sample.

F4 A portion of the treated sample stored in a damp room for six months smells musty and has formed lumps.

F5 The scholar writes: "Triphala converts iron oxide into an organic iron complex that is fully absorbed."

Q	Question	Marks	Domain	Level
(a)	Name the main phytoconstituents of Triphala and their functional groups.	4	CK	K
(b)	Explain F1 to F3 in terms of metal–ligand interaction and the reactivity of these functional groups.	6	CAN	KH
(c)	Explain F4. How do functional groups influence the stability and shelf life of such a product?	4	CAN	KH
(d)	Evaluate the statement in F5.	6	CE	KH

Model answer framework

(a) Triphala constituents — 4 marks

Constituent group	Examples	Functional groups
Hydrolysable tannins	Chebulagic acid, chebulinic acid	Many phenolic –OH; ester linkages
Phenolic acids	Gallic acid, ellagic acid	Phenolic –OH; –COOH (gallic acid); lactone (ellagic acid)
Flavonoids	Quercetin and related	Phenolic –OH; C=O; C=C
Organic acids, vitamin C (Amalaki)	Ascorbic acid	Enol –OH; lactone; easily oxidised
Sugars	Glucose and others	Alcoholic –OH; aldehyde (reducing)

(b) Metal–ligand interaction — 6 marks

- Neighbouring phenolic –OH groups (as in gallic acid) bind iron(III) as a chelate. Iron–polyphenol complexes are dark blue-black — this is the chemistry of old iron-gall ink. This explains the darkening (F1).
- Polyphenols and ascorbic acid are reducing agents. They can reduce some Fe(III) to Fe(II) at particle surfaces.
- FTIR (F2): the broad band near 3400 cm^{-1} is O–H stretch; near 1700 cm^{-1} is C=O of acids or esters; near 1610 cm^{-1} is aromatic C=C; near 1200 cm^{-1} is C–O stretch. These show that plant organic matter is now present on the Bhasma.
- F3: organic ligands and Fe(II) are more soluble in acid than crystalline hematite. So more iron dissolves in vitro. The organic layer also acts like a capping layer on particle surfaces.

(c) Stability and shelf life — 4 marks

- Many –OH and –COOH groups attract water (hygroscopic). In a damp room the sample takes up moisture, forms lumps and supports mould (F4).
- Phenols oxidise in air and light, so colour and composition can change on storage.
- Ester bonds of tannins can hydrolyse in moisture.
- So a Bhasma with an organic coat must be dried well, packed airtight with protection from light, and given a shelf life based on stability data.

(d) Evaluating F5 — 6 marks

Part of claim	Judgement
“converts iron oxide”	Only a surface fraction interacts. XRD would still show hematite as the main phase. “Converts” overstates it
“organic iron complex”	Supported in part: FTIR and colour show organic matter bound to iron. The exact complex is not identified
“fully absorbed”	Not supported. In-vitro solubility is not absorption. Absorption needs a study in animals or humans (iron levels in blood, ferritin, tissue iron)

Better wording: “Bhavana with Triphala adds polyphenol-rich organic matter to Lauha Bhasma and increases iron solubility in simulated gastric fluid. Its effect on absorption is not yet known.”

WHERE MARKS ARE LOST

- Listing Triphala constituents without their functional groups.
- Not using the FTIR data given in the case.
- Accepting “fully absorbed”.

Revise: Volume 0 — Module 3, Unit 3 (p. 163); Module 3, Unit 4 (p. 171).

ABQ 3.3.3 — Five Liquids for Hot Copper

Anchor: M3U3 (3.2, 3.3) · Also draws on: M4U2, M4U4 · Suggested time: 35 minutes

THE CASE

Copper sheets undergo Samanya Shodhana. Each sheet is heated red-hot and quenched seven times in each of five liquids: Tila Taila, Takra, Kanji, Gomutra and Kulattha Kwatha. The observations are as follows.

F1 Each quenching in Tila Taila gives thick smoke with an acrid smell and leaves a black, sooty layer on the sheets.

F2 Takra and Kanji turn bluish-green after a few quenchings. The pH of Kanji rises a little.

F3 Quenching in Gomutra gives a strong smell of ammonia. The liquid becomes dark blue-green.

F4 Kulattha Kwatha becomes turbid and dark, with a precipitate.

F5 At the end, the sheets are brittle, thinner and have lost 6 percent of their weight.

F6 One worker’s apron caught fire from oil splashing during quenching.

Q	Question	Marks	Domain	Level
(a)	Name the main chemical constituents and functional groups of each of the five liquids.	5	CK	KH
(b)	Explain the reactions behind F1 to F4.	7	CAN	KH
(c)	Explain why the sheets become brittle and lose weight (F5).	4	CC	KH
(d)	List the safety precautions for Nirvapa, with reference to F6.	4	CAP	KH

Model answer framework

(a) The five media — 5 marks

Medium	Main constituents	Functional groups
Tila Taila	Triglycerides of fatty acids (oleic, linoleic); lignans (sesamin)	Ester; C=C double bonds
Takra	Water, lactic acid, milk proteins, some fat	–COOH; –OH; amide (peptide); ester
Kanji	Lactic and acetic acids from fermentation of rice; sugars	–COOH; –OH
Gomutra	Urea, ammonia (from breakdown of urea), creatinine, salts	Amide (urea); amine / ammonia
Kulattha Kwatha	Polyphenols, phytic acid, proteins, carbohydrates	Phenolic –OH; phosphate; amide

(b) Reactions — 7 marks

- *Common first step.* Heating copper in air forms an oxide layer: $2Cu + O_2 \rightarrow 2CuO$ (black); some Cu_2O.
- *F1, Taila.* Hot metal breaks down the oil (thermal cracking). Glycerol gives acrolein, which causes the acrid smoke. Incomplete burning leaves carbon (soot). Fatty acids can react with copper oxide to form copper soaps.
- *F2, Takra and Kanji.* Organic acids dissolve the oxide: $CuO + 2\ RCOOH \rightarrow (RCOO)_2Cu + H_2O$. Copper lactate and acetate are blue-green. Acid is used up, so the pH of Kanji rises.
- *F3, Gomutra.* Heat releases ammonia from urea. Copper ions combine with ammonia to form a deep blue copper–ammine complex. This is complex formation by a nitrogen ligand.
- *F4, Kulattha.* Polyphenols and phytate bind copper ions and precipitate as dark complexes. Proteins denature with heat and add turbidity.

(c) Brittleness and weight loss — 4 marks

- Repeated heating and sudden cooling (thermal shock) cause stress and micro-cracks.
- Oxide forms at grain boundaries and along cracks; this weakens the metal.
- The media dissolve the oxide layer at each quenching. This removes metal as soluble salts, so the sheet becomes thinner and lighter.
- Brittleness is the practical aim: the metal can now be powdered for Marana. It also increases surface area for the reactions to come.

(d) Safety in Nirvapa — 4 marks

- Heat only small quantities of oil. Keep a lid at hand to smother an oil fire. Keep a dry-powder or CO_2 extinguisher. Never pour water on burning oil.
- Use long tongs, a face shield, heat-resistant gloves, and a cotton or flame-resistant apron (F6). Synthetic clothing melts and sticks to skin.
- Quench in a deep vessel to limit splashing; stand to one side.

- Work under an exhaust hood: acrolein and ammonia are irritant.
- Keep a first-aid kit for burns; cool burns with running water.

WHERE MARKS ARE LOST

- Naming the media without their chemistry.
- Writing only “copper reacts with the liquid” without equations or complexes.
- Missing the fire hazard, which the case clearly signals in F6.

Revise: Volume 0 — Module 3, Unit 3 (p. 163); Module 4, Unit 2 (p. 215) and Unit 4 (p. 232).

Module 4 — Shodhana and Marana: Concept, SOPs and Chemical Changes

SEMESTER 2 · AYPG-AB-RB

In the university paper (Table 6F)	
Marks range for this module	5 to 20
Units eligible for ABQ	Unit 1 — Shodhana and Marana Procedures; Unit 4 — Chemical Changes involved in Shodhana and Marana
Units eligible for LAQ	Units 2 and 3 (see Part B)
Cases in this module	ABQ 4.1.1 to ABQ 4.4.3

What the examiner is looking for. In Unit 1 cases, show that you understand *why* a step exists, and what is lost if it is skipped or changed. In Unit 4 cases, write the reactions. A Unit 4 answer without equations rarely scores above half marks.

Unit 1 — Shodhana and Marana Procedures

Syllabus points. 1.1 Overview of Shodhana and Marana procedures · 1.2 Importance of purification and incineration in Ayurvedic pharmaceutical manufacturing · 1.3 Principles of Shodhana and Marana procedures · 1.4 Applications of Shodhan and Maran procedures in Ayurvedic pharmaceutical manufacturing

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 4, Unit 1, page 207.

ABQ 4.1.1 — The Shortcut Pharmacy

Anchor: M4U1 (1.2, 1.3, 1.4) · Also draws on: M7U1 · Suggested time: 35 minutes

THE CASE

A small licensed unit wants to cut costs. The production head makes five proposals. The QA officer objects to all of them, and the owner asks you to advise.

F1 Use pharmacopoeial-grade sublimed sulphur directly in Kajjali, without Shodhana of Gandhaka in ghee and milk.

F2 Use commercial “reduced iron powder” in place of Lauha filings that have undergone Samanya and Vishesha Shodhana.

F3 Cut the Putas for Abhraka Bhasma from 60 to 10, because “the colour is already right”.

F4 Skip Bhavana with the prescribed plant juice between Putas, and triturate with water instead.

F5 Release Bhasmas after only two tests: Rekhapurnata and Varitara.

Q	Question	Marks	Domain	Level
(a)	Explain the importance of Shodhana and Marana in Ayurvedic pharmaceutical manufacturing, beyond the removal of impurities.	5	CC	KH
(b)	Explain the principles on which Shodhana and Marana work.	5	CC	KH
(c)	Evaluate each proposal F1 to F5. State what is lost, and what is open to debate.	6	CE	KH
(d)	What evidence and approvals would be needed before any change to a classical process can be accepted?	4	CS	KH

Model answer framework

(a) Importance beyond impurity removal — 5 marks

- *Removal of impurities (Mala) and harmful properties (Dosha)*: physical impurities, toxic constituents (for example in Visha dravya).
- *Change of form*: makes hard metals brittle and fit for powdering; reduces particle size.
- *Chemical transformation*: converts metals into oxides, sulphides or other stable compounds; removes free metal.
- *Samskara — imparting new properties*: “Samskaro hi gunantaradhanam” — processing transfers new qualities. Media add organic matter and may alter action.
- *Making the drug fit for the body*: improves suitability for oral use (classically, absorption and assimilation) and reduces dose.
- *Manufacturing value*: defines the identity of a classical product. A product made without the classical process is not the classical medicine.

(b) Principles — 5 marks

Principle	How it works	Example
Agni samyoga (heat)	Oxidation, decomposition, sublimation, sintering	Puta in Marana; Jarana of Naga
Jala / Drava samyoga (liquid media)	Dissolution, leaching, hydrolysis, complex formation	Nirvapa; Swedana; soaking Vatsanabha in Gomutra
Mardana and Bhavana (trituration with liquids)	Size reduction, mixing, mechanochemical reaction, coating with organic matter	Kajjali; Bhavana between Putas
Repetition (number of cycles)	Progressive, step-wise conversion	7 Nirvapas; many Putas
Kala (time) and Matra (proportion)	Completes reaction; controls product	Fixed media ratios and durations
Endpoint tests (Pariksha)	Confirms that the aim is achieved	Nishchandra; Apunarbhava

(c) Evaluating the proposals — 6 marks

Proposal	Judgement
F1 Sulphur without Shodhana	Sublimed sulphur is chemically pure. But the classical identity is “Shuddha Gandhaka”, processed in ghee and milk. The process changes physical form and adds organic matter. Whether this matters for the final product is debatable and not settled by published data. For a classical product, the classical process must be followed

Proposal	Judgement
F2 Reduced iron powder	Changes the starting material. Classical Shodhana makes Lauha brittle and changes its surface. A classical drug must be made "exclusively in accordance with the formulae" of First Schedule books (Section 3(a), Drugs and Cosmetics Act, 1940). This is not acceptable for a classical product
F3 Fewer Putas	Colour is not an endpoint. Transformation of mica is incomplete after 10 Putas; shine (Chandrika) remains; the product fails Nishchandra. Not acceptable
F4 Water instead of Bhavana dravya	Loses the plant constituents that coat particles, aid size reduction and are part of the formula. Not acceptable
F5 Two tests only	Rekhapurnata and Varitara test fineness and lightness only. They do not test Nishchandra, Apunarbhava, Niruttha, heavy metals or microbial quality. Not acceptable

(d) Evidence before any change — 4 marks

- A written change-control proposal with justification (QA system).
- Side-by-side batches by the old and new processes: XRD, elemental analysis, particle size, classical tests.
- Safety data (toxicity) where the change could affect safety.
- Stability data for the new product.
- Approval of the State Licensing Authority. If the product is no longer made as per the classical formula, it may fall into a different category under Rule 158-B.

WHERE MARKS ARE LOST

- Writing an essay on Shodhana with no reference to the five proposals.
- Not mentioning Section 3(a) or the identity of a classical drug.
- Rejecting F1 to F5 with the same one-line reason.

Revise: Volume 0 — Module 4, Unit 1 (p. 207); Module 1, Unit 1 for Section 3(a) (p. 9).

ABQ 4.1.2 — Two Rasashalas, One Abhraka Bhasma

Anchor: M4U1 (1.3, 1.4) · Also draws on: M3U2, M5U1 · Suggested time: 35 minutes

THE CASE

Two teaching Rasashalas make Abhraka Bhasma from the same lot of Krishna Vajrabhraka. Both do Shodhana and Dhanyabhraka in the same way. The differences are as follows.

F1 Unit A gives Bhavana with the prescribed plant juice before each Puta and fires 60 Putas in an electric muffle furnace with a fixed, recorded temperature cycle.

F2 Unit B gives 20 Gaja Putas with cow-dung cakes. Temperatures are not recorded. Bhavana is sometimes skipped.

F3 Unit A's product is brick-red and very fine. It passes Nishchandra.

F4 Unit B's product is blackish-brown and slightly gritty. Shining particles are seen in sunlight.

F5 XRD of Unit B's product still shows peaks of mica. Unit A's does not.

F6 Median particle size (D50) by laser diffraction is 5 µm for Unit A and 22 µm for Unit B.

Q	Question	Marks	Domain	Level
(a)	Explain the principles of Shodhana and Marana shown by Dhanyabhraka and repeated Puta.	5	CC	KH
(b)	Analyse why Unit B's product differs from Unit A's.	6	CAN	KH
(c)	Which product would you accept for use, and on what grounds?	4	CE	KH
(d)	Recommend the elements of a standard process that both units should follow.	5	CS	KH

Model answer framework

(a) Principles — 5 marks

- *Dhanyabhraka.* Abhraka is mixed with paddy grains, tied in cloth, and rubbed in water (or Kanji). Thin flakes pass out through the cloth. Principle: physical size reduction and separation of fine flakes; the grains act as an abrasive. It increases surface area for the reactions of Marana.
- *Bhavana.* Wet trituration with plant juice reduces size further, mixes organic matter in, and helps in forming pellets (Chakrika) for Puta.
- *Repeated Puta.* Each heating cycle takes the change a step further: dehydroxylation, oxidation of iron, breakdown of the mica sheets and sintering. Repetition gives a gradual, complete conversion rather than one harsh firing.
- *Endpoint tests* confirm completion: Nishchandra, Rekhapurnata, Varitara, Apunarbhava.

(b) Why Unit B differs — 6 marks

Factor	Unit A	Unit B	Effect
Number of Putas	60	20	Incomplete conversion in B (F5)
Temperature control	Fixed, recorded	Unknown, variable	B may not have reached or held the needed temperature
Bhavana	Every cycle	Irregular	Less size reduction; less organic matter; coarser product in B (F6)
Endpoint	Nishchandra passed	Shine remains	Mica sheets survive in B (F4, F5)

Colour (F3, F4): brick-red comes from hematite formed by full oxidation. Blackish-brown suggests incomplete oxidation and unconverted biotite.

(c) Which to accept — 4 marks

Accept Unit A’s product, provided it also passes the remaining tests (Apunarbhava, Niruttha, heavy metals, microbial limits). Reject Unit B’s batch: it fails Nishchandra, retains mica in XRD and is coarse. It may be reprocessed with more Putas and Bhavana, then retested. A batch is accepted on tests, not on colour or tradition of the unit.

(d) Elements of a standard process — 5 marks

- Raw material specification: identity of Krishna Vajrabhraka (physical tests; XRD or petrology).
- Written SOP for each step: media, ratios, number and duration of Bhavanas, size and drying of Chakrika.
- Puta specification: type of Puta or furnace programme, peak temperature, time at peak, cooling; temperature recorded with a calibrated thermocouple for every cycle.
- In-process checks at set stages (for example every 10 Putas): classical tests and, where available, XRD.
- Release specification: all classical Bhasma Pariksha plus physicochemical tests, heavy metals and microbial limits.
- Batch record of every step, signed and checked.

WHERE MARKS ARE LOST

- Writing “Unit A used a modern furnace, so it is better”. The furnace helps only because it gives a controlled, recorded temperature.
- Not using the XRD and particle size data given.
- Accepting a product without listing the release tests.

Revise: Volume 0 — Module 4, Unit 1 (p. 207) and Unit 3 (p. 224); Module 5, Unit 1 (p. 257).

ABQ 4.1.3 — Scaling Up Hingulottha Parada

Anchor: M4U1 (1.2, 1.4) · Also draws on: M4U4, M7U1 · Suggested time: 35 minutes

THE CASE

A GMP pharmacy decides to extract its own mercury from Hingula (Hingulottha Parada) instead of buying mercury. A trial batch uses 5 kg of Hingula. After Shodhana by Bhavana with Ardraka swarasa, the Hingula is heated in a Damaru yantra on an open fire in the general production hall. The trial shows the following.

F1 Recovered mercury is about 2.6 kg.

F2 Workers report headache, a metallic taste and irritation of the throat. There is no exhaust system, and no air monitoring.

F3 The clay seal of the Damaru yantra cracks after two hours. Fumes escape.

F4 The upper vessel holds a grey, sooty powder with fine mercury droplets, not clean globules.

F5 No temperature was recorded. The residue and the used clay were dumped in an open pit behind the unit.

Q	Question	Marks	Domain	Level
(a)	Explain the importance and principle of Hingula Shodhana and of extracting mercury by Urdhvapatana. Write the reaction and calculate the theoretical yield.	5	CAP	KH
(b)	Analyse the causes of F1, F3 and F4.	5	CAN	KH
(c)	Identify the safety and environmental failures in F2, F3 and F5, and give a corrective plan.	6	CS	KH
(d)	Which in-process and finished-product tests would confirm the quality of Hingulottha Parada?	4	CAP	KH

Model answer framework

(a) Principle, reaction and yield — 5 marks

- *Importance.* Classical texts value Hingulottha Parada as a pure source of mercury, free of the Doshas of market mercury. Shodhana of Hingula (Bhavana with Ardraka swarasa) is the classical preparation for this step.
- *Principle.* Heating mercury sulphide in air oxidises the sulphur. Mercury is released as vapour, which rises and condenses on the cool upper vessel (Urdhvapatana).
- *Reaction.* $HgS + O_2 \rightarrow Hg + SO_2$
- *Theoretical yield.* Mercury is 200.6 of the 232.7 mass units in HgS, that is about 86 percent. From 5 kg of pure HgS, about 4.3 kg of mercury is possible. The trial gave 2.6 kg, a yield of about 60 percent.

(b) Causes — 5 marks

Finding	Likely causes
F1 Low yield	Loss of vapour through the cracked seal (F3); incomplete heating; mercury trapped in soot and residue; poor cooling of the upper vessel; Hingula of low purity
F3 Cracked seal	Clay seal not dried in layers; uneven, open-fire heating with hot spots; no gradual heating schedule
F4 Grey sooty powder	Smoke and soot from the open fire mixed with condensing mercury; mercury broken into fine droplets coated with soot or sulphide (“sick” mercury). Needs washing and cleaning

(c) Safety and environmental failures, with corrections — 6 marks

Failure	Correction
Mercury vapour and SO_2 in the general production hall (F2)	Do this step only in a separate, segregated room with local exhaust and scrubbing of fumes; never in the general hall
No air monitoring	Measure mercury vapour in workroom air with a mercury vapour analyser; keep records
Workers with symptoms	Remove them from exposure; medical examination; urine or blood mercury tests; health surveillance of all exposed staff
Uncontrolled heating; cracked seal (F3)	Electric heating with temperature control; tested sealing method; gradual heating schedule; trained operator
Waste dumped in the open (F5)	Treat residue and clay as hazardous waste; store and dispose of through an authorised facility under the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016; consent of the State Pollution Control Board
No records	Batch record with weights, temperatures, times, yield and signatures

Add one line: the 2026 revision of Schedule T (G.S.R. 657(E)) adds specific requirements for Rasaushadhi manufacture, including segregated manufacturing, emission control and health surveillance of employees.

(d) Quality tests — 4 marks

- In-process: weight of Hingula and Ardraka swarasa; number of Bhavanas; temperature and time log; yield.
- Finished mercury: appearance (bright, mobile globules, no film or "tail" when it rolls); passes through cloth without residue; specific gravity; residue on evaporation; test for lead, tin and other metals by AAS or ICP; absence of sulphide.

WHERE MARKS ARE LOST

- Giving the reaction without the yield calculation.
- Writing safety as "wear a mask" only. The case needs engineering controls, monitoring, health checks and waste disposal.
- Missing the regulatory side of waste disposal.

Revise: Volume 0 — Module 4, Unit 1 (p. 207) and Unit 4 (p. 232).

Unit 4 — Chemical Changes involved in Shodhana and Marana

Syllabus points. 4.1 Chemical changes involved in Shodhan procedures · 4.2 Chemical Changes involved in Marana procedures

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 4, Unit 4, page 232.

ABQ 4.4.1 — From Copper Sheet to Tamra Bhasma

Anchor: M4U4 (4.1, 4.2) · Also draws on: M7U2 · Suggested time: 35 minutes

THE CASE

Copper sheets are taken through Samanya and Vishesha Shodhana and then Marana with Shuddha Gandhaka and Nimbu swarasa in Puta, following the department's classical SOP. Samples are tested at each stage.

F1 After Shodhana, the sheets are black on the surface and brittle.

F2 After the first Puta, XRD shows copper sulphides as the main phases, with some copper oxide.

F3 After the third Puta, the black powder is tested by Amla Pariksha: lemon juice placed on it turns blue-green within a few minutes.

F4 The teacher refuses to release this batch. She reminds the class that classical texts warn of vomiting, giddiness and burning from improperly prepared Tamra Bhasma.

F5 After Amritikarana and a further Puta, the Amla Pariksha is negative.

Q	Question	Marks	Domain	Level
(a)	Write the chemical changes during Shodhana of copper (F1).	5	CC	KH
(b)	Write the chemical changes during Marana with sulphur, and explain F2.	5	CAN	KH
(c)	Explain the chemistry of the Amla Pariksha (F3). Why is a positive test a safety concern (F4)?	5	CAN	KH
(d)	Explain the proposed chemistry of Amritikarana (F5). Mark what is established.	5	CE	KH

Model answer framework

(a) Changes during Shodhana — 5 marks

- Heating in air forms an oxide layer: $2Cu + O_2 \rightarrow 2CuO$; at higher temperature or in less air, $4Cu + O_2 \rightarrow 2Cu_2O$.
- Quenching in acidic media (Takra, Kanji, Nimbu) dissolves the oxide as copper salts: $CuO + 2H^+ \rightarrow Cu^{2+} + H_2O$.
- In Gomutra, ammonia forms a copper–ammine complex. In Taila, copper soaps form and carbon deposits.
- Thermal shock causes micro-cracks. Oxide forms at the grain boundaries. The sheet becomes brittle and black (F1).

(b) Changes during Marana — 5 marks

- Copper and sulphur combine on heating: $2Cu + S \rightarrow Cu_2S$; $Cu + S \rightarrow CuS$.
- Copper oxide also reacts with sulphur: $2CuO + 3S \rightarrow 2CuS + SO_2$ (one representative reaction).
- Excess sulphur burns off as SO_2. Where air is plentiful, some sulphide can oxidise back to oxide: $2CuS + 3O_2 \rightarrow 2CuO + 2SO_2$.
- So after the first Puta, sulphides dominate, with some oxide (F2). Organic acids of Nimbu swarasa also form copper citrate during trituration.
- Published characterisation of Tamra Bhasma reports copper sulphide as the main phase in many samples.

(c) Amla Pariksha — 5 marks

- Lemon juice contains citric acid. If the Bhasma still has free copper or copper oxide, the acid dissolves it: $CuO + 2H^+ \rightarrow Cu^{2+} + H_2O$. Copper(II) ions in the juice give a blue-green colour (F3). Copper sulphide dissolves very little in weak acid, so a well-made sulphide Bhasma gives no colour.
- So a positive Amla Pariksha means acid-soluble copper is present.
- *Safety concern (F4).* Soluble copper salts are gastric irritants and strong emetics. Copper sulphate was once used to cause vomiting. This matches the classical warnings of vomiting, giddiness and burning. The test is a simple, sensitive check for this specific danger.

(d) Amritikarana — 5 marks

Proposed	Established
Further trituration with sulphur and plant material, followed by Puta, converts the remaining free copper or oxide into sulphide	Amla Pariksha becomes negative after Amritikarana (F5), so acid-soluble copper has fallen
Organic matter coats the particles and lowers their reactivity	Classical texts describe Amritikarana as removing remaining Doshas
The product becomes better tolerated in the stomach	Controlled clinical comparison of tolerance before and after Amritikarana is limited

Conclude: the test result is established; the mechanism is a reasonable chemical model; clinical benefit needs further study.

WHERE MARKS ARE LOST

- Writing “Amla Pariksha detects toxicity” without saying it detects acid-soluble copper.
- No equations for Marana.
- Presenting Amritikarana as fully explained.

Revise: Volume 0 — Module 4, Unit 4 (p. 232); Module 7, Unit 2 for Bhasma Pariksha (p. 377).

ABQ 4.4.2 — Naga Bhasma: Lead, Fire and Realgar

Anchor: M4U4 (4.1, 4.2) · Also draws on: M3U1 · Suggested time: 35 minutes

THE CASE

Naga Bhasma is prepared in a teaching pharmacy. The steps and observations are as follows.

F1 *Shodhana by Dhalana.* Molten lead is poured seven times into Nirgundi swarasa. Granules with a dull, dark surface form.

F2 *Jarana.* The lead is heated in an iron pan and stirred for several hours with Ashvattha bark powder. A yellow-orange powder forms (Jarita Naga). XRD shows lead oxide.

F3 *Marana.* Jarita Naga is triturated with Shuddha Manahshila (As_4S_4) and plant juice, and fired in many Putas.

F4 Published characterisation studies of Naga Bhasma differ. Some report lead sulphide as the main phase; others report lead oxide or mixed phases. A trace of arsenic is often reported.

F5 Jarana was done in a closed room without exhaust. One worker later developed abdominal colic, and his blood lead was found to be high.

Q	Question	Marks	Domain	Level
(a)	Write the chemical changes during Dhalana and Jarana (F1, F2). What is the role of the plant powder?	5	CC	KH
(b)	Write the chemical changes expected during Marana with Manahshila (F3). What happens to the arsenic?	5	CAN	KH
(c)	Why does the identity of the final compound (F4) matter for safety? Compare lead oxide and lead sulphide.	5	CE	KH
(d)	Write an occupational safety plan for lead processing in this pharmacy (F5).	5	CS	KH

Model answer framework

(a) Dhalana and Jarana — 5 marks

- *Dhalana.* Molten lead (melting point 327 °C) is poured into a cold liquid. Sudden cooling breaks it into granules. The surface oxidises and reacts with plant juice constituents (acids, phenols), giving a dull, dark coat. The change is mainly physical (size reduction), with surface chemical change.
- *Jarana.* Prolonged heating with stirring in air oxidises lead: $2Pb + O_2 \rightarrow 2PbO$. Lead oxide is yellow (massicot) or red-orange (litharge).
- *Role of plant powder.* It keeps molten lead from forming a large mass, and it abrades the surface so fresh metal is exposed to air. It burns to carbon and ash; carbon can locally reduce some oxide. The ash adds minerals (for example potassium and calcium salts). It also helps powdering.

(b) Marana with Manahshila — 5 marks

- On heating, realgar supplies sulphur. Lead oxide reacts to form lead sulphide; sulphur is also oxidised to SO_2. A representative reaction: $2PbO + 3S \rightarrow 2PbS + SO_2$.
- Arsenic is oxidised to arsenic trioxide, which is volatile at Puta temperatures and largely escapes as fumes. A small amount may remain bound in the product (F4).
- Where air is plentiful, some lead sulphide may oxidise further (to oxide or sulphate). This may explain why studies report different phases.

(c) Why the identity matters — 5 marks

Point	Lead oxide (PbO)	Lead sulphide (PbS)
Solubility in gastric acid	Dissolves readily to give Pb^{2+}	Much less soluble
Expected absorption	Higher	Lower (but not zero)
Implication	A Bhasma rich in oxide carries more risk	A sulphide-rich product is expected to be less hazardous

- Total lead is the same; the risk differs. So batch-to-batch phase identity by XRD, plus an acid-solubility test, should be part of quality control.
- Mark the boundary: lower solubility does not mean safety. Lead has no safe threshold for some effects. Naga Bhasma must be used only under medical supervision, in the prescribed dose and duration.

(d) Occupational safety for lead — 5 marks

- *Engineering controls.* Do Dhalana and Jarana under local exhaust ventilation in a separate room. Lead fume and dust form at these temperatures.
- *Personal protection.* Respirator suited to metal fumes, gloves, apron, face shield; work clothes that stay in the unit.
- *Hygiene.* No eating, drinking or smoking in the area; wash hands and face before breaks; wet cleaning, never dry sweeping.

- *Health surveillance.* Pre-placement and periodic blood lead tests; haemoglobin; remove a worker from exposure if levels rise; medical review of symptoms such as colic (F5).
- *Records and training.* Written SOP; training records; exposure log.
- *Waste.* Lead-containing residues handled as hazardous waste.

WHERE MARKS ARE LOST

- Writing “lead becomes Bhasma” without equations.
- Ignoring the arsenic question in (b).
- Treating total lead as the only safety measure.

Revise: Volume 0 — Module 4, Unit 4 (p. 232); Module 3, Unit 1 (p. 145).

ABQ 4.4.3 — The Shankha Bhasma that Burned the Tongue

Anchor: M4U4 (4.1, 4.2) · Also draws on: M7U2 · Suggested time: 35 minutes

THE CASE

A pharmacy prepares three Sudha varga drugs. The QC chemist records the following.

F1 *Shodhana of Shankha.* When pieces of conch shell are boiled in Nimbu swarasa, the liquid fizzes and the pieces become softer.

F2 *Shankha Bhasma.* After Marana in a Gaja Puta, the Bhasma becomes warm when mixed with water. A 10 percent suspension has a pH of 12.4. Patients complain of a burning sensation in the mouth.

F3 The same Shankha Bhasma, after Bhavana with Kumari swarasa and a further low-heat Puta, gives a suspension of pH 9.5. There are no complaints.

F4 *Godanti Bhasma.* Gypsum heated in a mild Puta loses about one-fifth of its weight and becomes soft, white and chalky.

F5 *Mukta Shukti.* XRD of the Pishti (made by trituration with rose water, without heat) shows aragonite. XRD of the Bhasma (made with heat) shows calcite and a little calcium oxide.

Q	Question	Marks	Domain	Level
(a)	Write the chemical change during Shodhana of Shankha in Nimbu swarasa (F1).	4	CAP	KH
(b)	Explain the chemical changes during Marana of Shankha, and why F2 happened.	6	CAN	KH
(c)	Explain F3. How do Bhavana and re-heating change the product? Mark what is established.	4	CAN	KH
(d)	Explain F4 and F5 with equations. What do they teach about Pishti and Bhasma?	6	CC	KH

Model answer framework

(a) Shodhana of Shankha — 4 marks

- Shankha is mainly calcium carbonate (as aragonite) with a little organic matrix.
- Citric acid of lemon juice reacts with carbonate and releases carbon dioxide (the fizz): $3CaCO_3 + 2C_6H_8O_7 \rightarrow Ca_3(C_6H_5O_7)_2 + 3CO_2 + 3H_2O$
- The surface dissolves partly, organic matrix is loosened, and the pieces soften. Some calcium citrate stays on the surface.

(b) Marana and the burning Bhasma — 6 marks

- On heating, aragonite first changes to calcite (same formula, different crystal form).
- At high temperature (calcination, around 800 °C and above), calcium carbonate decomposes: $CaCO_3 \rightarrow CaO + CO_2$.
- Calcium oxide (quicklime) reacts with water with release of heat: $CaO + H_2O \rightarrow Ca(OH)_2$. This is why the Bhasma warms in water.
- Calcium hydroxide is strongly alkaline (pH above 12). It irritates and can burn the mucosa (F2).
- Cause: the Gaja Puta was too strong or too long, so the Bhasma was over-calcined. The aim is a fine carbonate-rich product, not quicklime.

(c) Effect of Bhavana and re-heating — 4 marks

- Kumari swarasa contains water, organic acids and polysaccharides. Acids neutralise part of the hydroxide.
- A mild re-heating with organic matter releases CO_2, which can convert some CaO and $Ca(OH)_2$ back to $CaCO_3$: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$.
- Result: lower pH (F3) and a less caustic Bhasma.
- Established: the pH falls and complaints stop. Proposed: the exact share of each reaction; the clinical role of the organic coat.

(d) Godanti and Mukta Shukti — 6 marks

- *Godanti (gypsum)* is calcium sulphate dihydrate. Heating removes water of crystallisation in steps: $CaSO_4 \cdot 2H_2O \rightarrow CaSO_4 \cdot \frac{1}{2}H_2O \rightarrow CaSO_4$ (anhydrite) Loss of $2H_2O$ from 172 g is 36 g, about 21 percent — the “one-fifth” weight loss in F4. The structure opens and the product becomes soft and chalky.
- *Mukta Shukti.* Pishti, made without heat, keeps the original aragonite. Bhasma, made with heat, converts aragonite to calcite, and over-heating produces some CaO (F5).

- *Lesson.* Pishti and Bhasma of the same raw material are different products, chemically and in crystal form. Heat-sensitive Sudha varga drugs need controlled, milder Puta. The pH of a suspension is a simple in-process test for over-calcination.

WHERE MARKS ARE LOST

- Writing “Shankha Bhasma is calcium oxide”. A good Shankha Bhasma should not be mainly quicklime.
- Missing the heat release in water as evidence of CaO.
- No calculation for the weight loss of gypsum.

Revise: Volume 0 — Module 4, Unit 4 (p. 232); Module 7, Unit 2 (p. 377).

Module 5 — Classical and Advanced Instruments: Principles, Specifications and SOPs

SEMESTER 2 · AYPG-AB-RB

In the university paper (Table 6F)	
Marks range for this module	5 to 20
Units eligible for ABQ	Unit 3 — Chromatography; Unit 4 — Spectroscopy; Unit 5 — Microscopy; Unit 6 — Calibration techniques; Unit 7 — Overview of characterisation instruments
Units eligible for LAQ	Units 1, 3, 4 and 7 (see Part B for Unit 1)
Cases in this module	ABQ 5.3.1 to ABQ 5.7.3

How instrument cases are marked. Three things earn marks: the right *choice* of instrument for the question asked, correct *interpretation* of the data in the case, and an honest statement of what the instrument *cannot* tell you. Long descriptions of instrument parts, without interpretation, earn little.

Unit 3 — Chromatography

Syllabus points. 3.1 Scope and applications of chromatography in herbal drug standardization (identification of active compounds, adulterant detection, phytochemical analysis, and quality assurance) · 3.2 Principle and types of chromatography, including TLC, HPTLC, HPLC, LC-MS, and GC-MS · 3.3 Instrumentation of chromatography, including equipment and tools used in different types of chromatography · 3.4 Data analysis in chromatography, including interpretation of chromatograms and spectra · 3.5 Chromatography's limitations, challenges, and prospects in pharmaceutical and phytochemical research

Table 6F. ABQ Yes · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 5, Unit 3, page 272.

ABQ 5.3.1 — Three Ashwagandha Churnas

Anchor: M5U3 (3.1, 3.2, 3.4, 3.5) · Also draws on: M7U3 · Suggested time: 35 minutes

THE CASE

A hospital purchase committee receives Ashwagandha (Withania somnifera root) Churna from three suppliers. The QC laboratory runs TLC using the method of the API monograph, with a withaferin A reference standard and an authenticated root powder. Then it runs HPLC.

F1 *Sample A:* the TLC pattern matches the authenticated root. HPLC marker content is close to that of the authenticated root.

F2 *Sample B:* the withaferin A band is much more intense than in the authenticated root. There are extra green-coloured bands near the solvent front, which the root reference does not show.

F3 *Sample C:* the TLC pattern matches the root reference, but every band is faint. HPLC shows marker content about one-third of the authenticated root. Microscopy shows many starch grains of a shape not seen in Withania root.

F4 The supplier of Sample B says: “More withaferin A means better quality.”

F5 The committee asks whether TLC alone is enough to approve a supplier.

Q	Question	Marks	Domain	Level
(a)	State the scope and applications of chromatography in herbal drug standardisation.	4	CC	KH
(b)	Explain the principle of TLC and HPTLC. Interpret the findings for samples A, B and C.	6	CAN	KH
(c)	Evaluate the supplier’s claim in F4.	5	CE	KH
(d)	Answer the committee’s question in F5. What are the limitations of chromatography here, and what else is needed?	5	CE	KH

Model answer framework

(a) Scope and applications — 4 marks

- *Identity:* fingerprint of the drug compared with a reference standard and an authenticated sample.
- *Adulterant and substitute detection:* extra or missing bands or peaks.
- *Assay:* quantification of marker compounds (HPTLC densitometry, HPLC).
- *Purity and safety:* residual solvents, pesticides, mycotoxins (GC, LC-MS).
- *Stability:* change in the profile over time.
- *Process control and batch consistency* in manufacturing.

(b) Principle and interpretation — 6 marks

- *Principle.* Components distribute between a stationary phase (silica gel layer) and a mobile phase (solvent) moving by capillary action. Components that bind more strongly to silica move less. Each spot has a retardation factor: R_f = distance moved by the spot ÷ distance moved by the solvent front.
- *HPTLC* uses finer, uniform plates, automatic sample application and a densitometric scanner. It gives better separation, reproducibility and quantification.

Sample	Interpretation
A	Pattern and marker content match the authenticated root. Probably genuine root powder
B	Withaferin A is generally richer in the aerial parts (leaves) of Withania than in the root. Green bands near the front suggest chlorophyll, which root does not contain. Likely adulteration or substitution with aerial parts. The official part in the API is the root
C	Right pattern, low intensity, foreign starch: the genuine root is diluted with a starchy filler

(c) "More withaferin A means better quality" — 5 marks

- Wrong. Quality means conformity to the specification of the official drug, not "more of one compound".
- The marker profile of the root (a balance of several withanolides and other constituents) is the identity of the drug. A different plant part gives a different profile, with different actions and possibly different safety.
- Withaferin A is a marker for identity and consistency. It is not, by itself, the measure of efficacy.
- The API monograph defines the root. Using leaves makes the product non-compliant (misbranded or adulterated under the Act), whatever its marker content.

(d) Is TLC alone enough? — 5 marks

- No. TLC is a qualitative fingerprint. It is less sensitive; similar compounds can share an R_f; and it does not quantify well. HPTLC or HPLC is needed for assay.
- Chromatography cannot detect inert fillers that have no bands (starch, chalk), so it misses sample C's dilution unless marker content is measured.
- It does not test heavy metals, microbial load, aflatoxins or pesticide residues.

- Approval should rest on the full API monograph: macroscopy, microscopy, foreign matter, ash and acid-insoluble ash, extractive values, loss on drying, TLC/HPTLC, assay, and the "other requirements" (microbial limits, aflatoxins, heavy metals, pesticides). Add supplier audit and testing of each lot.

WHERE MARKS ARE LOST

- Writing the principle of TLC but not interpreting the three samples.
- Agreeing that more marker means better quality.
- Saying "TLC is enough" or "TLC is useless". Both are wrong.

Revise: Volume 0 — Module 5, Unit 3 (p. 272); Module 7, Unit 3 (p. 385).

ABQ 5.3.2 — The New Peak in the Triphala Tablet

Anchor: M5U3 (3.3, 3.4) · Also draws on: M1U4 (stability) · Suggested time: 35 minutes

THE CASE

A company runs a stability study on Triphala tablets at 40 °C and 75 percent relative humidity. The HPLC method uses a C18 column, a gradient of acidified water and acetonitrile, and a photodiode array (PDA) detector at 270 nm. The analyst reports the following.

F1 At zero time, gallic acid elutes at 4.8 minutes. Chebulinic and chebulagic acid peaks elute later.

F2 At six months, the chebulinic and chebulagic acid peaks are smaller. Gallic acid and ellagic acid peaks are larger. A new peak appears at 3.9 minutes.

F3 In later runs, the gallic acid peak drifts from 4.8 to 5.3 minutes.

F4 The gallic acid peak now shows a tailing factor of 2.4. The system suitability test fails.

F5 Column back-pressure has risen steadily over the last two weeks.

Q	Question	Marks	Domain	Level
(a)	Describe the instrumentation of HPLC. Give the function of each part.	5	CC	KH
(b)	Interpret F1 and F2. What do they tell you about the stability of the tablets?	5	CAN	KH
(c)	Explain F3 to F5 and give troubleshooting steps for each.	6	CAP	KH
(d)	How would LC-MS help to identify the new peak in F2? What are its limits?	4	CE	KH

Model answer framework

(a) HPLC instrumentation — 5 marks

Part	Function
Solvent reservoirs and degasser	Hold the mobile phase; remove dissolved gas that causes bubbles and baseline noise
Pump (gradient)	Delivers mobile phase at constant, high pressure; mixes solvents in a set programme
Injector / autosampler	Introduces an exact sample volume
Guard column and analytical column (C18)	Separates the components (reversed-phase: non-polar stationary phase, polar mobile phase)
Column oven	Keeps temperature constant for reproducible retention
Detector (PDA)	Measures absorbance; the PDA records the full UV spectrum of each peak, which helps identity and peak purity
Data system	Records the chromatogram; integrates peak areas; calculates results and system suitability

(b) Interpreting F1 and F2 — 5 marks

- In reversed-phase HPLC, more polar compounds elute first. Gallic acid is small and polar, so it elutes early. Large tannins elute later.
- At six months, the tannin peaks fall while gallic and ellagic acids rise. Hydrolysable tannins (chebulinic, chebulagic acids) are esters of gallic acid and related acids. In heat and humidity, ester bonds hydrolyse and release gallic acid and ellagic acid.
- The new early peak is likely another polar degradation product.
- Conclusion: the tablets are not stable at accelerated conditions. Moisture is the main driver. Packaging and formulation need review, and shelf life must be set on these data.

(c) Troubleshooting F3 to F5 — 6 marks

Problem	Likely causes	Steps
Retention time drift (F3)	Change in mobile phase composition or pH; temperature change; column ageing; pump flow error; poor equilibration	Prepare fresh mobile phase with measured pH; use column oven; check pump flow; equilibrate longer before runs
Peak tailing (F4)	Secondary interaction of acids with column silanols; mobile phase pH not suppressing ionisation; void at column head; contamination	Acidify mobile phase properly; flush or replace guard column; reverse-flush or replace column
Rising pressure (F5)	Blocked inlet frit or guard column from particles; buffer precipitation; unfiltered samples	Filter samples and mobile phase (0.45 or 0.22 µm); replace guard column; flush salts with water before organic solvent

Add: no sample result is reported while system suitability fails. Fix the system and repeat.

(d) LC-MS for the new peak — 4 marks

- LC-MS gives the molecular mass of the compound in the peak, and MS/MS gives fragment ions. Comparing these with known degradation products (for example, partially hydrolysed tannins or gallic acid derivatives) can identify it.
- Limits: isomers have the same mass; ionisation may suppress some compounds; quantification needs reference standards; full structure may need NMR after isolation.

WHERE MARKS ARE LOST

- Describing HPLC parts without functions.
- Missing that a falling tannin peak and a rising gallic acid peak point to hydrolysis.
- Reporting results despite a failed system suitability test.

Revise: Volume 0 — Module 5, Unit 3 (p. 272); Module 1, Unit 4 (p. 56).

ABQ 5.3.3 — The “Arishta” that Blinded

Anchor: M5U3 (3.2, 3.3, 3.4) · Also draws on: M7U5, M7U1 · Suggested time: 35 minutes

THE CASE

Several people in one district are admitted to hospital with vomiting, blurred vision and severe metabolic acidosis. All had been drinking large amounts of a cheap product sold as an “Ayurvedic tonic — Arishta”. The label says “contains self-generated alcohol”. The State Drug Controller sends samples to the drug testing laboratory.

F1 Headspace gas chromatography with a flame ionisation detector shows ethanol at about 30 percent v/v.

F2 A second large peak matches the retention time of methanol. It is estimated at about 8 percent v/v.

F3 TLC shows almost none of the bands expected from the herbs named on the label.

F4 Total solids and reducing sugars are much lower than expected for a fermented herbal product.

F5 The manufacturer’s licence is for “Asava-Arishta”, but the premises have no fermentation vessels.

Q	Question	Marks	Domain	Level
(a)	Why is gas chromatography, and not HPLC, the method of choice for F1 and F2? Explain the principle of headspace GC.	5	CC	KH
(b)	Interpret F1 to F5. What kind of product is this?	5	CAN	KH
(c)	How would you confirm the identity of methanol and quantify it reliably? Explain the data analysis.	5	CAP	KH
(d)	What regulatory and public health actions follow?	5	CE	KH

Model answer framework

(a) Why GC; headspace GC — 5 marks

- GC separates compounds that are volatile and thermally stable. Ethanol and methanol are small, volatile alcohols — ideal for GC. In HPLC they give weak detector response and poor retention.
- *Principle of GC:* the sample is vaporised and carried by an inert gas (nitrogen or helium) through a column. Components partition between the gas and a liquid stationary phase coated inside the column. Each elutes at its own retention time. The flame ionisation detector responds to carbon compounds.
- *Headspace GC:* the sample is sealed in a vial and heated. Volatile components move into the gas above the liquid. Only this gas is injected. The non-volatile matrix (sugars, plant matter) does not enter the column. It gives clean, reproducible analysis of volatiles in complex liquids.

(b) Interpretation — 5 marks

Finding	Meaning
F1 Ethanol about 30 %	Far higher than natural fermentation in Asava-Arishta can produce. Yeast stops at much lower levels. Alcohol has been added
F2 Methanol about 8 %	Fermentation produces only traces of methanol. Percent levels mean illicit or industrial spirit was used. This explains the blurred vision and acidosis (methanol poisoning)
F3 Herb bands absent	The herbs named on the label are not present in meaningful amounts
F4 Low solids and sugars	Not a fermented herbal preparation
F5 No fermentation vessels	The product could not have been made as claimed

Conclusion: a spurious and adulterated product passed off as an Ayurvedic medicine — in practice, a spirit sold under a medicine label.

(c) Confirmation and quantification — 5 marks

- *Confirmation:* GC-MS. The mass spectrum of the peak is matched with a reference spectrum (library) and with an injected methanol standard. Retention time on a second column of different polarity adds confidence.
- *Quantification:* calibration with methanol standards across the expected range; an internal standard (for example n-propanol or tert-butanol) to correct for injection variation; blanks and a spiked control.
- *Data analysis:* identify peaks by retention time; integrate peak areas; calculate ratio to internal standard; read concentration from the calibration curve; report with units and uncertainty.

(d) Regulatory and public health action — 5 marks

- Immediate public warning and recall of all batches; stop sale through the State Licensing Authority.
- Seizure of stock and records by Drug Inspectors.

- Action under Chapter IVA of the Drugs and Cosmetics Act, 1940, which deals with ASU drugs that are misbranded (Section 33E), adulterated (Section 33EE) or spurious (Section 33EEA). Suspension or cancellation of the licence.
- Inform hospitals so that patients get antidote treatment (fomepizole or ethanol under medical care) and dialysis where needed.
- Report through the Ayush pharmacovigilance network. Inform the excise and police authorities, since illicit spirit is involved.

WHERE MARKS ARE LOST

- Choosing HPLC for alcohols.
- Not linking methanol to the clinical picture.
- Writing “action under the Act” without naming the relevant sections or steps.

Revise: Volume 0 — Module 5, Unit 3 (p. 272); Module 7, Unit 5 (p. 399); Module 1, Unit 1 (p. 9).

Unit 4 — Spectroscopy

Syllabus points. 4.1 Introduction, principles, and types (IR, UV, NMR, AAS, ICP, etc.) of spectroscopy · 4.2 Specifications of spectroscopy instruments · 4.3 SOPs for handling spectroscopy instruments · 4.4 Troubleshooting and maintenance of spectroscopy instruments

Table 6F. ABQ Yes · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 5, Unit 4, page 283.

ABQ 5.4.1 — The Rejected Export Consignment

Anchor: M5U4 (4.1, 4.3, 4.4) · Also draws on: M7U1 · Suggested time: 35 minutes

THE CASE

An Ayurvedic company exports Triphala Churna. The importing country's laboratory rejects a consignment. The company's own laboratory had passed it.

F1 In-house flame AAS: lead 6 ppm. Importer's ICP-MS: lead 14 ppm.

F2 In-house: mercury "not detected". Importer: mercury 0.8 ppm.

F3 In-house digestion: the sample is heated with nitric acid in an open beaker on a hot plate until nearly dry.

F4 In-house calibration standards were prepared three months ago and kept at room temperature. No certified reference material (CRM) or spiked sample is run with the batch.

F5 The in-house laboratory has no cold-vapour or hydride-generation attachment.

F6 The API limit for herbal formulations is lead 10 ppm and mercury 1 ppm.

Q	Question	Marks	Domain	Level
(a)	Explain the principles of AAS, ICP-OES and ICP-MS for elemental analysis.	5	CC	KH
(b)	Analyse the likely reasons for the difference in results (F1 to F5).	6	CAN	KH
(c)	Write the key steps of an SOP for heavy metal testing of herbal products.	5	CS	KH
(d)	Interpret the results against the limits (F6). How would the interpretation differ for a Rasaushadhi?	4	CE	KH

Model answer framework

(a) Principles — 5 marks

Technique	Principle	Typical use
AAS (flame)	Free ground-state atoms in a flame absorb light of their own specific wavelength from a hollow cathode lamp. Absorbance is proportional to concentration (Beer–Lambert)	One element at a time; ppm range
AAS (graphite furnace)	Same principle; sample atomised in a heated graphite tube; far more sensitive	Lead and cadmium at low levels
Cold-vapour AAS	Mercury reduced to vapour at room temperature and measured	Mercury
Hydride generation AAS	Arsenic converted to volatile hydride	Arsenic
ICP-OES	Argon plasma (about 6000 to 10,000 K) excites atoms; they emit light at specific wavelengths	Many elements at once
ICP-MS	Plasma ionises atoms; ions are separated by mass-to-charge ratio	Very low (trace) levels; many elements; isotopes

(b) Reasons for the difference — 6 marks

- *Loss of mercury (F2, F3).* Mercury is volatile. Open-beaker digestion to near dryness drives it off. Mercury needs closed-vessel (microwave) digestion and cold-vapour measurement (F5).
- *Under-recovery of lead (F1, F3).* Incomplete digestion of silica-rich plant matrix; losses by spattering or adsorption.
- *Weak calibration (F4).* Old, dilute standards change with time (adsorption on container walls, evaporation). No CRM, so bias is not detected.
- *Sensitivity.* Flame AAS is near its practical limit at a few ppm of lead in a digest. ICP-MS is far more sensitive and accurate at this level.
- Conclusion: the in-house result is likely biased low. The importer’s result is more credible.

(c) SOP — key steps — 5 marks

1. Representative sampling; homogenise; record weight.
2. Closed-vessel microwave digestion with high-purity nitric acid (with hydrogen peroxide as needed).
3. Reagent blanks, a CRM of similar matrix and a spiked sample in every batch.
4. Fresh multi-point calibration from traceable stock standards; check linearity.
5. Measure by the right technique: graphite furnace AAS or ICP for lead and cadmium; cold-vapour AAS for mercury; hydride AAS or ICP-MS for arsenic.
6. Accept the batch only if recovery and CRM values fall within set limits.
7. Report with LOQ; record raw data; second-person check.

(d) Interpretation — 4 marks

- Lead 14 ppm exceeds the limit of 10 ppm. Mercury 0.8 ppm is within the limit of 1 ppm, but close to it. The batch fails on lead.
- Investigate the source: soil on the raw fruits, grinding equipment, water, or adulterant.
- For a Rasaushadhi, metals such as mercury or lead are deliberate ingredients. The herbal limits do not apply to the intended metal. The questions become: is the correct compound formed (XRD), is free or soluble metal within safe levels, are *unintended* toxic metals absent, and is the label carrying the required caution for Schedule E(1) substances? The importing country's own rules decide whether such a product can be sold there at all.

WHERE MARKS ARE LOST

- Describing AAS with no mention of why mercury was "not detected".
- Ignoring the absence of a CRM.
- Applying herbal limits blindly to Rasaushadhi.

Revise: Volume 0 — Module 5, Unit 4 (p. 283); Module 7, Unit 1 (p. 368).

ABQ 5.4.2 — Five Mistakes in One Thesis Chapter

Anchor: M5U4 (4.1, 4.3) · Also draws on: M3U4 · Suggested time: 35 minutes

THE CASE

You are asked to review the "Analytical Study" chapter of a scholar's thesis on a herbal extract. You find the following.

F1 The FTIR spectrum (KBr pellet) shows a broad band at 3400 cm^{-1}. The scholar writes: "This proves phenolic compounds and hence antioxidant activity."

F2 The KBr powder is kept on an open shelf. A pellet of plain KBr also shows bands at 3400 and 1630 cm^{-1}.

F3 UV absorbance at 254 nm was measured in glass cuvettes. The reading was 3.2.

F4 Total phenolic content (Folin–Ciocalteu method at 760 nm) was calculated from a gallic acid curve. Several sample readings lie above the highest standard.

F5 In the ^{1}H-NMR spectrum (solvent $CDCl_3$) of an isolated compound, a sharp singlet at 7.26 ppm is assigned to "an aromatic proton of the compound".

Q	Question	Marks	Domain	Level
(a)	Explain the principles of IR, UV-visible and NMR spectroscopy, and the information each gives.	5	CC	KH
(b)	Identify the error in each of F1 to F5 and explain it.	7	CAN	KH
(c)	Give the key SOP points for handling a UV-visible spectrophotometer and an FTIR.	4	CAP	KH
(d)	What can these techniques prove, and not prove, about "antioxidant activity"?	4	CE	KH

Model answer framework

(a) Principles — 5 marks

Technique	Principle	Information
IR (FTIR)	Molecules absorb infrared energy at frequencies that match bond vibrations (stretching, bending)	Functional groups present (O–H, C=O, C–H, N–H, M–O)
UV-visible	Electrons in conjugated systems and chromophores absorb UV or visible light; absorbance follows Beer–Lambert law ($A = \varepsilon cl$)	Quantification; presence of chromophores; colour reactions
NMR	Nuclei with spin (^{1}H, ^{13}C) in a strong magnetic field absorb radiofrequency energy; their position (chemical shift) depends on chemical surroundings	Detailed structure: number and type of H and C atoms and their connections

(b) The five errors — 7 marks

Finding	Error and explanation
F1	Over-interpretation. A broad band at 3400 cm^{-1} shows O–H stretching from any hydroxyl group — alcohols, acids, sugars, phenols — or water. It cannot prove phenols, and FTIR cannot prove biological activity
F2	KBr is hygroscopic. Moist KBr itself gives water bands at about 3400 (O–H stretch) and 1630 cm^{-1} (H–O–H bend). So the band in F1 may be partly or wholly water. KBr must be dried and kept in a desiccator; a background or blank must be recorded
F3	Glass absorbs strongly below about 320 nm, so readings at 254 nm in glass cuvettes are meaningless. Use quartz cuvettes. Also, an absorbance of 3.2 is far outside the reliable linear range (roughly 0.1 to 1.0 for best accuracy); dilute the sample
F4	Results outside the calibration range are extrapolated. Dilute the samples so they fall within the range of the standards
F5	The singlet at 7.26 ppm is the well-known signal of residual $CHCl_3$ in $CDCl_3$ solvent. It is not from the compound. Solvent peaks must be identified before assigning signals

(c) SOP points — 4 marks

- *UV-visible:* warm up the lamp; run a baseline with the blank; use matched quartz cuvettes for UV; handle cuvettes by the frosted sides; check wavelength accuracy and absorbance with reference filters at set intervals; keep readings within the linear range.
- *FTIR:* keep the sample compartment dry (check the desiccant); record a background before samples; dry KBr and press uniform pellets, or use ATR; clean the ATR crystal between samples; record spectra at a set resolution and number of scans.

(d) What they prove — 4 marks

- They can show functional groups (FTIR), quantify a class of compounds by a colour reaction (UV-visible, for example total phenols), and identify structures (NMR).
- They cannot prove activity. Antioxidant activity must be measured by suitable assays (for example DPPH or FRAP in vitro), and its biological relevance needs cell or animal studies. Even then, in-vitro antioxidant activity does not prove a clinical effect.

WHERE MARKS ARE LOST

- Listing principles without addressing the five findings.
- Missing the $CDCl_3$ solvent peak — a classic examiner's trap.
- Accepting "proves antioxidant activity".

Revise: Volume 0 — Module 5, Unit 4 (p. 283); Module 3, Unit 4 (p. 171).

ABQ 5.4.3 — The New Laboratory's First Month

Anchor: M5U4 (4.2, 4.3, 4.4) · Also draws on: M5U6 · Suggested time: 35 minutes

THE CASE

A college sets up a new analytical laboratory for its drug testing unit. In the first month the following problems arise.

F1 *AAS:* the absorbance of the same copper standard falls steadily through the day. The burner slot has white deposits and black carbon. The hollow cathode lamp has been used continuously since installation.

F2 *UV-visible:* the check with a holmium oxide reference filter shows a peak shifted beyond the tolerance stated in the manual.

F3 *FTIR:* every spectrum shows sharp bands of water vapour and a band of carbon dioxide near 2350 cm^{-1}. The silica gel indicator inside the instrument has changed colour.

F4 *ICP-OES:* the plasma goes out when samples with high salt content are run.

F5 There is no instrument logbook, no maintenance schedule and no annual maintenance contract.

Q	Question	Marks	Domain	Level
(a)	Which specifications should the purchase committee have checked for a UV-visible spectrophotometer and an AAS?	5	CK	KH
(b)	Troubleshoot F1 to F4. Give the likely cause and the remedy for each.	7	CAN	KH
(c)	Prepare a maintenance and documentation plan for these instruments.	5	CS	KH
(d)	Why are these instrument problems a quality assurance issue, and not only a technical one?	3	CE	KH

Model answer framework

(a) Specifications — 5 marks

UV-visible spectrophotometer	AAS
Wavelength range (about 190 to 1100 nm)	Wavelength range and monochromator quality
Wavelength accuracy and repeatability	Number of lamp positions (turret)
Spectral bandwidth (resolution)	Background correction (deuterium lamp or Zeeman)
Stray light level	Atomisers available: flame, graphite furnace, hydride and cold-vapour systems
Photometric accuracy, range and noise	Detection limits for the elements of interest (Pb, Cd, As, Hg)
Single or double beam; software with audit trail	Gas safety interlocks; software with audit trail

(b) Troubleshooting — 7 marks

Problem	Likely cause	Remedy
F1 AAS signal falls	Burner slot blocked by salt and carbon; nebuliser partly blocked; lamp ageing; flame conditions drifting	Clean burner and nebuliser; aspirate water between samples; check aspiration rate; optimise flame; check lamp energy and replace if low; recalibrate during the day
F2 UV wavelength shift	Wavelength calibration out after transport or installation	Recalibrate wavelength by the service engineer; repeat the filter check; do not use for assays until it passes
F3 FTIR water and CO_2 bands	Saturated desiccant; humid room; background not recent	Replace or regenerate desiccant; purge with dry air or nitrogen; take a fresh background before each set
F4 ICP plasma goes out	High dissolved solids overload the plasma; blocked nebuliser or torch; low argon purity or pressure	Dilute samples; use a nebuliser and torch suited to high-salt samples; check argon supply; clean torch and cones

(c) Maintenance and documentation plan — 5 marks

- An instrument master list with identity number, location and responsible person.
- A logbook for each instrument: every use, every problem, every repair.
- Daily or pre-use checks (for example baseline and lamp energy); periodic performance checks (wavelength accuracy, photometric accuracy, resolution); annual preventive maintenance.
- Annual maintenance contract with the supplier; calibration by an accredited agency where needed.
- Status labels: “calibrated / due on”, “out of order”.
- SOPs for operation, cleaning and maintenance; training records of users.

(d) Why this is a QA issue — 3 marks

Every result issued from a faulty instrument is unreliable. Batches may be passed or failed wrongly; patients may be harmed. Data integrity — complete, accurate, traceable records — is a core QA requirement under GMP and laboratory accreditation. A laboratory that cannot show its instruments were working cannot defend its results.

WHERE MARKS ARE LOST

- Listing generic specifications for "spectroscopy" without separating UV-visible and AAS.
- Giving remedies without causes.
- Missing the documentation part of the case (F5).

Revise: Volume 0 — Module 5, Unit 4 (p. 283) and Unit 6 (p. 299).

Unit 5 — Microscopy

Syllabus points. 5.1 Introduction, principles, and types of microscopy (optical, petrological, electron, etc.) · 5.2 Specifications of microscopy instruments · 5.3 SOPs for handling microscopy instruments · 5.4 Use of micrometer in microscopy

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 5, Unit 5, page 292.

ABQ 5.5.1 — Which Abhraka Is This?

Anchor: M5U5 (5.1, 5.3) · Also draws on: M3U2 · Suggested time: 35 minutes

THE CASE

A Rasashala receives three samples sold as “Abhraka” and one sample of Gairika. Thin cleavage flakes and powder mounts are examined under a petrological (polarising) microscope.

F1 *Sample 1:* dark brown-black flakes. In plane-polarised light, the colour changes from dark brown to pale yellow as the stage is rotated. Between crossed polars, bright colours appear.

F2 *Sample 2:* colourless, transparent flakes. No colour change on rotation in plane-polarised light. Bright, high-order colours between crossed polars.

F3 *Sample 3:* golden-bronze flakes. When a flake is heated on a spatula, it swells into a long, worm-like thread many times its original thickness.

F4 *Gairika:* red particles, mixed with angular, colourless grains that are bright between crossed polars, and with fragments of fired clay.

F5 The pharmacy wants to use Krishna Abhraka for Abhraka Bhasma.

Q	Question	Marks	Domain	Level
(a)	Explain the principle of the petrological microscope and the optical properties used to identify minerals.	6	CC	KH
(b)	Identify samples 1 to 3 and comment on the Gairika sample. Which sample suits the pharmacy (F5)?	5	CAN	KH
(c)	Write the SOP for handling a petrological microscope.	5	CAP	KH
(d)	What are the limits of optical identification? Which tests would confirm your conclusion?	4	CE	KH

Model answer framework

(a) Principle and properties — 6 marks

- A petrological microscope is a compound microscope with a *polariser* below the stage, an *analyser* above the objective, and a *rotating stage*.
- With the analyser out, the sample is seen in *plane-polarised light* (PPL). With the analyser in, at right angles to the polariser, it is seen *between crossed polars* (XPL).
- Most minerals are anisotropic: light splits into two rays travelling at different speeds. Between crossed polars this gives *interference colours*.

Property	Seen in	Meaning
Colour and pleochroism	PPL	Colour change on rotation — strong in biotite
Cleavage and form	PPL	Perfect sheet cleavage in mica
Relief	PPL	How clearly the grain stands out from the mounting medium
Birefringence (interference colours)	XPL	Strength of double refraction
Extinction	XPL	Grain goes dark at set positions on rotation; angle helps identification
Isotropic behaviour	XPL	Stays dark (glass, some minerals)

(b) Identification — 5 marks

Sample	Identity	Reason
1	Biotite (black mica) — Krishna Abhraka	Dark colour, strong pleochroism, sheet cleavage, high interference colours
2	Muscovite (white mica) — Shweta Abhraka	Colourless, non-pleochroic, high interference colours
3	Vermiculite, not true Abhraka	Exfoliates into worm-like threads on heating — typical of vermiculite
Gairika	Adulterated	Angular quartz grains and fired clay suggest added brick or sand powder

Sample 1 suits the pharmacy (Krishna Abhraka, preferred for Bhasma).

(c) SOP — 5 marks

1. Check that the polariser and analyser are set at right angles (field dark with no sample).
2. Clean slides and cover slips; mount thin flakes or powder in a suitable medium.
3. Start with the low-power objective; focus with the coarse, then the fine, adjustment.
4. Centre the objective so that a grain stays in the field when the stage rotates.
5. Observe in PPL: colour, pleochroism, cleavage, relief. Then in XPL: interference colours, extinction.
6. Record observations and photographs with a scale bar.
7. Return to low power, lower the stage, switch off the lamp, cover the microscope; enter the logbook.

(d) Limits and confirmation — 4 marks

- Optical microscopy gives strong clues, but similar minerals can look alike, and fine powders are hard to examine.
- It cannot measure chemical composition or detect trace toxic elements.
- Confirm with XRD (crystal structure and phase), XRF or SEM-EDAX (elemental composition), and simple physical tests (specific gravity, hardness, behaviour on heating).

WHERE MARKS ARE LOST

- Writing the parts of a microscope without explaining PPL and XPL.
- Calling sample 3 a variety of Abhraka.
- Missing the Gairika adulteration.

Revise: Volume 0 — Module 5, Unit 5 (p. 292); Module 3, Unit 2 (p. 153).

ABQ 5.5.2 — “Nano-sized Under My Microscope”

Anchor: M5U5 (5.1, 5.4) · Also draws on: M8U1, M8U2 · Suggested time: 35 minutes

THE CASE

A scholar measures the particle size of a Bhasma with a trinocular microscope fitted with an eyepiece micrometer of 100 divisions. The stage micrometer is 1 mm divided into 100 divisions.

F1 With the 40× objective, she finds that 40 eyepiece divisions coincide with 10 stage divisions.

F2 She then switches to the 100× oil-immersion objective but uses the same calibration factor. She does not recalibrate.

F3 Later, on recalibration at 100×, 40 eyepiece divisions are found to coincide with 4 stage divisions.

F4 She reports: “Many particles measured 0.2 eyepiece division, that is 50 nm. The Bhasma is a nanomaterial.”

F5 Her photographs show many particles stuck together in clusters, which she counted as single particles.

Q	Question	Marks	Domain	Level
(a)	Explain the principle of micrometry. Calculate the calibration factor at 40× (F1) and at 100× (F3).	6	CAP	KH
(b)	Identify the errors in F2, F4 and F5. Explain the limit of resolution of a light microscope.	6	CAN	KH
(c)	Write an SOP for particle size measurement of a Bhasma by optical microscopy.	4	CS	KH
(d)	Which techniques should she use to study particles in the nanometre range?	4	CE	KH

Model answer framework

(a) Micrometry and calculation — 6 marks

- The eyepiece micrometer has divisions of no fixed real size. Its value depends on the magnification of the objective. It must be calibrated against a stage micrometer of known divisions, for each objective.
- One stage division = 1 mm ÷ 100 = 10 µm.
- Calibration factor = (number of stage divisions × 10 µm) ÷ number of eyepiece divisions that coincide.
- *At 40×:* (10 × 10) ÷ 40 = **2.5 µm per eyepiece division.**
- *At 100×:* (4 × 10) ÷ 40 = **1.0 µm per eyepiece division.**
- Particle size = number of eyepiece divisions × calibration factor.

(b) The errors — 6 marks

Finding	Error
F2	One calibration was used for two objectives. All measurements at 100× are wrong by a factor of 2.5
F4	(i) Using the 40× factor, 0.2 division = 0.5 µm, not 50 nm — an arithmetic error. At 100× it is 0.2 µm. (ii) A fraction of one division cannot be read reliably. (iii) Most importantly, 50 nm is far below what a light microscope can resolve
F5	Clusters (agglomerates) counted as single particles; size distribution is distorted. Samples must be dispersed properly before mounting

Resolution limit. By Abbe's relation, the smallest distance that can be resolved is about d = λ ÷ (2 × NA). With visible light (about 550 nm) and the best oil-immersion objective (NA about 1.4), d is about 0.2 µm (200 nm). Particles smaller than this cannot be sized by light microscopy. They may be seen as points of light, but not measured.

(c) SOP — 4 marks

1. Disperse a small amount of Bhasma in a suitable liquid (for example with a drop of dispersant), with brief sonication.
2. Place a thin film on a clean slide; cover with a cover slip.
3. Calibrate the eyepiece micrometer for the objective in use; record the factor.
4. Measure several hundred particles across random fields, using one fixed dimension (for example the longest dimension).
5. Report the size distribution (mean, range, D10, D50, D90) and the method; note the resolution limit.

(d) Techniques for the nanometre range — 4 marks

- *SEM:* surface shape and size of particles down to tens of nanometres; with EDAX, elemental composition.
- *TEM:* internal structure and size of individual nanoparticles; electron diffraction.
- *Dynamic light scattering (DLS):* size distribution of particles in suspension, from about a nanometre to a few micrometres.
- *XRD with the Scherrer equation:* crystallite size (not particle size).

• Laser diffraction particle size analysers measure from sub-micron to millimetre range.

WHERE MARKS ARE LOST

- Wrong or missing calculation — this part carries six marks.
- Not stating the resolution limit with Abbe’s relation.
- Accepting “nanomaterial” from light microscopy.

Revise: Volume 0 — Module 5, Unit 5 (p. 292); Module 8, Unit 1 (p. 421).

ABQ 5.5.3 — The Yashtimadhu Churna that Had Too Much Starch

Anchor: M5U5 (5.1, 5.2, 5.3) · Also draws on: M7U3 · Suggested time: 35 minutes

THE CASE

A drug testing laboratory examines a market sample of Yashtimadhu (Glycyrrhiza glabra root) Churna under a compound microscope.

F1 Yellowish cork cells and vessels with bordered pits are seen.

F2 The fibres expected to be surrounded by cells containing prismatic calcium oxalate crystals are scanty.

F3 Starch grains are abundant. Many are large, polygonal and have a central hilum. They are quite unlike the small, simple starch grains of liquorice root.

F4 The analyst used only water mounts. No clearing agent or stain was used.

F5 The laboratory's microscope has a 10× and a 40× objective only, no camera and no calibrated micrometer.

F6 The same laboratory has now been asked to examine Bhasma samples by SEM at a partner institute.

Q	Question	Marks	Domain	Level
(a)	Describe the types of microscopy used in drug testing and the principle of each.	5	CC	KH
(b)	Interpret F1 to F3. What do they suggest about the sample?	5	CAN	KH
(c)	How would you improve the examination in F4? Name the reagents and their purpose.	5	CAP	KH
(d)	Comment on F5. List the specifications of a microscope suited to a drug testing laboratory, and note the special sample preparation needed for SEM of a Bhasma (F6).	5	CK	KH

Model answer framework

(a) Types of microscopy — 5 marks

Type	Principle	Use in drug testing
Bright-field light	Visible light passes through the specimen; lenses magnify	Powder microscopy of crude drugs; cell types, crystals, starch
Polarising (petrological)	Polarised light; anisotropic materials show interference colours	Minerals; calcium oxalate crystals; starch (shows a dark cross)
Fluorescence	Specimen emits light when excited by UV or blue light	Certain constituents; some adulterants
SEM	Electron beam scans the surface; secondary electrons form an image	Surface shape and size of particles; EDAX for elements
TEM	Electrons pass through a very thin specimen	Internal structure; nanoparticles

(b) Interpretation — 5 marks

- F1 is consistent with liquorice root.
- F2: crystal fibres (fibres sheathed by cells with prismatic calcium oxalate crystals) are a key diagnostic feature of liquorice root. Scanty fibres suggest dilution.
- F3: large, polygonal starch grains with a central hilum are typical of maize starch, not of liquorice root.
- Conclusion: the Churna is probably adulterated with a starch filler such as maize starch. Confirm with TLC or HPTLC for glycyrrhizin and with assay.

(c) Better examination — 5 marks

Reagent	Purpose
Chloral hydrate	Clears cell contents so cell walls and crystals are seen clearly
Phloroglucinol with hydrochloric acid	Stains lignified walls (fibres, vessels, stone cells) pink-red
Iodine solution	Stains starch blue-black; makes starch grains easy to count and compare
Glycerin	Mounting medium; prevents drying
Polarised light	Calcium oxalate crystals and starch grains (dark cross) stand out

Compare with an authenticated reference powder side by side.

(d) Specifications and SEM preparation — 5 marks

- F5: with only 10× and 40× objectives and no calibrated micrometer or camera, the laboratory cannot measure or record findings. Results cannot be verified or defended.
- *Specifications:* objectives 4×, 10×, 40× and 100× oil immersion, with stated numerical aperture; Köhler illumination with LED or halogen source; mechanical stage; eyepieces 10× with an eyepiece micrometer and a stage micrometer; polarising attachment; trinocular head with digital camera and measuring software; good optics (achromatic or plan objectives).

- *SEM of a Bhasma (F6):* dry powder on a stub with conductive carbon tape; remove loose powder; coat non-conducting samples with a thin layer of gold or carbon to prevent charging; examine under vacuum. Record magnification and scale bar.

WHERE MARKS ARE LOST

- Describing microscopy in general without interpreting the starch.
- Missing crystal fibres as the key diagnostic feature.
- Ignoring the SEM preparation part.

Revise: Volume 0 — Module 5, Unit 5 (p. 292); Module 7, Unit 3 (p. 385).

Unit 6 — Calibration Techniques

Syllabus points. 6.1 Introduction, principles, and types of calibration (instrument calibration, method calibration, etc.) · 6.2 SOPs for calibration · 6.3 Troubleshooting and maintenance of calibration instruments

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 5, Unit 6, page 299.

ABQ 5.6.1 — The Balance that Drifted

Anchor: M5U6 (6.1, 6.2, 6.3) · Also draws on: M7U1 · Suggested time: 35 minutes

THE CASE

During a self-inspection of a pharmacy's QC and dispensing area, the inspector checks the analytical balance (readability 0.1 mg). The SOP allows a daily check error of ±0.5 mg.

F1 The daily check with a 100 g weight reads 100.0032 g.

F2 In the eccentricity test, the same weight placed at the corners of the pan gives readings that differ by 1.8 mg.

F3 The balance stands on an ordinary wooden table, beside the air-conditioner vent and close to the door.

F4 The last external calibration certificate expired four months ago. The check weights have no calibration certificate.

F5 In the last four months, the balance was used to weigh Shuddha Vatsanabha and Shuddha Kuchala for a batch of tablets.

Q	Question	Marks	Domain	Level
(a)	Explain the principles and types of calibration, including traceability.	5	CC	KH
(b)	Write an SOP for the calibration and routine checking of an analytical balance.	6	CS	KH
(c)	Troubleshoot F1 to F4.	4	CAN	KH
(d)	How should the pharmacy deal with the batches in F5?	5	CE	KH

Model answer framework

(a) Principles and types — 5 marks

- *Calibration* compares the reading of an instrument with a known reference standard, under stated conditions, and records the difference (error). It may lead to adjustment.
- *Traceability:* the reference must be linked, through an unbroken chain of calibrations, to national standards. In India this is the CSIR–National Physical Laboratory. In practice, calibration is done by a laboratory accredited by NABL.

Type	Meaning	Example
Instrument calibration	The instrument itself is checked against standards	Balance with certified weights; thermometer against a reference thermometer
Method calibration	The analytical method is calibrated with reference substances	HPLC calibration curve with gallic acid standards
Internal calibration / performance check	Done in-house, often daily	Daily check weight on a balance; pH buffers
External calibration	By an accredited agency, at set intervals	Annual balance calibration certificate

(b) SOP for an analytical balance — 6 marks

1. *Location:* firm, vibration-free table (stone or anti-vibration); away from doors, vents, sunlight.
2. *Before use:* check level bubble; warm up as per manual; clean pan and chamber.
3. *Daily check:* weigh certified check weights near the commonly used range; record; accept only within limits.
4. *Periodic checks:* repeatability (same weight ten times; standard deviation); eccentricity (weight at centre and corners); linearity (several weights across the range); minimum weight.
5. *External calibration:* yearly (or as per risk) by an accredited agency, using weights of known class with certificates.
6. *Records and labels:* logbook; calibration status label with date and due date; out-of-limit procedure (label “Do not use”, inform QA, investigate).

(c) Troubleshooting — 4 marks

Finding	Cause and remedy
F1 Error of 3.2 mg	Outside the ±0.5 mg limit. Stop use; recalibrate (internal adjustment, then external calibration)
F2 Eccentricity error	Pan or load cell problem, or the balance is not level. Level it; service engineer to check
F3 Location	Air draughts and temperature changes cause drift and unstable readings. Move to a proper bench; draught shield closed
F4 Expired certificate; uncertified weights	Results are not traceable. Calibrate now, and get certified weights

(d) Batches in F5 — 5 marks

- Treat as a deviation. QA opens an investigation.
- Estimate the possible weighing error and its effect on the dose of Vatsanabha and Kuchala per tablet. These are Schedule E(1) drugs with narrow safety margins, so even a small error matters.
- Review batch records and retained samples; test the tablets for uniformity and, where possible, alkaloid content.
- If safety cannot be assured, quarantine unsold stock and consider recall.
- CAPA: correct the balance; train staff; add a rule that the balance status is verified before every critical weighing; review the calibration schedule.

WHERE MARKS ARE LOST

- Writing “calibrate the balance” without the steps.
- No mention of traceability.
- Ignoring the batches — this is where most of the risk lies.

Revise: Volume 0 — Module 5, Unit 6 (p. 299); Module 7, Unit 1 (p. 368).

ABQ 5.6.2 — The Furnace Says 800 °C

Anchor: M5U6 (6.1, 6.2, 6.3) · Also draws on: M5U1 · Suggested time: 35 minutes

THE CASE

A pharmacy prepares Bhasmas in an electric muffle furnace programmed to hold 800 °C. Several batches fail Nishchandra and Apunarbhava, mainly the trays from the front of the furnace. An engineer places a calibrated reference thermocouple at different points inside the furnace.

F1 When the controller displays 800 °C, the reference thermocouple at the centre reads 742 °C.

F2 Near the door, it reads 705 °C.

F3 During heating, the temperature overshoots to 840 °C before settling.

F4 The furnace's own thermocouple is three years old, and its sheath looks scaled and oxidised.

F5 The door seal is worn. The trays are packed tightly, touching the walls.

Q	Question	Marks	Domain	Level
(a)	Explain the principles of temperature calibration and name the sensors used at Puta temperatures.	5	CC	KH
(b)	Write an SOP for calibration of a muffle furnace, including a temperature uniformity survey.	6	CS	KH
(c)	Analyse the causes of F1 to F5 and suggest remedies.	5	CAN	KH
(d)	What is the effect on the Bhasma batches, and what should be done with them?	4	CE	KH

Model answer framework

(a) Principles and sensors — 5 marks

- Temperature calibration compares the instrument's sensor and display with a reference sensor of known accuracy, placed at the same point, at several temperatures across the working range.
- The reference must be traceable to national standards (through an accredited calibration laboratory).
- *Sensors at Puta temperatures:* thermocouples — two dissimilar metals joined at a tip produce a small voltage that depends on temperature. Type K (nickel-chromium / nickel-aluminium) is common up to about 1200 °C; Types R and S (platinum-rhodium) are used for higher accuracy and higher temperatures. Infrared pyrometers measure without contact, but need the correct emissivity setting.

(b) SOP — 6 marks

1. Use a reference thermocouple and indicator with a valid certificate.
2. *Single-point check:* place the reference beside the furnace sensor; compare readings at three or more set points (for example 400, 600 and 800 °C) after stable holding.
3. *Temperature uniformity survey:* place several thermocouples at the corners and centre of the working space (with typical loading); record at each set point; calculate the spread.
4. Set acceptance limits (for example the centre within a stated tolerance; uniformity across the space within a stated range).
5. Adjust the controller offset if needed; repeat the check.
6. Record: date, set points, readings, error, adjustment, next due date; affix status label; QA review.

(c) Causes and remedies — 5 marks

Finding	Cause	Remedy
F1 Display 58 °C above actual	Drift of the ageing, oxidised thermocouple (F4); controller offset	Replace the thermocouple; recalibrate the controller
F2 Cold zone near door	Heat loss through the worn seal (F5); door area is naturally cooler	Replace the seal; define a usable zone and keep trays out of the cold zone
F3 Overshoot	Poor controller tuning (PID settings); fast ramp rate	Retune PID; use a slower ramp near the set point
F5 Tight packing	Blocks heat flow; uneven heating	Loading pattern in the SOP; space between trays and walls

(d) Effect on batches — 4 marks

- The Bhasma received less heat than the SOP intended, especially near the door. Incomplete conversion explains the failure of Nishchandra and Apunarbhava.
- The failed batches should not be released. Reprocess them with additional Putas in a calibrated furnace, then retest fully.

- Review batches released earlier from the same furnace: check retained samples. Treat as a deviation and record CAPA.

WHERE MARKS ARE LOST

- Writing about thermometers in general, not thermocouples at Puta temperatures.
- No temperature uniformity survey in the SOP.
- Not linking the cold zone to the front-row failures.

Revise: Volume 0 — Module 5, Unit 6 (p. 299) and Unit 1 (p. 257).

ABQ 5.6.3 — The Curve that Was Not Straight

Anchor: M5U6 (6.1, 6.2) · Also draws on: M5U3 · Suggested time: 35 minutes

THE CASE

An internal audit of a QC laboratory notes the following.

F1 An HPLC calibration curve for gallic acid uses five standards from 10 to 200 µg/ml. The correlation coefficient (r^2) is 0.972. The highest standard lies well below the line.

F2 Several sample results lie above the highest standard.

F3 The gallic acid reference standard's certificate shows that its retest date has passed. It is stored at room temperature.

F4 The pH meter is calibrated each morning with a pH 7.0 buffer only. It is then used to measure the pH of Arishtas, which lie near pH 3.5.

F5 The laboratory has no calibration schedule. Instruments carry no calibration status labels.

Q	Question	Marks	Domain	Level
(a)	Differentiate instrument calibration and method calibration. Define linearity, range, LOD and LOQ.	5	CC	KH
(b)	Analyse F1 to F4. State what is wrong and how to correct each.	6	CAN	KH
(c)	Design a calibration programme for this laboratory (F5).	6	CS	KH
(d)	How does this relate to validation of analytical procedures under ICH Q2?	3	CE	KH

Model answer framework

(a) Definitions — 5 marks

- *Instrument calibration:* checking the instrument's measurement against reference standards (for example HPLC flow rate, detector wavelength, pH meter with buffers).
- *Method calibration:* establishing the relationship between response and concentration for a specific analyte, using reference standards (the calibration curve).
- *Linearity:* the ability of a method to give results directly proportional to concentration within a given range.
- *Range:* the interval between the lowest and highest concentrations for which linearity, accuracy and precision are shown.
- *LOD (limit of detection):* the lowest amount that can be detected, but not necessarily quantified.
- *LOQ (limit of quantification):* the lowest amount that can be quantified with acceptable accuracy and precision.

(b) Analysis and corrections — 6 marks

Finding	What is wrong	Correction
F1 r^2 0.972; top point below the line	Non-linearity at high concentration (detector saturation or solubility); r^2 too low for an assay	Narrow the range; drop or re-prepare the top standard; check detector response; use more standards; inspect residuals, not only r^2
F2 Samples above top standard	Extrapolation beyond the validated range	Dilute samples to fall within the range
F3 Standard past retest date; wrong storage	Purity of the reference is uncertain; all results depend on it	Use a valid, properly stored reference standard; store as per certificate
F4 One-point pH calibration	Slope of the electrode is not checked; error grows away from pH 7	Two-point (or three-point) calibration with buffers bracketing the sample pH (pH 4.0 and 7.0 for Arishtas); check slope

(c) A calibration programme — 6 marks

- *Master list* of all instruments and measuring devices, with ID, location and owner.
- *Frequency* for each item, based on risk and manufacturer advice: daily checks (balance, pH meter), periodic in-house checks (HPLC performance), external calibration (yearly for balances, thermometers, furnaces).
- *SOPs* for each calibration, with acceptance criteria.
- *Status labels* on every instrument: date calibrated, due date, by whom.
- *Records* kept and reviewed by QA; certificates filed.
- *Out-of-calibration procedure:* stop use, label, investigate impact on past results, CAPA.
- *Reference standards register:* source, certificate, storage, retest dates.

(d) Link to ICH Q2 — 3 marks

ICH Q2(R2) describes validation of analytical procedures. Validation characteristics include specificity, linearity, range, accuracy, precision, and LOD/LOQ. Calibration of instruments is the foundation; validation shows that the whole method is fit for its purpose. A validated method run on uncalibrated instruments gives unreliable results.

WHERE MARKS ARE LOST

- Treating r^2 above 0.95 as "acceptable" without looking at the curve.
- Missing the bracketing rule for pH calibration.
- A calibration programme with no out-of-calibration procedure.

Revise: Volume 0 — Module 5, Unit 6 (p. 299) and Unit 3 (p. 272).

Unit 7 — Overview of Characterisation Instruments

Syllabus points. 7.1 Overview of advanced characterization instruments and their significance in the standardization of Ayurvedic medicines · 7.2 Principles and applications of working mechanism and applications of instrumental methods: PSA, XRD, XRF, SEM-E-Dax, TEM, TGA, DTA, VSM analysis

Table 6F. ABQ Yes · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 5, Unit 7, page 305.

ABQ 5.7.1 — The Complete Profile of Yashada Bhasma

Anchor: M5U7 (7.1, 7.2) · Also draws on: M8U2 · Suggested time: 35 minutes

THE CASE

A PhD scholar submits a characterisation chapter on Yashada Bhasma. The data are as follows.

F1 *XRD:* sharp peaks matching zinc oxide (hexagonal, wurtzite type). Two small peaks are not identified.

F2 *XRF:* zinc 76 percent by weight; small amounts of iron, silicon and calcium.

F3 *Laser diffraction PSA:* D10 0.8 µm, D50 3.2 µm, D90 12 µm.

F4 *SEM-EDAX:* clusters of sub-micron particles; EDAX shows zinc, oxygen and a little carbon.

F5 *TEM:* individual particles from about 30 nm to 200 nm.

F6 *TGA:* 2 percent mass loss up to 200 °C and a further 3 percent between 200 and 500 °C.

F7 The scholar writes: “DTA and VSM were not needed for this Bhasma.”

Q	Question	Marks	Domain	Level
(a)	Explain the significance of advanced characterisation in the standardisation of Bhasma.	4	CC	KH
(b)	State in a table the principle and the information given by PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA and VSM.	6	CC	KH
(c)	Interpret F1 to F6 together. What picture of the Bhasma emerges?	6	CAN	KH
(d)	Is the scholar right in F7? When are DTA and VSM needed?	4	CE	KH

Model answer framework

(a) Significance — 4 marks

- Classical tests confirm that a Bhasma is “made”; they do not define its chemical identity, purity or particle size.
- Instruments give identity (phase), composition, particle size and shape, thermal behaviour and magnetic nature.
- This allows batch-to-batch consistency, specification setting, detection of impurities, safety assessment and scientific communication.

(b) The instruments — 6 marks

Instrument	Principle	Information
PSA (laser diffraction)	Particles scatter laser light at angles related to their size	Size distribution (D10, D50, D90)
XRD	Diffraction of X-rays by crystal planes (Bragg’s law)	Crystalline phases; crystallite size
XRF	X-rays excite atoms, which emit characteristic secondary X-rays	Elemental composition (major and minor)
SEM-EDAX	Electron beam scans surface; X-rays emitted are analysed	Surface shape and size; local elemental composition
TEM	Electrons pass through thin sample	Size and structure of individual nanoparticles; lattice; diffraction
TGA	Mass recorded while heating	Moisture, organic matter, decomposition, stability
DTA	Temperature difference between sample and reference while heating	Melting, phase changes, reactions (endo- or exothermic)
VSM	Sample vibrates in a magnetic field; induced signal measured	Magnetic behaviour (saturation magnetisation, coercivity)

(c) Interpretation — 6 marks

- *Identity:* the main phase is ZnO (F1). Two minor unidentified peaks show a small crystalline impurity or a second phase — to be identified.
- *Composition:* pure ZnO contains about 80 percent zinc by weight. XRF shows 76 percent (F2). This fits ZnO with small amounts of other elements, carbon and moisture.
- *Organic matter and moisture:* carbon in EDAX (F4) and the TGA mass loss (F6) — about 2 percent moisture and volatile matter, about 3 percent organic matter burning between 200 and 500 °C. This agrees with organic residue from plant media.
- *Particle size:* primary particles of 30 to 200 nm (TEM, F5) are clustered into agglomerates; PSA (F3) measures these clusters, so its D50 of 3.2 µm is much larger. The two are not in conflict.
- *Picture:* a ZnO-based Bhasma made of nano-to-submicron primary particles, aggregated into micron-sized clusters, with a few percent organic matter and minor impurities.

(d) Are DTA and VSM needed? — 4 marks

- *VSM:* ZnO is not magnetic in the usual sense. VSM is not essential for Yashada Bhasma. It is important for iron-based Bhasmas (Lauha, Mandura) and for Abhraka, where magnetic phases (magnetite) show incomplete conversion.
- *DTA:* useful even here. It can show thermal events of impurities or organic matter and gives a thermal fingerprint for batch comparison. It is especially valuable for products with phase changes (for example Godanti, Kupipakwa preparations).
- So the scholar is right about VSM, and only partly right about DTA. A choice of instruments should be justified by the question, not omitted without reason.

WHERE MARKS ARE LOST

- Treating PSA and TEM results as contradictory.
- Not using the zinc percentage to check against ZnO stoichiometry.
- Saying “VSM is never needed”.

Revise: Volume 0 — Module 5, Unit 7 (p. 305); Module 8, Unit 2 (p. 429).

ABQ 5.7.2 — Same XRD, Different SEM

Anchor: M5U7 (7.2) · Also draws on: M4U1, M7U2 · Suggested time: 35 minutes

THE CASE

Two batches of Lauha Bhasma are compared. Batch 1 was made by the classical Puta method with Bhavana before each Puta. Batch 2 was made in a furnace with fewer cycles and without Bhavana.

F1 XRD of both batches is the same: hematite (α-Fe_2O_3).

F2 SEM of Batch 1 shows loose clusters of fine, sub-micron particles. Batch 2 shows dense, fused grains of 10 to 40 µm.

F3 PSA (D50): Batch 1, 2.1 µm; Batch 2, 8.4 µm.

F4 VSM: Batch 2 shows a clearly higher saturation magnetisation than Batch 1.

F5 Batch 2 fails Varitara.

Q	Question	Marks	Domain	Level
(a)	How can two batches have identical XRD but different SEM pictures (F1, F2)?	5	CAN	KH
(b)	Explain the principles of SEM and VSM.	5	CC	KH
(c)	Interpret F3 to F5. Which batch is better, and why?	5	CE	KH
(d)	Propose a release specification for Lauha Bhasma that combines classical and instrumental tests.	5	CS	KH

Model answer framework

(a) Same XRD, different SEM — 5 marks

- XRD identifies the crystal structure (phase). Both batches are hematite.
- SEM shows particle size, shape, surface and aggregation (morphology). The same compound can exist as fine, loose particles or as large, fused grains.
- Morphology depends on the process: Bhavana and repeated moderate firing give fine particles; fewer, harsher firings without Bhavana cause sintering — particles fuse into dense grains.
- So XRD alone cannot judge a Bhasma. “Same compound” does not mean “same product”.

(b) Principles — 5 marks

- *SEM:* a focused electron beam scans the sample surface in a vacuum. Secondary electrons from the surface are collected to form an image with high depth of field. Back-scattered electrons show contrast between heavier and lighter elements. EDAX analyses the X-rays emitted, giving elemental composition at chosen points.
- *VSM:* the sample vibrates in a uniform magnetic field. Its magnetic moment induces a voltage in pick-up coils. Plotting magnetisation against the applied field gives a hysteresis loop: saturation magnetisation, remanence and coercivity.

(c) Interpretation — 5 marks

- F3: Batch 2 is about four times coarser — consistent with SEM.
- F4: hematite is only weakly magnetic. A higher saturation magnetisation in Batch 2 means a strongly magnetic phase — most likely residual magnetite (or maghemite) — at a level XRD did not detect, or with peaks overlapping. So conversion in Batch 2 is incomplete.
- F5: coarse, dense grains do not float — failure of Varitara agrees with F2 and F3.
- *Better batch:* Batch 1 — finer, fully converted, passes classical tests. Batch 2 should be rejected or reprocessed.

(d) Release specification — 5 marks

Test	Requirement
Classical	Rekhapurnata, Varitara, Nishchandra (where applicable), Apunarbhava, Niruttha, colour, taste-free
Physicochemical	Loss on drying, total ash, acid-insoluble ash, total iron content
Phase (XRD)	Hematite as main phase; no metallic iron
Particle size (PSA)	D50 and D90 within limits set from good batches
Magnetic (VSM or a simple magnet test)	Saturation magnetisation below a set limit (indicates complete oxidation)
Safety	Heavy metals (lead, arsenic, cadmium, mercury); microbial limits

Limits are set from data on several good batches, and written into the specification.

WHERE MARKS ARE LOST

- Concluding that both batches are “the same” because XRD is identical.
- Not explaining why VSM is higher in Batch 2.
- A specification with only classical tests, or only instrumental tests.

Revise: Volume 0 — Module 5, Unit 7 (p. 305); Module 7, Unit 3 for Lauha QC (p. 385).

ABQ 5.7.3 — Rasasindura on the Thermal Balance

Anchor: M5U7 (7.2) · Also draws on: M3U1 · Suggested time: 35 minutes

THE CASE

A Rasasindura sample is characterised in a central instrumentation facility.

F1 *XRF:* mercury 85.6 percent, sulphur 13.6 percent, with traces of other elements.

F2 *XRD:* cinnabar (α-HgS) only.

F3 *TGA (nitrogen, 10 °C per minute):* less than 0.5 percent mass loss below 200 °C. Then one large mass-loss step at high temperature, ending with almost no residue.

F4 *DTA:* an endothermic peak over the same temperature range as the main TGA step.

F5 *SEM-EDAX:* mercury and sulphur, with about 1 percent carbon at some points.

F6 The exhaust of the TGA instrument is vented into the laboratory room. Staff work nearby.

Q	Question	Marks	Domain	Level
(a)	Explain the principles of TGA and DTA. What can they tell about a Kupipakwa product?	5	CC	KH
(b)	Interpret F1 to F5. Is free mercury likely to be present?	6	CAN	KH
(c)	Compare XRF, EDAX and ICP for elemental analysis of this product.	4	CAP	KH
(d)	What safety problem does F6 show? Write the corrective steps.	5	CS	KH

Model answer framework

(a) TGA and DTA — 5 marks

- *TGA:* the sample's mass is recorded continuously as temperature rises at a set rate in a chosen gas. Steps in the curve show moisture loss, burning of organic matter, decomposition or sublimation.
- *DTA:* the temperature difference between the sample and an inert reference is recorded during heating. Endothermic events (melting, sublimation, dehydration) and exothermic events (oxidation, crystallisation) appear as peaks.
- *For Kupipakwa products:* TGA shows moisture, organic residue and the temperature at which the product volatilises; it can reveal free mercury or free sulphur, which lose mass at lower temperatures than HgS. DTA gives a thermal fingerprint for batch comparison.

(b) Interpretation — 6 marks

- *Stoichiometry:* HgS contains about 86.2 percent mercury and 13.8 percent sulphur. XRF values (F1) match this closely. So the product is nearly pure HgS.
- *XRD (F2)* confirms the red cinnabar form.
- *TGA (F3):* very little mass loss at low temperature means little moisture, little organic matter and no noticeable free sulphur or free mercury. Free mercury would lose mass gradually from much lower temperatures. The single large step is the sublimation/decomposition of HgS, and the near-zero residue shows few non-volatile impurities.
- *DTA (F4):* the endotherm matches this volatilisation.
- *EDAX (F5):* traces of carbon at some points — organic residue at particle surfaces, in small amounts.
- *Free mercury:* unlikely in significant amounts. A direct test for free mercury would confirm this.

(c) XRF, EDAX and ICP — 4 marks

Point	XRF	EDAX (with SEM)	ICP-OES / ICP-MS
Sample	Powder, little preparation	Tiny area on the SEM stub	Sample must be digested to a solution
Information	Bulk composition	Local composition at chosen spots	Bulk composition; trace levels
Sensitivity	Moderate	Low (about 0.1 percent and above)	Very high (ppm to ppb)
Best for	Major elements	Linking composition to particles seen	Trace toxic elements

(d) Safety — 5 marks

- Heating HgS releases mercury and sulphur vapours. Venting into the room exposes staff to mercury vapour.

- *Corrective steps:* stop such runs until fixed; connect the exhaust to a fume extraction system with a mercury trap (for example activated carbon impregnated with sulphur); measure mercury vapour in the room; clean the area by proper methods; health check of exposed staff; write an SOP for thermal analysis of mercurial and arsenical samples, and train users; record the incident and CAPA.

WHERE MARKS ARE LOST

- Not checking the XRF numbers against the formula of HgS.
- Saying “TGA shows no mercury”. TGA shows mass loss, not identity.
- Missing the safety issue in F6.

Revise: Volume 0 — Module 5, Unit 7 (p. 305); Module 3, Unit 1 (p. 145).

Module 7 — Classical and Advanced Quality Control Techniques for Ayurveda Formulations

SEMESTER 2 · AYPG-AB-RB

In the university paper (Table 6F)	
Marks range for this module	5 to 20
Units eligible for ABQ	Unit 1 — QC and QA; Unit 3 — Solid dosage forms; Unit 4 — Semi-solid dosage forms; Unit 5 — Liquid and gaseous dosage forms
Unit eligible for LAQ	Unit 1
Unit eligible for SAQ only	Unit 2 — Classical aspects (Siddhi Lakshana, Bhasma Pariksha)
Cases in this module	ABQ 7.1.1 to ABQ 7.5.4 (the October 2026 paper's Avaleha ABQ is Q1 of the solved paper)

The most-examined module for ABQs. QC cases are the easiest to frame and the easiest to lose marks on. Answer every QC case in four layers: *what went wrong* (finding by finding), *which parameter would have caught it*, *which rule or standard applies*, and *what record would have prevented it*. Unit 2 (Siddhi Lakshana, Bhasma Pariksha) is not eligible for ABQ by itself, but its tests appear inside Unit 3 and Unit 4 cases as in-process checks.

Unit 1 — Quality Control and Quality Assurance

Syllabus points. 1.1 Importance of quality control in Ayurveda · 1.2 Quality Assurance of Ayurveda Products · 1.3 Quality control parameters (identity, purity, etc.) · 1.4 Regulatory requirements for quality control · 1.5 Documentation and record-keeping

Table 6F. ABQ Yes · SAQ No · LAQ Yes. **Textbook.** Volume 0, Module 7, Unit 1, page 368.

ABQ 7.1.1 — The Churna that Left Before Its Report

Anchor: M7U1 (1.1 to 1.5) · Also draws on: M7U3 · Suggested time: 35 minutes

THE CASE

During the festival rush, a licensed pharmacy runs short of Sitopaladi Churna. The following happens.

F1 The production head dispatches a new batch to distributors before the QC report is ready. He says, “We have made this for twenty years.”

F2 The QC report arrives five days later: total bacterial count 4×10^6 cfu/g, total yeast and mould 2×10^4 cfu/g, and *E. coli* present.

F3 The sugar (Sita) was bought from an open market and stored in jute bags in a damp corner of the store.

F4 Patients complain of lumps and a sour smell.

F5 No retained (control) sample of the batch was kept.

F6 The pharmacy has no written procedure for complaints, deviations or recalls.

Q	Question	Marks	Domain	Level
(a)	Explain the importance of quality control in Ayurveda, with reference to this case.	4	CC	KH
(b)	Which elements of a quality assurance system were missing? Link each to a finding.	6	CAN	KH
(c)	List the QC parameters (identity, purity, strength and safety) that should be tested for this Churna.	5	CAP	KH
(d)	Which regulatory requirements apply, and what documents should the pharmacy have had?	5	CE	KH

Model answer framework

(a) Importance of QC — 4 marks

- Ayurvedic medicines use natural raw materials that vary with source, season and storage, and are easily contaminated. QC is the only way to know that each batch is right.
- QC protects the patient (here, from a microbially contaminated product given to children and adults with cough), the practitioner's trust, and the reputation of Ayurveda.
- It is a legal requirement: products must meet pharmacopoeial standards.
- Tradition ("twenty years") is not evidence that *this* batch is safe.

(b) Missing QA elements — 6 marks

QA element	What it requires	Finding
Release by QA only	No batch leaves until QC tests pass and QA signs release	F1
Supplier qualification and raw material testing	Approved suppliers; each lot tested before use	F3
Storage controls	Dry, clean, pest-proof store; raised platforms; proper containers	F3, F4
Retained samples	Keep a sample of each batch for investigations	F5
Complaint handling	Record, investigate and respond to every complaint	F4, F6
Deviation and CAPA	Record departures from procedure; find root cause; prevent recurrence	F1, F6
Recall procedure	Ability to trace and withdraw a batch quickly	F6

(c) QC parameters for Sitopaladi Churna — 5 marks

Category	Parameters
Identity	Organoleptic (colour, odour, taste); microscopy of ingredients (Vamshalochana, Pippali, Ela, Twak); TLC or HPTLC fingerprint
Purity	Loss on drying; total ash; acid-insoluble ash; water- and alcohol-soluble extractives; foreign matter; fineness (sieve test)
Strength	Total sugar content (Sita is the major ingredient); marker assay where specified (for example piperine from Pippali)
Safety	Microbial limits — total bacterial count, total yeast and mould, absence of *E. coli*, *Salmonella*, *S. aureus* and *P. aeruginosa* (API Appendix 2.4); aflatoxins; heavy metals; pesticide residues

This batch fails microbial limits on all three counts: API limits are total bacterial count 10^5/g, total yeast and mould 10^3/g, and *E. coli* absent.

(d) Regulatory requirements and documents — 5 marks

- *Drugs Rules, 1945 — Schedule T (GMP for ASU drugs):* premises and storage, raw material control, quality control facility, batch records, release procedure, complaint and recall records. Schedule T was substantially revised in July 2026 (G.S.R. 657(E)), with a risk-based quality system; existing units must comply by 31 July 2029.
- *Ayurvedic Pharmacopoeia of India* standards for the formulation and its ingredients.
- *Rule 161:* labelling with batch number, dates and licence number — makes recall possible.
- *Documents the pharmacy should have had:* approved supplier list; raw material specifications and test reports; Batch Manufacturing Record; QC test report and Certificate of Analysis; QA release record; retained sample register; distribution record; complaint, deviation, CAPA and recall SOPs and registers.
- *Now:* recall the batch, inform the licensing authority, investigate, and record CAPA.

WHERE MARKS ARE LOST

- Writing only “QC is important” without tying it to the findings.
- Listing QC tests without saying the batch fails the microbial limits.
- Forgetting recall.

Revise: Volume 0 — Module 7, Unit 1 (p. 368) and Unit 3 (p. 385).

ABQ 7.1.2 — A Teaching Pharmacy Seeks a Licence

Anchor: M7U1 (1.2, 1.3, 1.4, 1.5) · Suggested time: 35 minutes

THE CASE

An Ayurveda college wants its teaching pharmacy licensed to manufacture 30 classical medicines: Churna, Vati, Avaleha, Asava-Arishta, Taila, Ghrita and four Bhasmas. The committee's draft plan says the following.

F1 Raw herbs will be bought from local herb sellers, whoever offers the lowest price.

F2 There will be no in-house QC laboratory. All testing will be "outsourced when needed".

F3 The pharmacy superintendent will also be in charge of QC and of release.

F4 Records will be kept in a general register.

F5 There is no plan for stability studies. Expiry dates will be "as per tradition".

Q	Question	Marks	Domain	Level
(a)	Design the quality assurance framework this pharmacy needs. Point out the weaknesses of the draft plan.	6	CS	KH
(b)	List the QC parameters for incoming raw herbs, under identity, purity and safety.	5	CAP	KH
(c)	Explain the regulatory requirements for quality control that apply to this pharmacy.	4	CC	KH
(d)	List the documents and records the pharmacy must maintain.	5	CS	KH

Model answer framework

(a) QA framework — 6 marks

Element	What to set up	Weakness in draft
Quality policy and manual	Written commitment; organisation chart	—
Independence of QC/QA	QC and release by a person independent of production	F3
Supplier qualification	Approved suppliers selected on quality, with audits; lowest price is not a criterion by itself	F1
Specifications	For every raw material, intermediate and finished product (from API, AFI)	F1, F2
Testing capability	In-house basic QC, plus a Government-approved laboratory for advanced tests	F2
SOPs and training	For every process and test	—
Batch documentation	BMR for each batch; QA review	F4
Change control, deviations, CAPA	Written procedures	—
Stability programme	Data to support shelf life	F5
Self-inspection	Periodic internal audits	—

(b) Raw herb QC — 5 marks

Category	Parameters
Identity	Macroscopy; microscopy (powder and section); TLC or HPTLC fingerprint against a reference; botanical authentication by a qualified person
Purity	Foreign matter; loss on drying; total ash; acid-insoluble ash; water- and alcohol-soluble extractives; volatile oil where relevant
Strength	Marker assay where the monograph specifies
Safety	Heavy metals (lead 10, arsenic 3, cadmium 0.3, mercury 1 ppm); microbial limits; aflatoxins; pesticide residues

(c) Regulatory requirements — 4 marks

- Licence under the Drugs and Cosmetics Act, 1940 and the Drugs Rules, 1945, granted by the State Licensing Authority. Licensing of ASU medicines follows Rule 158-B.
- GMP as per Schedule T, which requires a quality control facility. Testing may be done in the unit's own laboratory or through a Government-approved drug testing laboratory (approved under Rules 160-A to 160-J).
- Products must meet the standards of the Ayurvedic Pharmacopoeia of India.
- Labelling under Rule 161; expiry under Rule 161-B. Under the 2026 amendment (G.S.R. 657(E)), shelf life must be supported by scientific data — "as per tradition" (F5) is not acceptable.
- The revised Schedule T (2026) applies, with compliance due by 31 July 2029 for existing units.

(d) Documents and records — 5 marks

Site master file; organisation chart and job descriptions; SOPs; specifications; approved supplier list; raw material receipt and test records; Batch Manufacturing Records; QC test records, raw data and Certificates of Analysis; release records; equipment logbooks, cleaning and calibration records; stability reports; retained sample register; distribution records; complaint, deviation, change control, CAPA and recall records; training records; self-inspection reports.

WHERE MARKS ARE LOST

- Writing a list of tests only, with no QA system.
- Allowing the superintendent to control production, QC and release.
- Missing the 2026 changes on shelf life and Schedule T.

Revise: Volume 0 — Module 7, Unit 1 (p. 368); Module 1, Unit 4 (p. 56).

ABQ 7.1.3 — The Export Complaint and the Pencil Entries

Anchor: M7U1 (1.2, 1.4, 1.5) · Also draws on: M5U4 · Suggested time: 35 minutes

THE CASE

An importer complains that a consignment of Kaishora Guggulu tablets contains lead at 18 ppm. The importer asks for the Certificate of Analysis (CoA), the Batch Manufacturing Record (BMR) and the raw material test records. The company's investigation finds the following.

F1 Three months earlier, the purchase department changed the Guggulu supplier to save cost. No one in QA was informed.

F2 The new lot of Guggulu resin was not tested for heavy metals before use.

F3 Two pages of the BMR are missing. Several entries are in pencil; some are overwritten.

F4 The CoA was signed on the day the tablets were packed — two days before the heavy metal test was completed.

F5 The raw data printout of the AAS run cannot be found.

Q	Question	Marks	Domain	Level
(a)	Which quality assurance concepts were violated? Explain each with reference to the findings.	6	CAN	KH
(b)	Explain the principles of good documentation practice (ALCOA+). Show how each finding breaches them.	5	CC	KH
(c)	Outline the root-cause investigation and the CAPA.	5	CS	KH
(d)	What are the regulatory implications for the company in India and for export?	4	CE	KH

Model answer framework

(a) QA concepts violated — 6 marks

Concept	What it means	Finding
Change control	Any change in supplier, material, process or equipment is assessed and approved by QA before it is made	F1
Supplier qualification	New suppliers are audited and their material tested before approval	F1, F2
Raw material release	No material is used until QC tests pass	F2
Batch record integrity	The BMR is complete, original and reviewed	F3
Product release	QA releases only after all results are available and reviewed	F4
Data integrity	Raw data are kept and traceable	F5

(b) ALCOA+ — 5 marks

Principle	Meaning	Breach
Attributable	Who did it, and when, is recorded	Unsigned, overwritten entries (F3)
Legible	Readable and permanent	Pencil entries (F3)
Contemporaneous	Recorded at the time of the activity	CoA signed before the test (F4)
Original	The first record, or a certified copy	Missing AAS printout (F5)
Accurate	True and correct; corrections made by single line, initialled and dated, with reason	Overwriting (F3)
“+” Complete, Consistent, Enduring, Available	Nothing missing; consistent in time; lasts; can be retrieved	Missing BMR pages (F3); missing raw data (F5)

(c) Investigation and CAPA — 5 marks

- Quarantine remaining stock; test retained samples of this and nearby batches; test the remaining Guggulu lot.
- Root cause analysis (for example “five whys” or a fishbone diagram): likely root causes — no change-control system; purchase acting outside QA; release before results; weak documentation culture.
- *Corrective actions:* recall affected batches; reject the supplier lot; reissue a correct CoA only on verified data.
- *Preventive actions:* written change-control SOP; approved supplier list controlled by QA; raw material release before use; QA release checklist; GDP training; periodic data-integrity audits; secure storage of instrument raw data.

(d) Regulatory implications — 4 marks

- The batch fails the API heavy metal limit for lead (10 ppm).
- Signing a CoA before results exist is a serious GMP breach under Schedule T, and may be treated as falsification. The licensing authority may suspend or cancel the licence.
- The importer may reject future consignments and alert its national regulator. Export certification (for example WHO-GMP certification and the Certificate of Pharmaceutical Product, CoPP) depends on documented GMP compliance and may be withdrawn.

WHERE MARKS ARE LOST

- Talking only about lead testing and missing the documentation failures.
- Not naming change control.
- A CAPA with only corrective actions.

Revise: Volume 0 — Module 7, Unit 1 (p. 368); Module 5, Unit 4 (p. 283).

Unit 3 — Quality Control Parameter for Solid Dosage Forms

Syllabus points. 3.1 Churn, kwatha churna, lepa · 3.2 Kshara, Lavana, Satva · 3.3 Tablets, pills, and Varti · 3.4 Khalveeya, Parpati, and Kupipakwa Rasayana · 3.5 Lauha/Mandura

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 7, Unit 3, page 385.

ABQ 7.3.1 — Gritty Churna, Cloudy Kwatha, Stinging Lepa

Anchor: M7U3 (3.1) · Also draws on: M7U1 · Suggested time: 35 minutes

THE CASE

A hospital pharmacy receives three powders from the same supplier.

F1 *Triphala Churna* feels gritty between the teeth.

F2 Its total ash and acid-insoluble ash are both well above the monograph limits.

F3 Its loss on drying is high, and the powder has caked in the container. The microbial count exceeds the API limit.

F4 A large part of the Triphala Churna does not pass through the sieve specified for Churna.

F5 *Dashamula Kwatha Churna* is supplied as a fine powder. When decocted, it gives a muddy decoction that clogs the filter cloth.

F6 A *Lepa Churna* for acne causes stinging on application. Its aqueous suspension is strongly alkaline, and coarse particles are seen under the microscope.

Q	Question	Marks	Domain	Level
(a)	Tabulate the QC parameters for a Churna and the significance of each.	6	CAP	KH
(b)	Interpret F1 to F4 and trace each to a fault in manufacture or storage.	5	CAN	KH
(c)	How do QC requirements of a Kwatha Churna and a Lepa Churna differ from those of an oral Churna? Use F5 and F6.	5	CAN	KH
(d)	Suggest corrective measures for the supplier.	4	CS	KH

Model answer framework

(a) QC parameters for Churna — 6 marks

Parameter	Significance
Organoleptic (colour, odour, taste, texture)	First check of identity and freshness
Fineness (sieve test)	Uniform particle size; dose uniformity; mouthfeel
Microscopy	Identity of each ingredient; detects adulterants
Loss on drying	Moisture; risk of caking and microbial growth
Total ash	Inorganic content; earth and sand
Acid-insoluble ash	Silica — sand and soil specifically
Water- and alcohol-soluble extractives	Amount of extractable constituents; detects exhausted or poor material
TLC / HPTLC; assay	Identity; strength
Microbial limits, aflatoxins, heavy metals, pesticides	Safety

(b) Interpretation — 5 marks

Finding	Meaning	Likely fault
F1, F2	Sand and earth present (acid-insoluble ash is silica)	Fruits not washed or sorted; dried on bare ground; stones not removed; grinding stones wearing
F3	Too much moisture; supports microbes; causes caking	Incomplete drying; poor packing; humid store
F4	Coarse powder	Inadequate grinding; sieve damaged or skipped

(c) Kwatha Churna and Lepa Churna — 5 marks

Point	Oral Churna	Kwatha Churna (F5)	Lepa Churna (F6)
Particle size	Fine	Coarse (Yavakuta) — fine powder makes a muddy decoction and blocks filtering	Very fine and smooth — coarse particles irritate the skin
Key extra tests	—	Sieve test for coarseness; extractive values; yield and clarity of the decoction	pH of the suspension; microbial limits (applied to broken skin); absence of irritant particles
F6 cause	—	—	Alkaline ingredient or contamination (for example lime or alkaline ash); poor grinding

(d) Corrective measures — 4 marks

Clean, sort and wash raw materials; dry on clean raised trays or in driers to a set moisture; validate grinding and sieving with the correct sieve for each product type; check sieves for damage; pack in moisture-proof containers with liners; store in a dry, controlled area; test each batch before dispatch; label Kwatha Churna and Lepa Churna with their correct particle size specification.

WHERE MARKS ARE LOST

- Not separating total ash from acid-insoluble ash.
- Treating Kwatha Churna like an oral Churna.
- Missing pH and skin safety for Lepa.

Revise: Volume 0 — Module 7, Unit 3 (p. 385).

ABQ 7.3.2 — The Vati that Would Not Break

Anchor: M7U3 (3.3) · Also draws on: M1U3 · Suggested time: 35 minutes

THE CASE

A QC laboratory tests three solid oral and ophthalmic products.

F1 *Chandraprabha Vati* (machine-made, 500 mg): 4 of 20 tablets deviate by more than 5 percent from the average weight; 1 tablet deviates by more than 10 percent.

F2 Its hardness is very high, and the disintegration time is over 90 minutes.

F3 The tablet surface shows sticky patches. The Guggulu was added as a hot, soft mass during granulation.

F4 *Sanjivani Vati* (hand-rolled pills) varies from 90 mg to 160 mg per pill.

F5 A *Varti* for eye application (rubbed with water and applied as Anjana) shows hard, coarse particles when rubbed, and a high microbial count.

Q	Question	Marks	Domain	Level
(a)	Describe the QC tests for tablets and pills, with the usual limits.	6	CAP	KH
(b)	Interpret F1 to F4 and suggest causes.	5	CAN	KH
(c)	Describe the QC parameters for a Varti. Interpret F5.	4	CC	KH
(d)	Suggest process remedies for the Vati and the pills.	5	CS	KH

Model answer framework

(a) QC tests for tablets and pills — 6 marks

Test	Purpose	Usual requirement
Uniformity of weight	Dose uniformity	IP limits for uncoated tablets: up to 80 mg ±10%; 80 to 250 mg ±7.5%; above 250 mg ±5%. Not more than 2 of 20 may deviate by the stated percentage, and none by double that percentage
Hardness	Strength for handling	Within the range set in the specification
Friability	Resistance to chipping	Loss usually not more than 1%
Disintegration	Breaks up in time for release	As stated in the API monograph for the Vati (IP uncoated tablets: 15 minutes)
Dissolution	Release of marker	Where a method is specified
Identity and assay	TLC/HPTLC; marker content	As per monograph
Physicochemical	Loss on drying, ash, acid-insoluble ash, extractives	As per monograph
Safety	Microbial limits, heavy metals, aflatoxins	API limits

(b) Interpretation — 5 marks

- F1: for a 500 mg tablet, limit is ±5%; 4 tablets outside (only 2 allowed) and 1 beyond ±10% (none allowed). *Fails.* Causes: poor flow of sticky granules; uneven die fill.
- F2: over-hard tablets with slow disintegration. Causes: high compression force; Guggulu acting as a strong binder; no disintegrant.
- F3: hot Guggulu mass did not mix evenly; it formed sticky lumps — causes weight variation and sticking.
- F4: hand-rolled pills vary widely (90 to 160 mg). Causes: no weighing or measuring of the mass per pill; variable skill.

(c) Varti — 4 marks

- Varti is a solid elongated dosage form, used here for the eye. QC: uniform shape and weight; fineness (must rub smoothly, with no gritty particles); loss on drying; identity tests; microbial limits — for eye use, as low as achievable, with absence of pathogens; heavy metals where minerals are used.
- F5: coarse particles can scratch the cornea; a high microbial count risks eye infection. The batch must be rejected. Grinding must be finer, and production and packing must be hygienic.

(d) Remedies — 5 marks

- *Vati:* purify and soften Guggulu by the classical method, then mix it uniformly as a paste or dried powder; granulate to a set size; dry to a set moisture; add a disintegrant if the formula allows; set compression force to give target hardness; in-process checks of weight and hardness every set interval.

- *Pills:* use a pill-cutting or pill-rolling machine, or weigh the mass for each batch of pills; check weight of 20 pills at intervals.

WHERE MARKS ARE LOST

- Quoting weight limits without applying them to the data.
- Ignoring how Guggulu was added.
- Treating Varti like an oral tablet, with no eye-safety tests.

Revise: Volume 0 — Module 7, Unit 3 (p. 385); Module 1, Unit 3 (p. 44).

ABQ 7.3.3 — Four Rasaushadhi Samples on the QC Bench

Anchor: M7U3 (3.2, 3.4, 3.5) · Also draws on: M7U2, M5U7 · Suggested time: 35 minutes

THE CASE

The QC section of a Rasaushadhi unit receives four samples for release.

F1 *Rasasindura:* under a hand lens, a few tiny shining globules are seen. A test for free mercury is positive.

F2 *Mandura Bhasma:* acid-insoluble ash is 18 percent. A magnet attracts it strongly. Its colour is blackish.

F3 *Rasa Parpati:* the flakes bend rather than break with a snap. They are dull, not glossy black.

F4 *Apamarga Kshara:* grey-white, wet lumps; a 1 percent solution is only mildly alkaline; much water-insoluble matter remains.

F5 *Saindhava Lavana:* black specks are visible; water-insoluble matter is high; sodium chloride content is 88 percent.

Q	Question	Marks	Domain	Level
(a)	List the classical and modern QC parameters for Kupipakwa Rasayana and Parpati.	5	CAP	KH
(b)	Interpret F1, F2 and F3.	6	CAN	KH
(c)	List the QC parameters for Kshara and Lavana, and interpret F4 and F5.	5	CAN	KH
(d)	What does F1 mean for patient safety? What decision should QA take on each sample?	4	CE	KH

Model answer framework

(a) QC for Kupipakwa and Parpati — 5 marks

Product	Classical tests	Modern tests
Kupipakwa (Rasasindura)	Colour (bright red), lustre, crystalline form; no free mercury globules; fineness after grinding	XRD (cinnabar); mercury and sulphur content; test for free mercury; heavy metals other than mercury; particle size
Parpati	Thin flakes; glossy black; breaks with a snap (crisp); no free mercury	XRD (mainly black HgS); mercury and sulphur content; free sulphur; free mercury

(b) Interpretation — 6 marks

- *F1 Rasasindura:* free mercury is present — incomplete reaction or too little sulphur, insufficient heating time, or poor sublimation. The product is not acceptable.
- *F2 Mandura Bhasma:* very high acid-insoluble ash means sand or silica (from dirty raw Mandura or from grinding). Strong magnetic response and black colour mean incomplete oxidation — residual magnetite or iron. Needs further Putas or rejection.
- *F3 Parpati:* failure to snap and lack of gloss mean improper Paka — wrong temperature or time when the molten Kajjali was pressed, or impure ingredients. Structure and composition are not as intended.

(c) Kshara and Lavana — 5 marks

Product	Parameters	Interpretation
Kshara (F4)	Colour (white), texture; pH and total alkalinity; water-soluble matter; moisture; carbonate and chloride content	Grey colour and insoluble matter: incomplete filtration of the ash solution (Kshara jala), so unburnt ash remains. Mild alkalinity: poor yield of alkali salts or dilution. Wet lumps: hygroscopic salts absorbing moisture — poor drying and packing
Lavana (F5)	Colour, crystal form; sodium chloride assay; water-insoluble matter; moisture; heavy metals; other salts	Black specks and insoluble matter: earth and dirt; low NaCl: impure or mixed salt. Purify by dissolving, filtering and recrystallising

(d) Safety and decisions — 4 marks

- Free mercury in Rasasindura is a direct safety risk. Mercury metal can vaporise and be inhaled during grinding and handling, and it shows that the process is not complete.

- *Decisions:* Rasasindura — reject or reprocess, investigate the Kupipaka log. Mandura Bhasma — reject; investigate raw material and reprocess. Parpati — reject; review the Paka procedure. Kshara and Lavana — reject or reprocess (re-filter, re-crystallise), then retest. Record each as a deviation.

WHERE MARKS ARE LOST

- Writing the classical tests only, or the modern tests only.
- Missing that a strong magnetic response means incomplete conversion.
- Releasing any sample “with caution”.

Revise: Volume 0 — Module 7, Unit 3 (p. 385) and Unit 2 (p. 377).

Unit 4 — Quality Control Parameter for Semi-solid Dosage Forms

Syllabus points. 4.1 Rasakriya/Avaleha/Paka · 4.2 Anjana

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 7, Unit 4, page 394. The October 2026 preliminary ABQ on Chyavanaprasha Avaleha belongs to this unit and Unit 1; it is solved in full as Q1 in the chapter “The October 2026 Preliminary Paper — Solved”.

ABQ 7.4.1 — Rasanjana: Too Hard, Too Soft

Anchor: M7U4 (4.1) · Also draws on: M7U2 · Suggested time: 35 minutes

THE CASE

A pharmacy makes Rasanjana (Rasakriya of Daruharidra) by the classical method. It is used both orally and as an eye application. Three batches are tested.

F1 Batch 1 is hard and brittle, with a burnt smell.

F2 Batch 2 is soft and sticky. Its moisture content is 28 percent. Mould appears after six weeks.

F3 Batch 3 looks normal.

F4 HPTLC shows berberine content of 0.5 percent in Batch 1, 1.4 percent in Batch 2 and 2.5 percent in Batch 3.

F5 The pharmacy has no written endpoint test for the Paka. The operator decides by experience.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for a Rasakriya.	5	CAP	KH
(b)	Explain why the three batches differ (F1 to F4).	5	CAN	KH
(c)	Propose a specification and an in-process endpoint for Rasanjana.	6	CS	KH
(d)	How should Rasanjana be packed and stored, given its dual use (oral and eye)?	4	CC	KH

Model answer framework

(a) QC parameters for Rasakriya — 5 marks

Organoleptic (colour, odour, consistency); loss on drying or moisture; total solids; total ash and acid-insoluble ash; water-soluble extractive; pH of an aqueous solution; solubility in water; identity by TLC/ HPTLC; marker assay (berberine for Daruharidra); microbial limits; heavy metals. For eye use, microbial quality must be especially strict.

(b) Why the batches differ — 5 marks

Batch	Explanation
1 (hard, burnt, low berberine)	Over-heating and over-concentration. Charring degraded constituents, so berberine fell
2 (soft, sticky, wet, mouldy)	Paka stopped too early. High moisture lowered the marker per gram (dilution) and allowed mould
3 (normal)	Correct endpoint

The root cause is F5: no defined endpoint, no temperature control, and no in-process testing.

(c) Specification and endpoint — 6 marks

In-process endpoint (all recorded in the BMR): consistency — the mass draws into a thread between fingers and does not stick; a drop does not spread on a plate; moisture by a rapid moisture balance within a set range; temperature of the mass recorded; heating on a controlled source with continuous stirring.

Finished-product specification (limits set from good batches):

Test	Limit
Description	Dark brown-yellow, solid mass, bitter
Loss on drying	Not more than a set value
pH (aqueous solution)	Within a set range
Water-soluble matter	Not less than a set value
Berberine (HPTLC)	Within a set range
Microbial limits	API limits; stricter for ophthalmic use
Heavy metals	API limits

(d) Packing and storage — 4 marks

Airtight, light-resistant containers (amber glass or suitable plastic); small unit packs to limit repeated opening; separate packs and labels for oral use and for eye use; store in a cool, dry place; label with batch, dates and directions. For eye use, a hygienically produced, sealed pack with a short in-use period.

WHERE MARKS ARE LOST

- Listing tests without explaining why berberine falls in both bad batches.
- No defined endpoint in (c).
- Ignoring the eye use.

Revise: Volume 0 — Module 7, Unit 4 (p. 394) and Unit 2 (p. 377).

ABQ 7.4.2 — The Surma Camp

Anchor: M7U4 (4.2) · Also draws on: M5U7, M7U1 · Suggested time: 35 minutes

THE CASE

A voluntary group distributes free “Ayurvedic Surma (Anjana)” at a child health camp. Mothers are told to apply it daily to their infants’ eyes. The district drug inspector draws samples.

F1 XRD shows galena (lead sulphide, PbS) as the main phase.

F2 XRF shows lead at about 60 percent by weight.

F3 The powder feels gritty. Microscopy shows particles of up to 100 µm.

F4 The total microbial count is high.

F5 The label has no manufacturer, no licence number and no batch number.

F6 The group says it is “Sauviranjana as described in the classics”.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for an Anjana, with the reason for each.	6	CAP	KH
(b)	Interpret F1 to F4. Comment on the claim in F6.	4	CAN	KH
(c)	Evaluate the safety of this product for infants.	5	CE	KH
(d)	What regulatory and corrective action should follow?	5	CS	KH

Model answer framework

(a) QC parameters for Anjana — 6 marks

Parameter	Reason
Identity of mineral (XRD; XRF or ICP)	Confirms the correct mineral; detects substitution
Fineness / particle size (microscopy, sieve)	Must be extremely fine and smooth — coarse particles injure the cornea
Microbial quality (sterility or very low counts, absence of pathogens)	Eye infection risk
Heavy metals (especially lead)	Toxic absorption through the eye and by hand-to-mouth in children
pH of an aqueous suspension	Irritation
Moisture	Stability and microbial growth
Packaging and labelling	Hygiene; traceability

(b) Interpretation — 4 marks

- F1, F2: the product is galena, a lead mineral. 60 percent lead fits nearly pure PbS (lead is about 87 percent of PbS), diluted with other matter.
- F3: gritty, coarse — unsafe for the eye.
- F4: contaminated.
- F6: classical Anjana minerals such as Sauviranjana and Srotonjana have been identified by different authors with antimony sulphide (stibnite) or with lead sulphide (galena). The identification is debated. Whatever the classical identity, a lead-rich product applied daily to infants is not acceptable.

(c) Safety for infants — 5 marks

- Lead-based kohl and surma are a recognised cause of raised blood lead in children. Lead is absorbed from the eye surface and, much more, from hand-to-mouth contact as infants rub their eyes.
- Young children absorb lead more readily than adults. Lead harms the developing brain; there is no known safe blood lead level for children.
- Gritty particles damage the cornea; contamination causes infection.
- Conclusion: unsafe. Use must stop at once.

(d) Regulatory and corrective action — 5 marks

- Seize stock; stop distribution; issue a public warning through the health department.
- The product carries no manufacturer or licence details — a misbranded and possibly spurious drug under Chapter IVA of the Drugs and Cosmetics Act, 1940 (Sections 33E, 33EEA). Trace and prosecute the source.
- Screen exposed children for blood lead and refer for medical care.

- Health education for mothers: no surma or kajal for infants unless it is a licensed, tested product meant for that use.
- Report through the Ayush pharmacovigilance network.

WHERE MARKS ARE LOST

- Writing “Anjana is a classical dosage form, so it is safe”.
- Not calculating or commenting on the lead content.
- No public health action.

Revise: Volume 0 — Module 7, Unit 4 (p. 394); Module 5, Unit 7 (p. 305).

ABQ 7.4.3 — The Grainy Kushmanda Avaleha and the Three-Year Label

Anchor: M7U4 (4.1) · Also draws on: M7U1, M7U2 · Suggested time: 35 minutes

THE CASE

A company makes Kushmanda Avaleha. Two batches made in the same month behave differently.

F1 *Batch 1:* after two months, sugar crystals form throughout the jar; the texture becomes grainy.

F2 *Batch 2:* froth appears on the surface, with an alcoholic smell.

F3 Test results — Batch 1: total sugar 72%, reducing sugar 12%. Batch 2: total sugar 58%, reducing sugar 45%, loss on drying 26%.

F4 In Batch 2, ghee has separated at the top. The peroxide value of the extracted fat rises over three months.

F5 The company wants to print a shelf life of three years on the label, based on a three-month accelerated stability study.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for an Avaleha, with significance.	5	CAP	KH
(b)	Explain the chemistry of F1 to F3.	6	CAN	KH
(c)	Explain F4. How can it be prevented?	4	CAN	KH
(d)	Evaluate the proposal in F5 against the current rules on shelf life.	5	CE	KH

Model answer framework

(a) QC parameters for Avaleha — 5 marks

Parameter	Significance
Organoleptic; consistency	Signs of Paka quality and spoilage
Loss on drying	Water content — key to microbial stability
Total, reducing and non-reducing sugar	Degree of inversion; risk of crystallisation or fermentation
pH	Stability; spoilage
Fat content; acid value; peroxide value (of the ghee or oil)	Rancidity
Total ash; acid-insoluble ash	Inorganic and sand content
TLC / HPTLC; marker	Identity
Microbial limits; aflatoxins; heavy metals	Safety

(b) Chemistry of F1 to F3 — 6 marks

- During Paka, heat and plant acids split part of the sucrose into glucose and fructose (*inversion*). Glucose and fructose are reducing sugars.
- *Batch 1:* little inversion (reducing sugar only 12% of 72% total). The concentrated sucrose solution is supersaturated on cooling, so sucrose crystallises — graininess (F1). Inverted sugars help to prevent crystallisation.
- *Batch 2:* high moisture (LOD 26%) and lower total sugar mean higher water activity. Yeasts can grow, fermenting sugars to alcohol and carbon dioxide — froth and alcoholic smell (F2). Very high reducing sugar here may partly reflect over-inversion and microbial action.
- Both show a poorly controlled Paka endpoint (Batch 1 too concentrated with too little inversion; Batch 2 under-cooked).

(c) Fat separation and rancidity — 4 marks

- Under-cooked, wet Avaleha does not hold the ghee in a uniform mass, so fat separates (F4). Exposed fat oxidises; peroxide value rises (rancidity).
- Prevention: correct Paka endpoint; add ghee at the right stage with thorough mixing; fill hot into clean, dry, airtight containers with little headspace; protect from light and heat; set acid and peroxide value limits.

(d) The three-year label — 5 marks

- Rule 161-B of the Drugs Rules, 1945 sets the shelf life of ASU medicines; for Avaleha the notified period is three years.

- The 2026 amendment (G.S.R. 657(E)) requires shelf life to be supported by scientific data. As reported, shelf life based on accelerated studies is limited to one year for a three-month study (two years for six months), and a longer shelf life must rest on real-time data. Check the Gazette text.
- So a three-year label on a three-month accelerated study is not acceptable. The company may label a shorter shelf life now and extend it as real-time data accumulate.
- In any case, these batches failed basic QC and must not be released.

WHERE MARKS ARE LOST

- Explaining graininess without the word “inversion”.
- Missing that high moisture drives fermentation.
- Accepting the three-year label without data.

Revise: Volume 0 — Module 7, Unit 4 (p. 394) and Unit 2 (p. 377).

Unit 5 — Quality Control Parameter for Liquid and Gaseous Dosage Forms

Syllabus points. 5.1 Kashaya · 5.2 Arka · 5.3 Sharkara · 5.4 Asava-arishta · 5.5 Netra Bindu/Karna Bindu · 5.6 Aerosols/sprays

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 7, Unit 5, page 399.

ABQ 7.5.1 — Two Dashamularishta Batches

Anchor: M7U5 (5.4) · Also draws on: M7U2, M5U3 · Suggested time: 35 minutes

THE CASE

A pharmacy's QC section tests two batches of Dashamularishta.

F1 *Batch X:* alcohol 2 percent v/v; pH 3.1; strong vinegar smell; a thin film on the surface.

F2 Batch X was fermented in a vessel whose lid had been left loosely closed.

F3 *Batch Y:* alcohol 16 percent v/v. The fermentation log records no gas formation, and the batch was "ready" in two days.

F4 The Dhataki flowers for Batch Y were not received; the batch record shows a blank against them.

F5 Neither batch has records of temperature or of specific gravity during fermentation.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for Asava-Arishta, with significance.	5	CAP	KH
(b)	Interpret the findings for Batch X and Batch Y.	6	CAN	KH
(c)	Describe the in-process controls that should be applied during Sandhana.	5	CS	KH
(d)	What is the quality and regulatory problem with Batch Y?	4	CE	KH

Model answer framework

(a) QC parameters — 5 marks

Parameter	Significance
Organoleptic (colour, odour, taste, clarity)	Spoilage; identity
Alcohol content (by distillation and specific gravity, or GC)	Confirms fermentation; acts as preservative; must be within the monograph range
Test for methanol	Should be at most traces; high levels mean adulteration
pH; total acidity	Acetic spoilage
Specific gravity; total solids	Consistency between batches
Reducing and non-reducing sugars	Extent of fermentation
TLC / HPTLC	Identity of herbal constituents
Microbial limits	Safety

(b) Interpretation — 6 marks

- *Batch X:* fermentation produced alcohol, but air entered through the loose lid (F2). Acetic acid bacteria oxidised ethanol to acetic acid — low alcohol, low pH, vinegar smell, surface film (F1). This is *acetification*: the batch is spoiled. Reaction: $C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$.
- *Batch Y:* natural fermentation in two days without gas is not possible. Yeast fermentation releases carbon dioxide continuously over days to weeks. Dhataki, the classical source of fermenting organisms, was not added (F4). An alcohol of 16% that appears without fermentation means alcohol was added from outside.

(c) In-process controls — 5 marks

- Clean, suitable fermentation vessels; tight closure with a gas outlet that keeps air out.
- Correct proportions of drugs, jaggery or sugar, and Dhataki; record weights.
- Temperature monitoring (fermentation is slow in cold and spoils in excess heat).
- Daily observation of gas release; specific gravity or Brix at set intervals; endpoint when gas stops, the flame test is negative, and sugar falls to a set level.
- Alcohol and pH checked at the end; filtration and maturation; record everything in the BMR.

(d) The problem with Batch Y — 4 marks

- A classical Arishta is defined by fermentation; its alcohol must be self-generated. Adding alcohol means it is not made as per the formula in the authoritative book. It is non-conforming — treated as adulterated or misbranded under Chapter IVA of the Drugs and Cosmetics Act, 1940.
- It also misleads the patient: high alcohol content, and none of the products of fermentation that are part of the medicine.

• Reject the batch; investigate; CAPA.

WHERE MARKS ARE LOST

• Calling Batch X “under-fermented” — it fermented and then spoiled.

• Missing the missing Dhataki.

• Not stating that added alcohol is not permitted for a classical Arishta.

Revise: Volume 0 — Module 7, Unit 5 (p. 399) and Unit 2 (p. 377).

ABQ 7.5.2 — Eye Drops from the College Pharmacy

Anchor: M7U5 (5.5) · Also draws on: M1U3 · Suggested time: 35 minutes

THE CASE

The college pharmacy prepares a Triphala-based Netra Bindu (eye drop) for the Shalakya OPD, and a Karna Bindu (medicated oil ear drop).

F1 The Netra Bindu is made by boiling, then filtering through cloth. It is filled into plastic dropper bottles on an open bench. It is not sterilised and contains no preservative.

F2 Its pH is 5.2. Fine particles are visible when a bottle is held against light.

F3 Patients use one bottle for about two months. Several report redness, discharge and stinging.

F4 The Karna Bindu has a rancid smell after four months.

F5 Neither product has a QC specification.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for a Netra Bindu, with the reason for each.	6	CAP	KH
(b)	Interpret F1 to F3.	4	CAN	KH
(c)	Design a corrected process and pack for the Netra Bindu.	6	CS	KH
(d)	How do the QC parameters of a Karna Bindu (oil) differ? Interpret F4.	4	CC	KH

Model answer framework

(a) QC parameters for Netra Bindu — 6 marks

Parameter	Reason
Sterility	The eye has weak defences against infection
Clarity; particulate matter	Particles irritate and scratch
pH (close to tear pH, about 7.4; tolerated over a range)	Comfort; stability
Tonicity	Avoid stinging and tissue damage
Viscosity	Contact time; drop size
Preservative content and preservative efficacy (for multi-dose packs)	Protects after opening
Identity and marker	TLC; marker assay
Fill volume; container integrity	Dose; leaks

(b) Interpretation — 4 marks

- F1: no sterilisation, open filling, no preservative — product is not sterile at the start, and becomes contaminated after opening.
- F2: acidic pH and visible particles cause stinging and irritation.
- F3: using a multi-dose, unpreserved bottle for two months invites infection — the symptoms fit contamination and irritation.

(c) Corrected process and pack — 6 marks

1. Prepare the decoction; clarify by settling and filtration through progressively finer filters.
2. Adjust pH with a suitable buffer and tonicity with an agent such as sodium chloride.
3. Add a preservative suited to eye use (permitted for the product), unless packed as single-dose units.
4. Sterilise: filter through a 0.22 µm membrane into sterile containers in an aseptic area (or terminally sterilise if constituents are stable to heat).
5. Fill into sterile, tamper-evident dropper bottles; or single-dose units.
6. Test each batch for sterility, pH, clarity and fill volume.
7. Label: “Sterile until opened; discard a set period after opening” — commonly about four weeks for multi-dose eye drops — and storage directions.

(d) Karna Bindu — 4 marks

- An oil-based product. QC: acid value, peroxide value, saponification value, iodine value, refractive index, specific gravity, viscosity; identity by TLC of the oil; microbial limits; moisture (water promotes rancidity and microbes).

- F4: oxidation of unsaturated fatty acids — rancidity, shown by a rising peroxide and acid value. Prevent with a proper Sneha Paka endpoint (no residual water), airtight, light-resistant packs, cool storage, and an antioxidant if permitted.

WHERE MARKS ARE LOST

- Forgetting sterility — the single most important test here.
- No in-use (after opening) period.
- Applying Netra Bindu tests to an oil.

Revise: Volume 0 — Module 7, Unit 5 (p. 399).

ABQ 7.5.3 — Kashaya, Arka and Sharkara on One Shelf

Anchor: M7U5 (5.1, 5.2, 5.3) · Also draws on: M5U3 · Suggested time: 35 minutes

THE CASE

A hospital store holds three liquid products from one supplier.

F1 *Ready-to-use Kashaya* (bottled decoction): within a month, opened bottles show a cloudy layer and a sour smell. Sediment forms in unopened bottles.

F2 *Yavani Arka* (Ajwain distillate): has hardly any Ajwain smell. No oil droplets are visible. GC shows very little thymol. The bottles are thin plastic; the Arka smells faintly of plastic.

F3 *Sharkara* (a medicinal syrup): crystals form at the bottom; the Brix value is lower than the specification in some bottles and higher in others.

F4 The supplier provides one CoA for all three products, with only “complies” written against each test.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for a Kashaya and interpret F1.	5	CAP	KH
(b)	List the QC parameters for an Arka and interpret F2.	6	CAN	KH
(c)	List the QC parameters for a Sharkara and interpret F3.	5	CAN	KH
(d)	Comment on the CoA in F4. What should a proper CoA contain?	4	CE	KH

Model answer framework

(a) Kashaya — 5 marks

- *Parameters:* organoleptic; pH; specific gravity; total solids; TLC/HPTLC and marker; microbial limits; preservative content (for bottled forms); fill volume; sediment.
- *F1:* a decoction is a rich medium for microbes. A bottled ready-to-use Kashaya needs a validated preservation method (preservative, pasteurisation or hot filling in sealed containers) and a short in-use period after opening. Sediment shows unstable colloids — clarify before filling, or label "shake well" if it re-disperses.

(b) Arka — 6 marks

- *Parameters:* organoleptic (characteristic odour); clarity; pH; specific gravity; volatile oil content or presence; GC fingerprint and marker (thymol for Yavani Arka); residue on evaporation; microbial limits.
- *F2:* weak odour, no oil droplets and low thymol mean poor distillation (too much water, too little drug, or loss of volatiles), or dilution with water. The plastic smell suggests the volatile oil interacts with thin plastic: oil constituents can be absorbed into or pass through the plastic, while plastic additives may leach into the Arka. Use glass or a suitable, tested container.

(c) Sharkara — 5 marks

- *Parameters:* organoleptic; Brix or total solids (sugar concentration); specific gravity; viscosity; pH; total and reducing sugars; identity of drug component; microbial limits; fill volume.
- *F3:* crystallisation at the bottom means supersaturation of sucrose, with too little inversion. Varying Brix between bottles means the syrup was not mixed uniformly, or evaporation and filling were not controlled. Low-Brix bottles are at risk of fermentation.

(d) The CoA — 4 marks

- "Complies" without numbers is not acceptable, and one CoA cannot cover three products.
- A proper CoA is product- and batch-specific: product name, batch number, dates of manufacture and expiry, specification reference, each test with its method, limit and *actual result*, conclusion, date, and signature of the authorised QC person.
- The hospital should ask for proper CoAs and test samples itself or through an approved laboratory.

WHERE MARKS ARE LOST

- One common list of tests for all three liquids.
- Missing the container issue for Arka.
- Accepting "complies".

Revise: Volume 0 — Module 7, Unit 5 (p. 399).

ABQ 7.5.4 — The Nasal Spray that Squirted

Anchor: M7U5 (5.6) · Also draws on: M1U3 · Suggested time: 35 minutes

THE CASE

A company converts Anu Taila into a metered-dose nasal spray (label: 100 µl per actuation). Pilot testing shows the following.

F1 Delivered volume per actuation varies from 50 to 140 µl.

F2 Instead of a fine mist, the spray comes out as a narrow stream with large drops.

F3 The nozzle clogs after a few days of use. Some bottles leak at the pump.

F4 The oil is thick in winter; the spray works better in summer.

F5 The development team also wants to make an “inhaler” from the same oil for asthma patients.

Q	Question	Marks	Domain	Level
(a)	List the QC parameters for nasal sprays and aerosols, with the reason for each.	6	CAP	KH
(b)	Analyse the causes of F1 to F4.	5	CAN	KH
(c)	Suggest formulation and device changes to correct them.	5	CS	KH
(d)	Explain the aerodynamic basis of where droplets deposit. Comment on the idea in F5.	4	CE	KH

Model answer framework

(a) QC parameters — 6 marks

Parameter	Reason
Delivered (shot) volume or weight; dose uniformity across the pack	Each spray gives the labelled dose
Number of actuations per container	Label claim
Priming and re-priming	Dose after first use or after storage
Spray pattern and plume geometry	Coverage of the nasal surface
Droplet size distribution (laser diffraction)	Decides site of deposition
Viscosity	Affects atomisation
Leak test; pump and valve function	Pack integrity
Microbial limits; preservative	Safety in repeated use
Identity and assay; physicochemical tests of the oil	Quality of the medicine

(b) Causes — 5 marks

- F1, F2: high viscosity of oil prevents proper atomisation; the pump is designed for watery liquids; poor priming.
- F3: thick oil and its oxidation products deposit in the fine nozzle; the pump seal is not compatible with the oil.
- F4: viscosity of oil changes greatly with temperature, so spray performance changes with season.

(c) Corrections — 5 marks

- Select a pump and actuator designed for viscous or oily liquids; check seal compatibility with the oil.
- Consider a lower-viscosity form: an oil-in-water nanoemulsion or microemulsion of Anu Taila (with suitable surfactants) — then test for stability and nasal tolerance.
- Add an antioxidant if permitted, to limit gumming.
- Specify viscosity limits; test spray performance at low and high temperatures.
- Validate dose uniformity across the life of the pack.

(d) Aerodynamics and F5 — 4 marks

- Deposition depends on aerodynamic droplet size. Large droplets (above about 10 µm) deposit mainly in the nose by impaction. Small droplets (about 1 to 5 µm) can pass the nose and reach the lungs.
- A nasal spray is designed to *avoid* lung deposition. An inhaler for asthma needs particles in the respirable range, with a different device.

- F5 is not appropriate: inhaling oil droplets into the lungs can cause lipoid pneumonia. Nasya is a nasal therapy, not a pulmonary one. Any inhaled product would need a separate development and safety programme, and approval as a new product.

WHERE MARKS ARE LOST

- Listing “aerosol tests” with no link to oil viscosity.
- Not giving droplet size ranges.
- Supporting the inhaler idea.

Revise: Volume 0 — Module 7, Unit 5 (p. 399); Module 1, Unit 3 (p. 44).

Module 8 — Nano-science, Nano-technology and its Application in Ayurveda

SEMESTER 2 · AYPG-AB-RB

In the university paper (Table 6F)	
Marks range for this module	5 to 20
Units eligible for ABQ	Unit 2 — Ayurvedic Formulations and Nanotechnology; Unit 4 — Regulatory Aspects and Safety Concerns
Unit eligible for LAQ	Unit 1 (see Part B)
Cases in this module	ABQ 8.2.1 to ABQ 8.4.3

The one skill this module tests. Nearly every nano case contains a claim — “nano”, “100 percent”, “ancient nanomedicine”, “ten times better”. The marks go to the scholar who separates what the data show from what the claim says. Present the evidence fully, and mark the boundary honestly.

Unit 2 — Ayurvedic Formulations and Nanotechnology

Syllabus points. 2.1 Overview of Ayurvedic formulations and their limitations · 2.2 Potential of nanotechnology in Ayurvedic formulations · 2.3 Characterization micro/nanoparticles in Bhasma · 2.4 Characterization micro/nanoparticles in Kupipakwa Rasayana and Pottali · 2.5 Characterization of micro-nano sized particles generated during Bhavana · 2.6 Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method and grasp the fundamental principles underlying nanoparticle synthesis · 2.7 Market Survey of Nano-Based Herbal Formulations

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 8, Unit 2, page 429.

ABQ 8.2.1 — “100% Nano Swarna Bhasma”

Anchor: M8U2 (2.1, 2.2, 2.3, 2.7) · Also draws on: M5U7, M8U4 · Suggested time: 35 minutes

THE CASE

A company advertises its Swarna Bhasma as “100% nano — ancient nanomedicine”. You obtain its technical file.

F1 The brochure shows one TEM image of about 20 particles, 20 to 40 nm in size, with the caption “100% nanoparticles”.

F2 DLS of the Bhasma in water gives a Z-average of 1,800 nm with a polydispersity index (PDI) of 0.9.

F3 SEM shows agglomerates of 1 to 10 µm.

F4 XRD shows the face-centred cubic pattern of metallic gold. Crystallite size by the Scherrer equation is about 35 nm.

F5 The file does not state how the TEM sample was prepared, or how many particles were counted.

Q	Question	Marks	Domain	Level
(a)	State the limitations of conventional Ayurvedic formulations that nanotechnology is said to address, and its potential.	4	CC	KH
(b)	Explain what each technique in F1 to F4 measures in a Bhasma.	5	CAN	KH
(c)	Interpret the data together and evaluate the claim “100% nano”.	6	CE	KH
(d)	Design a sound protocol to characterise the nano fraction of this Bhasma.	5	CS	KH

Model answer framework

(a) Limitations and potential — 4 marks

Limitations of conventional formulations	Potential of nanotechnology (claimed or shown)
Poor solubility and absorption of many plant constituents (for example curcumin)	Higher surface area; faster dissolution; better absorption
Large doses; bulky dosage forms	Lower dose for the same effect (where shown)
Batch variability; poor characterisation	Defined particle size and surface
Instability of some constituents	Protection inside carriers
Limited targeting	Carriers that reach particular tissues

Add: for Bhasma, “nano” is a description of particle size that published studies report in part of the material. It is not proof of benefit.

(b) What each technique measures — 5 marks

Technique	Measures	Caution
TEM (F1)	Size and shape of individual particles in a tiny sample	Very small field; can be selective; needs counting of many particles
DLS (F2)	Hydrodynamic size of particles moving in liquid (intensity-weighted); PDI shows breadth	Dominated by large particles and agglomerates; PDI above about 0.7 means the sample is too broad for reliable DLS
SEM (F3)	Surface shape and size of particles and agglomerates	Dry state; agglomeration visible
XRD (F4)	Phase; crystallite size (Scherrer)	Crystallite size is not particle size

(c) Interpretation and evaluation — 6 marks

- The Bhasma is metallic gold (F4). Its crystallites are nano-sized (about 35 nm).
- TEM (F1) confirms that nano-sized particles *exist*. But about 20 particles, with unknown preparation and selection, cannot represent the whole sample.
- DLS (F2) and SEM (F3) show that, as used, the material is mostly micron-sized agglomerates with a very broad distribution.
- “100% nano” is not supported. Defensible statement: “The Bhasma contains nano-sized primary particles and crystallites, largely present as micron-sized agglomerates.”
- For comparison, the European Commission’s recommended definition of a nanomaterial (2022) requires that 50 percent or more of particles, by number, have at least one external dimension of 1 to 100 nm. The file has no number-based distribution, so even this cannot be judged.
- “Ancient nanomedicine” goes further — it implies that nano-size explains the therapeutic effect. That is not established.

(d) A sound protocol — 5 marks

1. Representative sampling from the batch.
2. Standard dispersion method (medium, dispersant, sonication energy and time), stated in the report.
3. TEM: count several hundred particles from many fields chosen at random; report a number-based size distribution.
4. Laser diffraction (whole size range) and DLS (sub-micron fraction) for the bulk; report D10, D50, D90 and PDI.
5. SEM for agglomeration; XRD for phase and crystallite size; specific surface area (BET) as a supporting measure.
6. Repeat on at least three batches.
7. Report the claim in words that match the data.

WHERE MARKS ARE LOST

- Accepting one TEM image as proof.
- Treating DLS, TEM and XRD sizes as interchangeable.
- Calling the claim “false” without saying what the data *do* support.

Revise: Volume 0 — Module 8, Unit 2 (p. 429); Module 5, Unit 7 (p. 305).

ABQ 8.2.2 — How Small Does Trituration Make It?

Anchor: M8U2 (2.4, 2.5) · Also draws on: M3U1, M5U5 · Suggested time: 35 minutes

THE CASE

A scholar studies particle size in three Rasaushadhi.

F1 *Kajjali:* samples are taken after 4, 12, 24 and 48 hours of trituration. Laser diffraction D50: 18 µm, 7 µm, 4.2 µm and 3.9 µm. Nishchandra is achieved at 24 hours.

F2 TEM of the 48-hour sample shows a few particles of 50 to 100 nm, mostly stuck to larger grains.

F3 *Rasasindura (Kupipakwa):* after grinding, particles of 1 to 20 µm by SEM; XRD crystallite size in the nanometre range.

F4 *Rasa Pottali:* SEM of a broken surface shows a dense, fused mass. A dose is prepared by rubbing the Pottali on a stone in a liquid; the rubbed material has particles of a few micrometres.

F5 The scholar concludes: “Longer trituration always gives nanoparticles and better efficacy.”

Q	Question	Marks	Domain	Level
(a)	Explain why particle size falls quickly and then levels off during trituration (F1, F2).	6	CAN	KH
(b)	How are micro- and nano-particles characterised in Kupipakwa Rasayana and Pottali? Interpret F3 and F4.	5	CC	KH
(c)	Evaluate the scholar’s conclusion in F5.	5	CE	KH
(d)	State two limitations of these formulations that nanotechnology research could address, and one caution.	4	CC	KH

Model answer framework

(a) Why size falls, then levels off — 6 marks

- Early trituration breaks large particles easily. Size falls fast (18 to 7 µm).
- As particles get smaller, each one has fewer flaws to crack along, so breaking needs more energy.
- Fine particles have high surface energy. They stick together (agglomerate) and, under pressure, can even fuse (cold welding). Breaking and re-joining reach a balance — the *comminution limit*. So D50 levels off (4.2 to 3.9 µm).
- Nano-sized particles do form (F2), but they cling to larger grains. Laser diffraction sees these as part of larger particles.
- Nishchandra at 24 hours (F1) is reached at about the same time as the plateau — the mechanochemical reaction of mercury and sulphur is near completion.

(b) Kupipakwa and Pottali — 5 marks

- *Methods:* XRD (phase, crystallite size), SEM-EDAX (morphology, elements), TEM (individual particles), laser diffraction and DLS (size distribution), BET (surface area), and test for free mercury.
- *F3:* grinding gives micron-sized particles, each made of nano-sized crystallites. So “nano” applies to the crystal domain, not to the particle as used.
- *F4:* Pottali is a fused, dense mass. Particle size depends on how the dose is prepared — rubbing generates the particles taken by the patient. So Pottali must be characterised both as a solid and as the rubbed dose.

(c) Evaluating the conclusion — 5 marks

Part	Judgement
“Longer trituration always gives nanoparticles”	Not supported. Size reaches a plateau; only a small nano fraction appears, mostly attached to larger grains
“and better efficacy”	Not tested. No pharmacological or clinical data are given. Smaller is not automatically better — it may also change safety

Better conclusion: “Trituration reduced D50 from 18 to about 4 µm, with a plateau after 24 hours that coincided with Nishchandra. A small nano-sized fraction was seen by TEM. The effect of particle size on activity was not studied.”

(d) Limitations, and a caution — 4 marks

- *Limitations:* variable particle size between batches; poor characterisation and standardisation; uncertain absorption; large doses in some forms.
- *Caution:* for products that contain mercury, arsenic or lead, smaller particle size may increase dissolution and absorption of the metal. Any move towards "nano" must be matched by safety studies.

WHERE MARKS ARE LOST

- Saying particles "keep getting smaller" with more trituration.
- Treating crystallite size as particle size in F3.
- Accepting "better efficacy" without data.

Revise: Volume 0 — Module 8, Unit 2 (p. 429); Module 3, Unit 1 (p. 145).

ABQ 8.2.3 — Silver Nanoparticles and Rajata Bhasma

Anchor: M8U2 (2.6, 2.7, 2.3) · Also draws on: M5U4 · Suggested time: 35 minutes

THE CASE

In a practical class, students make silver nanoparticles (AgNPs) by chemical reduction and compare them with Rajata Bhasma.

F1 *Batch 1:* dilute silver nitrate is reduced with ice-cold sodium borohydride, added drop by drop with stirring, with trisodium citrate as stabiliser. The solution turns clear yellow. UV-visible spectrum shows a sharp peak near 400 nm.

F2 *Batch 2:* borohydride is added all at once at room temperature, without citrate. The solution turns grey-black and particles settle. The UV-visible peak is broad and shifted to longer wavelength.

F3 *Rajata Bhasma:* XRD shows mainly metallic silver; some published samples also report silver chloride. SEM shows micron-sized aggregates.

F4 One student spilled silver nitrate on his hand; dark stains appeared next day. Sodium borohydride had been kept near a water bath; the bottle was found hissing.

F5 In a market survey, several “nano silver” herbal products give no particle size data on the label or in their literature.

Q	Question	Marks	Domain	Level
(a)	Explain the fundamental principles of nanoparticle synthesis by chemical reduction.	5	CC	KH
(b)	Interpret F1 and F2.	5	CAN	KH
(c)	Compare the synthesised AgNPs with Rajata Bhasma (F3).	5	CAN	KH
(d)	State the safety lessons from F4, and comment on the market survey finding (F5).	5	CS	KH

Model answer framework

(a) Principles — 5 marks

- *Bottom-up synthesis:* atoms are built up into particles. (Top-down methods break bulk material down — grinding, as in Bhasma.)
- *Reduction:* $Ag^{+} + e^{-} \rightarrow Ag^{0}$. A reducing agent (borohydride, citrate, or plant extracts in green synthesis) supplies electrons.
- *Nucleation and growth:* when silver atoms reach supersaturation, many tiny nuclei form at once; then they grow by adding atoms. A short, sharp burst of nucleation followed by controlled growth gives small, uniform particles.
- *Stabilisation (capping):* citrate or polymers adsorb on the surface and keep particles apart by charge or by steric hindrance. Without it, particles aggregate.
- Variables: concentration, temperature, rate of addition, ratio of reductant to silver, stabiliser, pH.

(b) Interpretation — 5 marks

- *Batch 1:* a sharp peak near 400 nm is the *surface plasmon resonance* of small, spherical, well-separated silver nanoparticles (typically a few tens of nanometres). Cold, slow addition and citrate gave controlled nucleation and good stabilisation.
- *Batch 2:* fast addition at room temperature with no stabiliser led to uncontrolled growth and aggregation. Large aggregates scatter light, appear grey-black and settle. The broad, red-shifted peak shows larger and aggregated particles.

(c) AgNPs versus Rajata Bhasma — 5 marks

Point	Synthesised AgNPs (Batch 1)	Rajata Bhasma
Approach	Bottom-up chemical reduction	Top-down processing: Shodhana, Marana with media and Puta
Phase	Metallic silver	Mainly metallic silver; silver chloride in some samples
Size	Nano, narrow distribution	Micron-sized aggregates; nano fraction possible
Surface	Citrate cap	Organic matter from plant media; inorganic residues
Form	Colloid in water	Dry powder
Characterisation	UV-vis SPR, TEM, DLS	XRD, SEM-EDAX, PSA

Conclusion: both contain metallic silver, but they are different materials. Results from AgNP research cannot be transferred directly to Rajata Bhasma.

(d) Safety lessons and market survey — 5 marks

- *Silver nitrate* is corrosive and oxidising; it stains skin black as silver is reduced by light. Wear gloves and goggles; wash spills at once.

- *Sodium borohydride* reacts with water and acids to release flammable hydrogen gas. Store dry, tightly closed, away from water and acids; use in a fume hood; no flames.
- Collect silver-containing waste separately; do not pour it down the drain (silver nanoparticles are toxic to aquatic life).
- *F5:* a "nano" claim without particle size data cannot be checked. It may mislead buyers. Products should state the size data and test method, and should be supported by safety data for the nano form.

WHERE MARKS ARE LOST

- No mention of surface plasmon resonance.
- Treating Rajata Bhasma as "the same as AgNPs".
- Missing the hydrogen hazard of borohydride.

Revise: Volume 0 — Module 8, Unit 2 (p. 429).

Unit 4 — Regulatory Aspects and Safety Concerns

Syllabus points. 4.1 Regulatory frameworks for nanotechnology-based Ayurvedic formulations · 4.2 Safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations

Table 6F. ABQ Yes · SAQ Yes · LAQ No. **Textbook.** Volume 0, Module 8, Unit 4, page 446.

ABQ 8.4.1 — The Nano-Haridra Start-up

Anchor: M8U4 (4.1, 4.2) · Also draws on: M1U1, M1U5 · Suggested time: 35 minutes

THE CASE

A start-up makes polymer-coated nanoparticles of Haridra (turmeric) extract. It plans the following.

F1 Sell the product as an “Ayurvedic proprietary medicine” under an Ayush licence.

F2 Do no new safety studies, because “turmeric is a food”.

F3 The particles are about 80 nm. They use a synthetic biodegradable polymer and polysorbate 80 as surfactant.

F4 Claim “10 times better bioavailability”, based on one in-vitro solubility test.

F5 Export the product to the European Union later.

Q	Question	Marks	Domain	Level
(a)	Describe the regulatory frameworks that apply to nanotechnology-based products in India and abroad.	6	CC	KH
(b)	Does the product fit the definition of an ASU patent or proprietary medicine? Discuss, marking what is settled and what is not.	4	CE	KH
(c)	What data package should the start-up prepare?	6	CS	KH
(d)	Evaluate the claim in F4.	4	CE	KH

Model answer framework

(a) Regulatory frameworks — 6 marks

Framework	Key points
India — Guidelines for Evaluation of Nanopharmaceuticals in India (DBT, ICMR and CDSCO, 2019)	Cover nanoscale active ingredients and formulations regulated as drugs; address quality, safety and efficacy assessment — characterisation, stability, non-clinical and clinical data
India — ASU drugs	Drugs and Cosmetics Act, 1940, Chapter IVA; Drugs Rules, 1945 — licensing under Rule 158-B, GMP under Schedule T. At the time of writing, no separate Ayush guideline for nano-based Ayurvedic formulations was identified — check the current position
European Union	Commission Recommendation on the definition of nanomaterial (2022): 50% or more of particles by number with an external dimension of 1 to 100 nm. Medicinal products follow EMA rules; food and food supplements follow their own rules, including novel food requirements
USA	FDA guidance (2014) on whether a product involves nanotechnology: about 1 to 100 nm, or up to 1 µm if the material shows size-dependent properties; case-by-case review by product category

(b) Does it fit ASU proprietary medicine? — 4 marks

- *Settled:* under Section 3(h), an ASU patent or proprietary medicine contains only ingredients from the formulae of the First Schedule books. Haridra qualifies as an ingredient. Excipients are allowed.
- *Not settled:* whether a nano-engineered form, with a synthetic polymer carrier designed to change absorption, is still an Ayurvedic medicine, or a new drug delivery system that needs evaluation as a nanopharmaceutical. The State Licensing Authority and the Ministry of Ayush would decide. The start-up should seek a formal opinion before investing.

(c) Data package — 6 marks

- *Characterisation:* particle size distribution (number-based), PDI, zeta potential, morphology (TEM), drug loading and encapsulation efficiency, residual solvent and polymer content.
- *Quality:* specifications; batch-to-batch consistency over at least three batches; stability (size, loading, degradation) under real-time and accelerated conditions.
- *Safety:* nano-specific toxicology — acute and repeated-dose studies as per OECD test guidelines, biodistribution and accumulation, genotoxicity; safety of polymer and surfactant at the dose used. “Turmeric is a food” (F2) does not apply to a new nano form.
- *Bioavailability:* a pharmacokinetic study comparing the nano form with a conventional extract (Cmax, Tmax, AUC).
- *Efficacy:* data appropriate to the claims made.
- *Labelling and claims* supported by the data.

(d) “10 times better bioavailability” — 4 marks

- In-vitro solubility is not bioavailability. Bioavailability is the rate and extent to which the active substance reaches the circulation, measured in vivo.
- A ten-fold claim needs a controlled pharmacokinetic study in humans (or at least in animals, stated as such), with the comparator and the analyte named.
- Until then, the honest claim is: “improved solubility in vitro”.

WHERE MARKS ARE LOST

- Saying “there is no regulation for nano, so anything can be sold”.
- Naming the 2019 Indian guidelines without the issuing bodies.
- Accepting in-vitro data as bioavailability.

Revise: Volume 0 — Module 8, Unit 4 (p. 446); Module 1, Unit 5 (p. 66).

ABQ 8.4.2 — A Safety Signal in the Rats

Anchor: M8U4 (4.2) · Also draws on: M6U5, M5U4 · Suggested time: 35 minutes

THE CASE

A 28-day repeated-dose oral toxicity study compares “nano Lauha Bhasma” (D50 about 90 nm) with conventional Lauha Bhasma (D50 about 4 µm), at the same dose of iron.

F1 In the nano group, liver enzymes (ALT, AST) are raised at the high dose. The conventional group is normal.

F2 Liver histology of the nano group shows iron deposits (positive Perls’ stain) and small foci of inflammation.

F3 ICP-MS shows higher iron in the liver and spleen of the nano group.

F4 In vitro, the nano form dissolves faster in simulated gastric fluid and produces more reactive oxygen species (ROS) in a cell assay.

F5 The sponsor wants the report to conclude: “Both forms are safe because no animal died.”

Q	Question	Marks	Domain	Level
(a)	Explain the safety concerns and toxicity issues specific to nano-sized materials.	6	CC	KH
(b)	Interpret F1 to F4.	5	CAN	KH
(c)	Evaluate the sponsor’s proposed conclusion (F5).	4	CE	KH
(d)	What further studies and ethical requirements apply?	5	CS	KH

Model answer framework

(a) Nano-specific safety concerns — 6 marks

Concern	Explanation
Higher surface area and reactivity	More surface per unit mass; more dissolution; more catalytic and ROS-generating activity
Different uptake	Can cross epithelia and enter cells by routes not open to larger particles
Biodistribution and accumulation	Taken up by liver and spleen macrophages; may persist
Crossing barriers	Possible passage across biological barriers (placenta, blood-brain barrier) — studied case by case
Protein corona	Proteins coat the particle in body fluids and change how it behaves
Oxidative stress and inflammation	ROS generation; inflammatory response
Standard tests may miss effects	Dose metrics by mass may not reflect effect; surface area or particle number may matter

(b) Interpretation — 5 marks

- F1 and F2: liver injury — raised enzymes, iron overload and inflammation — at the high dose of the nano form only.
- F3: the nano form distributes more iron to liver and spleen, consistent with greater absorption and macrophage uptake.
- F4: faster dissolution and more ROS offer a plausible mechanism — more free iron, which catalyses ROS formation.
- Together: at equal iron dose, the nano form shows a hepatotoxicity signal that the conventional form does not. Particle size changes the safety profile.

(c) The sponsor's conclusion — 4 marks

- Unacceptable. Absence of deaths is not absence of toxicity. A repeated-dose study looks for target organ effects, and the liver is clearly affected.
- The report must state the findings truthfully, identify the no-observed-adverse-effect level (NOAEL) where possible, and conclude that the nano form shows dose-related hepatotoxicity in this study.
- Selective reporting is scientific misconduct, and it would mislead regulators and patients.

(d) Further studies and ethics — 5 marks

- 90-day repeated-dose study (OECD TG 408) with a recovery group, to check if changes reverse.
- Toxicokinetics: blood and tissue iron over time; biodistribution.
- Genotoxicity battery; oxidative stress markers.
- Characterise the dosed material (size in the vehicle; stability).

- *Ethics:* approval of the Institutional Animal Ethics Committee under CCSEA guidelines; follow the 3Rs (Replace, Reduce, Refine); use humane endpoints; justify group sizes.

WHERE MARKS ARE LOST

- Listing generic nano risks without interpreting the data.
- Accepting “no deaths, so safe”.
- Forgetting IAEC and CCSEA.

Revise: Volume 0 — Module 8, Unit 4 (p. 446); Module 6, Unit 5 (p. 350).

ABQ 8.4.3 — Dust in the Sunbeam

Anchor: M8U4 (4.2) · Also draws on: M4U1, M8U2 · Suggested time: 35 minutes

THE CASE

An institute runs a Rasashala and a small nanotechnology laboratory in the same building.

F1 Workers grind and sieve Bhasmas in an open hall. Fine dust is visible in the beams of sunlight. No one wears a mask.

F2 In the nano laboratory, students make silver and zinc oxide nanoparticles. Waste solutions are poured down the sink.

F3 A spray-drier used to make powder from a nanoemulsion vents its exhaust into the room.

F4 There is no measurement of airborne dust, and no safety data sheets are available for the nanomaterials.

F5 One worker who has ground Bhasmas for many years now has a chronic cough and breathlessness on exertion.

Q	Question	Marks	Domain	Level
(a)	Explain the safety concerns of fine and nano-sized particles for workers.	5	CC	KH
(b)	Apply the hierarchy of controls to F1 and F3.	6	CS	KH
(c)	Explain the environmental concern in F2 and the correct waste handling.	5	CAN	KH
(d)	What regulatory and policy gaps does this case show, and what do you recommend?	4	CE	KH

Model answer framework

(a) Safety concerns for workers — 5 marks

- Inhalation is the main route of exposure at work. The finer the particle, the deeper it travels into the lung: very fine and nano-sized particles reach the alveoli.
- Particles that are not cleared can cause inflammation and, over years, lung disease (F5 fits chronic dust exposure; needs medical evaluation).
- Metal-containing dusts (lead, mercury, arsenic, zinc, silver) can be absorbed from the lung into the blood.
- Nano-sized particles have high surface area and reactivity; some may move from the lung to other organs.
- Skin and eye contact; swallowing through dirty hands.

(b) Hierarchy of controls — 6 marks

Level	Grinding and sieving (F1)	Spray-drier (F3)
Elimination / substitution	Wet grinding where the process allows; buy pre-ground material only from controlled suppliers	Use a process that does not make dry nano powder, where possible
Engineering controls	Enclosed grinders and sieves; local exhaust ventilation with HEPA filtration	Duct exhaust outside through HEPA filters; enclosure
Administrative controls	SOPs; restricted access; wet cleaning (no dry sweeping); job rotation; training; exposure monitoring; health surveillance	SOPs; maintenance; access control
Personal protective equipment (last line)	Fitted particulate respirators (for example N95 or higher), gloves, goggles, work clothes kept on site	Same

(c) Environmental concern and waste — 5 marks

- Silver nanoparticles release silver ions that are toxic to aquatic organisms and to the bacteria of sewage treatment. Zinc oxide nanoparticles are also toxic to aquatic life. Drains carry them to water bodies.
- *Correct handling:* collect nanoparticle waste in labelled, closed containers; do not pour into sinks; treat (for example precipitate or aggregate the particles) and dispose of through an authorised hazardous waste agency under the Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016; keep records; follow the institution’s environmental clearance conditions.

(d) Gaps and recommendations — 4 marks

- *Gaps:* no exposure monitoring; no safety data sheets; no health surveillance; no nano-specific SOPs; mixing of a manufacturing area and a research laboratory.

- *Recommendations:* a written safety policy and risk assessment for each process; air monitoring; safety data sheets for all materials; annual medical examination including lung function for exposed workers (the 2026 revision of Schedule T also requires employee health surveillance in Rasaushadhi units); training; a safety committee; refer the worker in F5 for medical evaluation at once.

WHERE MARKS ARE LOST

- Answering with "wear masks" only — PPE is the last line, not the first.
- Ignoring the environmental route in F2.
- Not linking F5 to chronic exposure.

Revise: Volume 0 — Module 8, Unit 4 (p. 446); Module 4, Unit 1 (p. 207).

PART B — Case-based Analytical Structured LAQs

Ten analytical LAQs of 10 marks each, from the units that Table 6F marks "Yes" for LAQ and that have no ABQ in Part A. With the solved paper, this is how Modules 2 and 6 enter this book. Each LAQ is numbered by its module and unit: LAQ B6.4 is on Module 6, Unit 4.

Each LAQ starts on a new page. The framework follows on the same page: cover it with a sheet while you attempt the question in about 18 minutes.

Module 1 · Unit 1 — Introduction to Pharmaceutics

Syllabus points. 1.1 Definition and scope of pharmaceutics · 1.2 Historical development of pharmaceutics · 1.3 Pharmaceutical industry overview · 1.4 Regulatory frameworks (FDA, etc.) · 1.5 Pharmaceutical product development process

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 1, Unit 1, page 9.

LAQ B1.1 — One Product, Three Markets

Anchor: M1U1 (1.4, 1.5) · Semester 2 · Suggested time: 18 minutes

THE CASE

An Indian company has developed a tablet containing standardised extracts of Shallaki (Boswellia serrata) and Haridra for joint pain. It wants to sell the product in India, in the USA and in the European Union, with the claim “relieves osteoarthritis pain”. Its regulatory team assumes that one dossier will serve all three markets.

Q	Question	Marks	Domain	Level
(a)	Analyse the regulatory pathway for this product in India.	4	CAN	KH
(b)	Compare the pathways open to it in the USA and the EU, and what the claim means in each.	4	CAN	KH
(c)	Is the team’s assumption correct? Give a one-paragraph recommendation.	2	CE	KH

Model answer framework

(a) India — 4 marks

- Ingredients are classical drugs, so the product can be an ASU *patent or proprietary medicine* (Section 3(h), Drugs and Cosmetics Act, 1940), if it contains only ingredients from First Schedule books.
- Licence from the State Licensing Authority under Rule 158-B of the Drugs Rules, 1945, with the proof of safety and effectiveness that Rule 158-B specifies for this category of medicine.
- GMP under Schedule T (revised in 2026); standards of the API for ingredients; labelling under Rule 161; shelf life under Rule 161-B, supported by stability data.

(b) USA and EU — 4 marks

Point	USA	European Union
Route 1	Dietary supplement (Dietary Supplement Health and Education Act, 1994): no pre-market approval, but it may not claim to treat or cure a disease. “Relieves osteoarthritis pain” is a disease claim — not allowed	Food supplement: no medicinal claims allowed
Route 2	Botanical drug: FDA’s Botanical Drug Development guidance (2016); needs an Investigational New Drug application and clinical trials for the disease claim	Traditional herbal medicinal product (Directive 2004/24/EC): simplified registration based on at least 30 years of traditional use, including 15 years in the EU, for conditions suitable for self-medication; or full marketing authorisation with clinical data
Meaning for the claim	Disease claim needs the drug route	Disease claim needs a medicinal route; traditional-use evidence must exist in the EU

(c) Recommendation — 2 marks

The assumption is wrong. Regulation follows the market where the product is sold. One core technical dossier (quality, stability, safety, clinical data) can be built, but each market needs its own legal route, claims and labelling. The company should decide the claim for each market first, then the route, then the studies.

Revise: Volume 0 — Module 1, Unit 1 (p. 9).

Module 1 · Unit 5 — Biopharmaceutics

Syllabus points. 5.1 Biopharmaceutics and pharmacokinetics · 5.2 Bioavailability and bioequivalence · 5.3 Pharmaceutical dissolution and absorption · 5.4 Pharmaceutical product development and optimization

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 1, Unit 5, page 66.

LAQ B1.5 — Haridra, Trikatu and the AUC

Anchor: M1U5 (5.1, 5.2, 5.3) · Semester 2 · Suggested time: 18 minutes

THE CASE

A scholar studies curcumin given to rats alone and with piperine (the main alkaloid of Pippali and Maricha). Plasma results (illustrative values) are as follows.

Group	C_{max} (ng/ml)	T_{max} (h)	AUC_{0-t} (ng·h/ml)
Curcumin alone	120	1.0	300
Curcumin + piperine	480	1.0	1,350

She writes: “This proves the classical concept of Yogavahi, and every Haridra formulation should contain Trikatu.”

Q	Question	Marks	Domain	Level
(a)	Define C_{max}, T_{max} and AUC. Calculate the relative bioavailability of curcumin with piperine.	3	CAP	KH
(b)	Explain why curcumin has poor bioavailability, and how piperine increases it.	4	CAN	KH
(c)	Evaluate the scholar’s conclusion.	3	CE	KH

Model answer framework

(a) Definitions and calculation — 3 marks

- C_{max}: highest plasma concentration. T_{max}: time to reach C_{max} (rate of absorption). AUC: area under the plasma concentration–time curve (extent of exposure).
- Relative bioavailability = (AUC_{test} ÷ $AUC_{reference}$) × 100 = (1,350 ÷ 300) × 100 = **450 percent**, that is 4.5 times. C_{max} rose four-fold; T_{max} is unchanged.

(b) Poor bioavailability and the role of piperine — 4 marks

- Curcumin is poorly soluble in water; it is unstable at intestinal pH; it is rapidly conjugated (glucuronidation, sulphation) in the gut wall and liver; and it is quickly eliminated.
- Piperine inhibits glucuronidation and some drug-metabolising enzymes, and inhibits efflux transporters such as P-glycoprotein. More unchanged curcumin reaches the circulation.
- A human study published in 1998 (Shoba and colleagues, *Planta Medica*) reported a large increase in curcumin bioavailability when 20 mg piperine was given with curcumin.

(c) Evaluating the conclusion — 3 marks

- *Supported:* piperine increases curcumin exposure in this model. This fits the classical idea of Yogavahi (a substance that enhances the action of others) as one possible mechanism.
- *Not supported:* "proves" — one rat study on one marker does not prove a classical concept as a whole; higher exposure does not by itself prove better effect; and piperine can also raise blood levels of other medicines the patient takes (drug interactions).
- *Not supported:* "every Haridra formulation should contain Trikatu" — formulations are designed for a purpose and a patient; adding Trikatu changes the formula (and its Guna), and would change the identity of a classical product.

Revise: Volume 0 — Module 1, Unit 5 (p. 66).

Module 2 · Unit 2 — Metal and Mineral Databases

Syllabus points. 2.1 Overview of metal and mineral databases · 2.2 USGS Mineral Database · 2.3 Mineral Database · 2.4 Webmineral Mineral Database · 2.5 Introduction to Metal Databases

Table 6F. ABQ No · SAQ No · LAQ Yes — this unit can appear *only* as an LAQ. **Textbook.** Volume 0, Module 2, Unit 2, page 102. Question 2.1 of the October 2026 paper, on this unit, is solved in the chapter "The October 2026 Preliminary Paper — Solved".

LAQ B2.2 — Is This Swarna Makshika?

Anchor: M2U2 (2.2 to 2.5) · Also draws on: M3U5 · Suggested time: 18 minutes

THE CASE

A Rasashala buys "Swarna Makshika" for Makshika Bhasma. The sample is golden-yellow. A scholar tests it and records: it scratches glass easily; specific gravity about 5.0; cubic crystals with striated faces; XRD peaks that match pyrite in an online reference database. Copper is not detected by XRF. Swarna Makshika is generally identified with chalcopyrite.

Q	Question	Marks	Domain	Level
(a)	Using data from mineral databases, compare chalcopyrite and pyrite.	4	CAP	KH
(b)	Interpret the scholar's findings.	3	CAN	KH
(c)	Describe how the databases are searched to reach this conclusion, and their limits.	3	CAP	KH

Model answer framework

(a) Chalcopyrite and pyrite — 4 marks

Property	Chalcopyrite (Swarna Makshika)	Pyrite
Formula	$CuFeS_2$	FeS_2
Copper content	About 35 percent	None
Colour	Brass-yellow; tarnishes to iridescent colours	Pale brass-yellow
Hardness (Mohs)	3.5 to 4 — does not scratch glass	6 to 6.5 — scratches glass
Specific gravity	About 4.1 to 4.3	About 4.9 to 5.2
Crystal system and habit	Tetragonal; often massive	Cubic; cubes with striated faces
Streak	Greenish-black	Greenish- to brownish-black

(b) Interpretation — 3 marks

Every finding — hardness, specific gravity, cubic striated crystals, XRD match and absence of copper — fits pyrite, not chalcopyrite. The sample is pyrite sold as Swarna Makshika. It should be rejected; Bhasma made from it would differ in composition (no copper) and properties.

(c) Searching the databases, and limits — 3 marks

- Search Webmineral or Mindat by name (chalcopyrite, pyrite) to get formula, physical properties and crystal data. Search RRUFF or the Crystallography Open Database for reference XRD patterns and Raman spectra. Overlay the sample's XRD peaks on the reference pattern; check that all major peaks match.
- *Limits:* databases describe pure minerals; natural samples contain mixtures and impurities; a single physical property can mislead; matching needs good-quality XRD; and the link between a classical name and a modern mineral is an identification made by scholars, which may be debated for some drugs.

Revise: Volume 0 — Module 2, Unit 2 (p. 102); Module 3, Unit 5 (p. 183).

Module 3 · Unit 4 — Phytochemistry

Syllabus points. 4.1 Definition and significance of phytochemistry · 4.2 Overview of plant-based bioactive compounds · 4.3 Classes of Phytochemicals · 4.4 Extraction methods · 4.5 Phytochemical screening

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 3, Unit 4, page 171.

LAQ B3.4 — Guduchi in the Soxhlet

Anchor: M3U4 (4.3, 4.4, 4.5) · Semester 2 · Suggested time: 18 minutes

THE CASE

A scholar extracts Guduchi (Tinospora cordifolia) stem with methanol in a Soxhlet apparatus for 72 hours, then concentrates the extract on a water bath at 90 °C. The extract is dark and tarry. Its HPTLC shows fewer bands than a cold-macerated extract. She then screens the extract with only two tests and concludes “Guduchi contains alkaloids and flavonoids only”.

Q	Question	Marks	Domain	Level
(a)	Analyse what went wrong in the extraction, and suggest a better method with reasons.	4	CAN	KH
(b)	Explain how a solvent is chosen for extraction.	2	CC	KH
(c)	Tabulate a complete phytochemical screening scheme, and comment on her conclusion.	4	CE	KH

Model answer framework

(a) What went wrong, and a better method — 4 marks

- 72 hours of continuous heating at the solvent’s boiling point, then concentration at 90 °C in open air: heat-sensitive constituents (some glycosides, diterpene lactones) can degrade, and the extract chars — dark, tarry, fewer bands.
- Better: cold maceration or percolation; or ultrasound-assisted extraction (short time, low temperature); concentrate under reduced pressure in a rotary evaporator at about 40 to 50 °C. For a classical correlation, also prepare an aqueous extract (as in Kwatha).
- Report yield (percent w/w) and the conditions.

(b) Choice of solvent — 2 marks

“Like dissolves like”: non-polar solvents (hexane, petroleum ether) for fats, waxes and some terpenoids; medium polarity (chloroform, ethyl acetate) for aglycones and many alkaloid bases; polar (methanol, ethanol, water) for glycosides, phenolics, sugars and salts of alkaloids. Successive extraction from non-polar to polar separates classes. Also weigh safety, cost, residue limits and the relevance to traditional use (water, hydro-alcoholic).

(c) Screening scheme and her conclusion — 4 marks

Class	Test	Positive result
Alkaloids	Dragendorff's; Mayer's; Wagner's	Orange-red; cream; reddish-brown precipitate
Flavonoids	Shinoda (magnesium and HCl); alkaline reagent	Pink to red; yellow that fades with acid
Phenols and tannins	Ferric chloride; lead acetate	Blue-black or green; precipitate
Saponins	Foam test	Persistent froth
Steroids and triterpenoids	Liebermann–Burchard; Salkowski	Green-blue; red-brown at the interface
Glycosides	Keller–Kiliani (cardiac); Borntrager's (anthraquinone)	Brown ring / pink
Carbohydrates	Molisch's; Fehling's	Violet ring; red precipitate
Proteins and amino acids	Biuret; ninhydrin	Violet; purple

Her conclusion is not supported: two tests can only show what they test for. Guduchi is reported to contain alkaloids (including berberine-type), diterpenoid lactones, glycosides, steroids and polysaccharides. Screening tests are qualitative and can give false results with coloured extracts; HPTLC and specific assays must follow.

Revise: Volume 0 — Module 3, Unit 4 (p. 171).

Module 3 · Unit 5 — Metallurgy

Syllabus points. 5.1 Introduction to metallurgy and general physicochemical methods of extraction of metals from ores · 5.2 Common ores used in Rasashastra and their Satvapatana from metallurgical point of view

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 3, Unit 5, page 183.

LAQ B3.5 — The Satva that Stayed in the Slag

Anchor: M3U5 (5.1, 5.2) · Also draws on: M2U2 · Suggested time: 18 minutes

THE CASE

A scholar attempts Satvapatana of Swarna Makshika (chalcopyrite) by mixing it with the prescribed flux and plant materials, making pellets and heating them strongly in a crucible furnace. Yield of metallic Satva is very low. Small metal beads are trapped inside a glassy slag. White fumes with a garlic-like smell were noticed during heating.

Q	Question	Marks	Domain	Level
(a)	Describe the general physicochemical steps of extracting a metal from a sulphide ore.	4	CC	KH
(b)	Analyse Satvapatana of Swarna Makshika from a metallurgical point of view, and explain the low yield.	4	CAN	KH
(c)	What do the white fumes indicate? State the safety measures.	2	CAP	KH

Model answer framework

(a) General steps — 4 marks

Step	Purpose	Example
Concentration (ore dressing)	Remove gangue (rock)	Washing, gravity separation, froth flotation for sulphides
Roasting	Partly convert sulphide in air; drive off sulphur as SO_2	$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$
Smelting and reduction	Iron is oxidised and removed with flux as slag; copper is freed by reduction (by carbon, or by self-reduction of the sulphide with oxide)	$FeO + SiO_2 \rightarrow FeSiO_3$ (slag); $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$
Refining	Purify crude metal	Electrolytic refining of copper

Pyrometallurgy (heat), hydrometallurgy (leaching in solution) and electrometallurgy (electrolysis) are the three broad routes.

(b) Satvapatana as metallurgy; low yield — 4 marks

- Satvapatana is a small-scale pyrometallurgical extraction: the plant materials supply carbon (reducing agent) and organic acids; the flux (for example Tankana, borax) lowers the melting point of the gangue and forms a fluid slag; strong heating roasts and reduces the ore, and the Satva (a copper-rich metal) separates.
- *Low yield:* temperature not high enough or held too short for the metal to melt and coalesce; slag too viscous (wrong flux or too little), so droplets stay trapped; incomplete roasting (sulphur remains, forming matte rather than metal); poor mixing; loss as fumes.
- *Remedy:* control temperature and time; adjust flux; roast first; crush the slag to recover beads.

(c) White fumes — 2 marks

White fumes with a garlic-like smell suggest arsenic trioxide from arsenic-bearing impurities in the ore (common in sulphide minerals). Work only in a fume hood or with local exhaust; wear a respirator, gloves and face shield; analyse the ore for arsenic before use; handle slag and dust as hazardous waste.

Revise: Volume 0 — Module 3, Unit 5 (p. 183).

Module 4 · Unit 2 — SOP of Shodhana Processes

Syllabus points. 2.1 Types of Shodhana procedures (physical, chemical, biological) · 2.2 Methods of Shodhana (filtration, sedimentation, distillation, etc.) · 2.3 SOPs for Shodhana procedures

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 4, Unit 2, page 215.

LAQ B4.2 — Rewriting the Kupilu SOP

Anchor: M4U2 (2.1, 2.2, 2.3) · Also draws on: M3U3 · Suggested time: 18 minutes

THE CASE

The “SOP” for Shodhana of Kupilu (Strychnos nux-vomica seeds) in a pharmacy reads, in full: “Soak Kupilu in Gomutra for some days. Boil in milk. Remove skin and germ. Dry and powder.” There is no title, no batch quantity, no safety instruction, no record format and no test of the end product.

Q	Question	Marks	Domain	Level
(a)	Classify the steps of this Shodhana as physical, chemical or biological, with reasons.	3	CAN	KH
(b)	Identify the defects of this SOP.	2	CAN	KH
(c)	Write a corrected SOP outline, including safety and acceptance criteria.	5	CS	KH

Model answer framework

(a) Classification — 3 marks

Step	Type	Reason
Soaking in Gomutra (several days, medium changed daily)	Chemical (and physical leaching)	Softens the seed; alkaline medium and time promote chemical change in alkaloids; some alkaloid leaches out
Boiling in milk (Dolayantra Swedana)	Chemical and physical	Heat with proteins and fat; further change and leaching of alkaloids
Removing seed coat and embryo	Physical	Mechanical removal of parts
Drying and powdering	Physical	Size reduction

Published HPLC studies report lower strychnine and brucine after such Shodhana. Biological Shodhana would involve fermentation or microbial action — not a step here.

(b) Defects — 2 marks

Vague quantities and durations (“some days”); no ratio of drug to medium; no specification for raw material or media; no equipment; no responsibility; no safety measures for a Schedule E(1) poison; no in-process checks; no acceptance criteria; no record or batch format; no reference to the classical text.

(c) Corrected SOP outline — 5 marks

1. *Title, SOP number, version, effective date, approval signatures.*
2. *Purpose and scope:* Shodhana of Kupilu for use in formulations.
3. *Reference:* the classical text followed (chapter and verse).
4. *Responsibility:* operator, supervisor, QA.
5. *Materials:* Kupilu seeds (tested for identity and quality), fresh Gomutra, cow’s milk — each with a specification; quantities per batch.
6. *Equipment:* stainless steel or glass vessels, Dolayantra, cloth, drier, pulveriser — clean and labelled.
7. *Safety:* Kupilu is a Schedule E(1) poison — gloves, mask, apron; powdering in a closed system or under exhaust; separate labelled storage under lock; issue and return register; spill and first-aid instructions (strychnine poisoning causes convulsions — urgent hospital care).
8. *Procedure:* soak in Gomutra for the stated number of days, changing it daily; wash; boil in milk in Dolayantra for the stated time; wash; remove seed coat and embryo; dry at a set temperature to a set moisture; powder and sieve.
9. *In-process checks:* softness; removal of all seed coats and embryos; weights at each stage.

10. *Acceptance criteria:* description; loss on drying; TLC/HPTLC of strychnine and brucine compared with raw seed (reduction recorded); microbial limits.
11. *Records:* batch record format; yield; deviations; signatures; label “Shuddha Kupilu” with batch number and date.

Revise: Volume 0 — Module 4, Unit 2 (p. 215); Module 3, Unit 3 (p. 163).

Module 4 · Unit 3 — SOP of Marana Process

Syllabus points. 3.1 Types of Marana procedures (incineration, calcination, etc.) · 3.2 Methods of Marana (oxidation, thermal decomposition, etc.) · 3.3 SOPs for Marana procedures

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 4, Unit 3, page 224.

LAQ B4.3 — White Smoke from the Yashada Furnace

Anchor: M4U3 (3.1, 3.2, 3.3) · Also draws on: M4U4 · Suggested time: 18 minutes

THE CASE

A pharmacy prepares Yashada Bhasma. After Shodhana, zinc is heated with plant powder (Jarana) in an iron pan. An operator raises the flame “to finish faster”. Thick white fumes rise, and a white fluffy powder settles on nearby surfaces. The yield of Jarita Yashada is low. The operator later reports fever, chills and muscle aches that settle in a day.

Q	Question	Marks	Domain	Level
(a)	Describe the types and methods of Marana, with examples.	3	CC	KH
(b)	Analyse the events in this case: the fumes, the low yield and the operator’s illness.	3	CAN	KH
(c)	Write the key elements of an SOP for Jarana and Marana of Yashada, with safety.	4	CS	KH

Model answer framework

(a) Types and methods — 3 marks

Type / method	Chemistry	Example
Puta-based incineration (Putapaka)	Oxidation, decomposition and sintering in repeated heating cycles	Lauha, Tamra, Abhraka Bhasma
Jarana (calcination with plant powder and stirring)	Oxidation of a low-melting metal in air	Naga, Vanga, Yashada
Marana with sulphur or sulphide minerals	Sulphide formation	Tamra (with Gandhaka); Naga (with Manahshila)
Marana with mercury or Kajjali	Amalgamation, then removal of mercury	Swarna, Rajata
Thermal decomposition	Breakdown of carbonate or hydrate	Shankha, Godanti
Kupipakwa (sublimation in a glass bottle)	Formation and sublimation of a compound	Rasasindura

(b) Analysis — 3 marks

- Zinc melts at about 420 °C and boils at about 907 °C. Strong heating vaporises zinc; the vapour burns in air to zinc oxide, seen as white fumes and fluffy “philosopher’s wool” ($2Zn + O_2 \rightarrow 2ZnO$).
- Low yield: the product escaped as fume instead of forming in the pan.
- The illness — fever, chills and aches that settle in about a day after breathing fresh zinc oxide fume — is typical of *metal fume fever*. It is a warning of excessive exposure.

(c) SOP elements — 4 marks

- Batch quantity; ratio of plant powder; type of pan; heating source with temperature control; temperature limit kept well below the boiling point of zinc; continuous stirring; duration; endpoint (fine, yellowish-white powder; no metallic lustre).
- Subsequent Marana: Bhavana with the prescribed juice, Chakrika, number and type of Puta with temperature records.
- *Safety:* local exhaust over the pan; respirator suitable for metal fumes; heat-resistant gloves, face shield, apron; trained operator; no raising of heat beyond the SOP; first aid and medical reporting of symptoms; air monitoring.
- *Records:* temperatures, times, weights, yield, deviations, signatures; tests on the product (Rekhapurnata, Varitara, Apunarbhava; XRD for ZnO; zinc content; heavy metals).

Revise: Volume 0 — Module 4, Unit 3 (p. 224); Module 4, Unit 4 (p. 232).

Module 5 · Unit 1 — Puta and its Modern Aspects

Syllabus points. 1.1 Introduction, principles, and specifications of Puta · 1.2 SOPs for handling Puta · 1.3 Modern aspects of Puta (automation, computerization, etc.) · 1.4 Troubleshooting and maintenance of Puta

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 5, Unit 1, page 257.

LAQ B5.1 — From Gaja Puta to Programmable Furnace

Anchor: M5U1 (1.1 to 1.4) · Also draws on: M5U6 · Suggested time: 18 minutes

THE CASE

A pharmacy records the temperature of a traditional Gaja Puta with a thermocouple placed beside the Sharava. It rises to a peak of about 900 °C in roughly three hours, stays above 600 °C for about four hours, and cools naturally overnight. The pharmacy wants to replace the Gaja Puta with a programmable electric furnace for consistency. The first furnace batch, heated straight to 900 °C and cooled with the door open, gives a harder, darker product.

Q	Question	Marks	Domain	Level
(a)	Explain the principles and specifications of Puta.	3	CC	KH
(b)	Analyse why the first furnace batch differed, and how the furnace should be programmed.	4	CAN	KH
(c)	List the modern aspects of Puta and the key troubleshooting and maintenance points.	3	CAP	KH

Model answer framework

(a) Principles and specifications — 3 marks

- Puta is a measured quantum of heat — defined classically by the size of the pit, the quantity and type of fuel, and the method of heating and cooling. Types range from mild (for example Kapota, Kukkuta) to strong (Gaja, Maha Puta), chosen according to the drug.
- Principle: a defined time–temperature cycle (heating, peak, soaking, slow cooling — Svangasheeta) brings about oxidation, decomposition and sintering in a controlled, repeatable way.
- Specifications: pit dimensions and fuel as per the text; Sharava sealing; placement of the Samputa; recorded peak temperature and duration.

(b) Why the batch differed; programming — 4 marks

- The Gaja Puta gives a gradual rise, a long soak above 600 °C and slow natural cooling. The furnace batch rose abruptly, may have overshot, and was cooled fast with the door open.
- Rapid heating and fast cooling change reaction and sintering: grains fuse (harder), oxidation may be incomplete or uneven (darker), and thermal shock may crack the Sharava.
- *Programme:* copy the recorded profile — ramp to peak over about three hours, soak to match the time above 600 °C, then controlled slow cooling with the door closed; validate by comparing products (classical tests, XRD, particle size) over several batches.

(c) Modern aspects; troubleshooting and maintenance — 3 marks

- *Modern aspects:* electric and gas furnaces with PID controllers; programmable multi-step cycles; thermocouples and data loggers; PLC and computer control with recorded batch data; uniform heating chambers; exhaust and emission control.
- *Troubleshooting:* uneven heating (overloading, cold zones near the door); overshoot (controller tuning); slow heating (failing elements); wrong readings (thermocouple drift).
- *Maintenance:* calibration of thermocouples and temperature uniformity surveys; inspection of elements, insulation and door seals; cleaning of chamber and exhaust; logbook.

Revise: Volume 0 — Module 5, Unit 1 (p. 257) and Unit 6 (p. 299).

Module 6 · Unit 4 — Animal Health and Disease Prevention

Syllabus points. 4.1 Common diseases in laboratory animals · 4.2 Vaccination and immunization programs · 4.3 Parasite control and management

Table 6F. ABQ No · SAQ Yes · LAQ Yes — Module 6 has no ABQ unit. **Textbook.** Volume 0, Module 6, Unit 4, page 343. Question 2.2 of the October 2026 paper, on this unit, is solved in the chapter "The October 2026 Preliminary Paper — Solved".

LAQ B6.4 — Pinworms Before the 90-Day Study

Anchor: M6U4 (4.1, 4.2, 4.3) · Also draws on: M6U1 · Suggested time: 18 minutes

THE CASE

Two weeks before a 90-day oral toxicity study, routine screening of the mouse colony finds pinworm eggs on perianal tape tests, and adult worms in the caecum of a sentinel mouse. Some guinea pigs in the next room have circular patches of hair loss with scaling; one animal attendant has an itchy ring-shaped rash on the forearm.

Q	Question	Marks	Domain	Level
(a)	Identify the likely infections and explain how each is diagnosed.	3	CAN	KH
(b)	Plan a parasite control programme for the mouse colony before the study.	4	CS	KH
(c)	Explain the zoonotic risk, and the place of vaccination and immunisation programmes in a laboratory animal facility.	3	CC	KH

Model answer framework

(a) Likely infections and diagnosis — 3 marks

Infection	Agent	Diagnosis
Pinworms in mice	*Syphacia obvelata* (eggs laid around the anus) and *Aspiculuris tetraptera* (eggs passed in faeces)	Perianal cellophane tape test for *Syphacia*; faecal flotation for *Aspiculuris*; adults in caecum and colon at necropsy
Ringworm (dermatophytosis) in guinea pigs	Dermatophyte fungi, commonly *Trichophyton* species	Skin scrapings and hair plucks for microscopy and fungal culture

(b) Parasite control programme — 4 marks

- Do not start the study with infected animals: treat or replace the colony.
- Treatment as advised by the veterinarian (for example an anthelmintic such as fenbendazole given in medicated feed over a set period), treating all animals in the room at the same time.
- Pinworm eggs are light, resistant and spread through the air and on surfaces: change and sanitise cages, bedding and racks; decontaminate the room; repeat testing after treatment.
- Prevent re-entry: quarantine new animals; test incoming stock; keep sentinel monitoring.
- Record everything; inform the study director and the ethics committee, since treatment may affect study animals.

(c) Zoonosis; vaccination and immunisation — 3 marks

- Dermatophytes spread from animals to people; the attendant's rash is likely zoonotic ringworm. Isolate and treat affected guinea pigs; staff to wear gloves and gowns; medical treatment for the attendant; disinfect the room.
- *Vaccination* is used mainly for larger laboratory species (for example dogs and some other species, as per veterinary advice and local disease risk). In rodent colonies, disease prevention relies on biosecurity and health monitoring, not on vaccines.
- *Staff immunisation* (for example tetanus, and others as advised by occupational health according to the species handled) is part of a facility's occupational health programme.

Revise: Volume 0 — Module 6, Unit 4 (p. 343).

Module 8 · Unit 1 — Introduction to Nanotechnology

Syllabus points. 1.1 Definition and history of nanotechnology · 1.2 Basic principles and applications of nanotechnology · 1.3 Understanding Nanoscale

Table 6F. ABQ No · SAQ Yes · LAQ Yes. **Textbook.** Volume 0, Module 8, Unit 1, page 421.

LAQ B8.1 — Why Gold Turns Red

Anchor: M8U1 (1.1, 1.2, 1.3) · Semester 2 · Suggested time: 18 minutes

THE CASE

In a class, the teacher shows a gold leaf (yellow, shining) beside a flask of colloidal gold (deep red). A student asks how the same metal can have two colours. Another asks, "If a Bhasma particle is 10 µm and a nanoparticle is 50 nm, how much more surface does the nanoparticle have?"

Q	Question	Marks	Domain	Level
(a)	Define nanotechnology and the nanoscale. Outline its history with key milestones.	3	CK	K
(b)	Calculate the surface-area-to-volume ratio for spheres of 10 µm and 50 nm diameter, and explain its significance.	4	CAP	KH
(c)	Explain why colloidal gold is red, and give three applications of nanotechnology in medicine.	3	CC	KH

Model answer framework

(a) Definition and history — 3 marks

- *Nanoscale:* about 1 to 100 nanometres (1 nm = 10^{-9} m). *Nanotechnology:* understanding and control of matter at this scale, where size-dependent properties allow new applications.
- *Milestones:* Roman Lycurgus Cup (4th century CE) — glass coloured by gold–silver nanoparticles, though unknown to its makers; Richard Feynman's lecture "There's Plenty of Room at the Bottom" (1959); Norio Taniguchi coins "nano-technology" (1974); scanning tunnelling microscope (1981); discovery of fullerenes (1985); carbon nanotubes described (1991).

(b) Calculation and significance — 4 marks

- For a sphere, surface area ÷ volume = $4\pi r^2 \div (4/3)\pi r^3 = \mathbf{3 \div r}$.
- 10 µm diameter: r = 5 µm → 3 ÷ 5 = **0.6 µm^{-1}**.
- 50 nm diameter: r = 0.025 µm → 3 ÷ 0.025 = **120 µm^{-1}**.
- The nanoparticle has **200 times** more surface per unit volume (120 ÷ 0.6).
- *Significance:* a far larger share of atoms sits at the surface. Surface atoms are more reactive. This changes dissolution rate, chemical reactivity, catalytic activity, interaction with proteins and cells, and — for a medicine — absorption and possibly toxicity.

(c) Red gold; applications — 3 marks

- In gold nanoparticles, light makes the free electrons at the surface oscillate together — *surface plasmon resonance*. Particles of a few tens of nanometres absorb green light (around 520 nm), so the colloid looks red. Bulk gold reflects light from its continuous electron sea and looks yellow. Colour changes with particle size and shape — a nanoscale property.
- *Applications:* drug delivery carriers (liposomes, polymeric nanoparticles) for better solubility and targeting; diagnostics (gold nanoparticle-based rapid tests, imaging contrast agents); antimicrobial coatings (silver nanoparticles); and nano-formulations of poorly soluble plant constituents.

Revise: Volume 0 — Module 8, Unit 1 (p. 421).

Ready Reckoner — Facts Used in the Answers

This section collects the limits, rules and numbers that recur in the answer frameworks. Each was checked against a primary or reliable secondary source in October 2026. Rules and limits change: confirm the current text before relying on any of them.

Pharmacopoeial limits (Ayurvedic Pharmacopoeia of India)

Test (API appendix)	Limit
Heavy metals (Appendix 2.3)	Lead 10 ppm; Arsenic 3 ppm; Cadmium 0.3 ppm; Mercury 1 ppm
Microbial limits (Appendix 2.4)	Total bacterial count 10^5/g; total yeast and mould 10^3/g; *E. coli*, *Salmonella* spp., *S. aureus* and *P. aeruginosa* absent
Pesticide residues (Appendix 2.5)	As listed in the appendix for each pesticide
Aflatoxins (Appendix 2.7)	As stated in the appendix and the monograph
Example of a semi-solid monograph — Vasavaleha	Loss on drying not more than about 12%; total sugar, reducing and non-reducing sugar, pH and ash within stated ranges

Heavy metal limits apply to herbal drugs and formulations. For a Rasaushadhi, the metal that is a deliberate ingredient is controlled by the monograph and by the classical tests; unintended toxic metals are still contaminants.

The law: Act, Rules and Schedules

Provision	Subject
Drugs and Cosmetics Act, 1940, Section 3(a)	Definition of an Ayurvedic, Siddha or Unani drug — made exclusively as per the formulae of the authoritative books in the First Schedule
Section 3(h)	Patent or proprietary medicine (ASU) — ingredients only from the authoritative books; *excludes medicines given by parenteral route*
Chapter IVA, Sections 33E, 33EE, 33EEA	Misbranded, adulterated and spurious ASU drugs
Sections 33EEB, 33EEC	Regulation of manufacture for sale; prohibition of manufacture and sale of certain ASU drugs
Sections 33G, 33-I	Inspectors; penalties
Drugs Rules, 1945	Earlier titled the Drugs and Cosmetics Rules, 1945; cosmetics are now governed by the Cosmetics Rules, 2020
Rule 157 and Schedule T	Good Manufacturing Practices for ASU drugs

Provision	Subject
Rule 158-B	Licensing of ASU medicines; proof of safety and effectiveness by category
Rules 160-A to 160-J	Approval of laboratories for testing ASU drugs
Rule 161	Labelling; products with Schedule E(1) substances carry “Caution: to be taken under medical supervision”
Rule 161-B	Shelf life of ASU medicines (Avaleha: three years in the 2016 notification)
Schedule E(1)	Poisonous substances of ASU origin

Updates in 2026

Notification	Main points
Drugs (Eleventh Amendment) Rules, 2026 — Ministry of Ayush, G.S.R. 657(E), dated 24 July 2026	Manufacturing licences for ASU drugs valid without periodic renewal; shelf life must be supported by scientific data (as reported: accelerated studies support one or two years; longer shelf life needs real-time data; a change of dosage form needs its own stability data); standard product code to appear on labels; long ingredient lists and other label information may be given by QR code, GTIN or barcode; risk-based inspection; revised qualifications for inspectors and testing staff
Revised Schedule T (same notification)	Risk-based quality management system; detailed requirements for premises, stores, QC section, training and self-inspection; additional requirements for Rasaushadhi — segregated manufacturing, emission control, employee health surveillance, controlled release, adverse event investigation and recall. Existing manufacturers to comply by 31 July 2029

Sources for 2026: PRS Legislative Research, Monthly Policy Review, July 2026, and published legal summaries. Read the Gazette text for exact wording.

Other frameworks named in the answers

Framework	Use in this book
New Drugs and Clinical Trials Rules, 2019	New drugs, including a parenteral herbal product (ABQ 1.3.4)
Guidelines for Evaluation of Nanopharmaceuticals in India — DBT, ICMR and CDSCO, October 2019	Nano-based products (ABQ 8.4.1)
CCSEA (Committee for Control and Supervision of Experiments on Animals; formerly CPCSEA)	Animal ethics; IAEC approval

Framework	Use in this book
OECD Test Guidelines 407 (28-day), 408 (90-day), 423 (acute oral)	Toxicity studies
ICH Q1A(R2) (stability) and ICH Q2(R2) (validation of analytical procedures)	Stability; method validation
Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016	Disposal of mercury, lead, arsenic and nanoparticle waste
European Commission Recommendation on the definition of nanomaterial (2022)	50% or more of particles by number with an external dimension of 1 to 100 nm
US FDA guidance on products involving nanotechnology (2014)	About 1 to 100 nm, or up to 1 µm with size-dependent properties
Dietary Supplement Health and Education Act, 1994 (USA); FDA Botanical Drug Development guidance (2016); EU Directive 2004/24/EC (traditional herbal medicinal products)	Export pathways (LAQ B1.1)

Pharmaceutical numbers

Item	Value
IP uniformity of weight, uncoated tablets	Up to 80 mg ±10%; 80 to 250 mg ±7.5%; above 250 mg ±5% (not more than 2 of 20 outside; none outside double)
Friability	Loss usually not more than 1%
IP disintegration, uncoated tablets	15 minutes (each Vati has its own limit in its API monograph)
Nasal spray droplets	Above about 10 µm deposit in the nose; about 1 to 5 µm can reach the lungs

Chemistry numbers and formulae

Item	Value
Mercury in HgS	86.2% (sulphur 13.8%)
Zinc in ZnO	80.3%
Lead in PbS	86.6%

Item	Value
Copper in chalcopyrite ($CuFeS_2$)	34.6%
Water lost by gypsum ($CaSO_4 \cdot 2H_2O \rightarrow CaSO_4$)	20.9%
Melting points (approx.)	Lead 327 °C; zinc 420 °C (boils about 907 °C); gold 1064 °C; copper 1085 °C; iron 1538 °C
Bragg's law	$n\lambda = 2d \sin \theta$
Scherrer equation	$D = K\lambda / (\beta \cos \theta)$, K about 0.9 — crystallite size, not particle size
Abbe resolution limit	$d = \lambda / (2\ NA)$ — about 0.2 µm for the best light microscope
Surface-area-to-volume of a sphere	3 / r
Beer–Lambert law	$A = \varepsilon cl$
Micrometry	Calibration factor = (stage divisions × 10 µm) ÷ coinciding eyepiece divisions
Relative bioavailability	$(AUC_{test} \div AUC_{reference}) \times 100$

Choosing the instrument

The question asked	Instrument
What compound (phase) is it?	XRD
Which elements, and how much (major)?	XRF; SEM-EDAX for a spot
How much toxic metal (trace)?	ICP-MS, ICP-OES, AAS (graphite furnace, cold-vapour for Hg, hydride for As)
What size are the particles?	Laser diffraction (bulk); DLS (sub-micron, in liquid); SEM and TEM (direct); optical micrometry (above about 1 µm)
What shape and surface?	SEM; TEM for internal structure
How much moisture and organic matter? Thermal behaviour?	TGA; DTA
Is it magnetic (residual magnetite or iron)?	VSM
Which functional groups?	FTIR

The question asked	Instrument
Identity and assay of plant constituents	TLC, HPTLC, HPLC; GC for volatiles; LC-MS and GC-MS for confirmation
Which mineral, from optical properties?	Petrological (polarising) microscope

Topic Finder

Use this table to find every case on a topic, across modules. P1, P2.1 … P3.8 are the questions of the October 2026 preliminary paper, solved in its own chapter.

Topic	Questions
Heavy metals — testing and limits	5.4.1, 7.1.3, 7.4.2, P1, P2.1
Mercury — chemistry, safety, Kupipakwa	3.1.2, 4.1.3, 5.7.3, 7.3.3, 8.2.2
Arsenic compounds	3.1.3, 4.4.2, B3.5
Lead and Naga	4.4.2, 7.4.2, 5.4.1, 7.1.3
Microbial contamination and preservation	1.3.2, 1.3.3, 7.1.1, 7.4.1, 7.4.3, 7.5.2, 7.5.3, P1
Stability and shelf life	1.3.1, 1.3.2, 5.3.2, 7.4.3, P1
QA systems, documentation, CAPA	7.1.1, 7.1.2, 7.1.3, 5.6.1, 5.4.3, 4.1.3, P1
Regulation and licensing	1.3.4, 4.1.1, 5.3.3, 7.1.2, 7.4.2, 8.4.1, B1.1
Safety in the Rasashala and laboratory	3.1.2, 3.1.3, 3.3.1, 3.3.3, 4.1.3, 4.4.2, 5.7.3, 8.2.3, 8.4.3, B3.5, B4.2, B4.3
Shodhana and Marana — principles and chemistry	4.1.1, 4.1.2, 4.4.1, 4.4.2, 4.4.3, 3.3.1, 3.3.3, B4.2, B4.3, P2.3
Chemistry principles in Rasashastra	3.1.1, 3.1.2, 3.1.3, P3.1
Phytochemistry and extraction	B3.4, P2.4
Musha, crucibles and Puta	P3.2, B3.5, B5.1
Classical tests (Bhasma Pariksha, Siddhi Lakshana)	3.2.3, 4.1.2, 4.4.1, 4.4.3, 5.7.2, 7.3.3, 7.4.1, P1
XRD and crystal forms	3.1.2, 3.2.1, 3.2.3, 4.1.2, 5.7.1, 5.7.2, 5.7.3, 7.4.2, 8.2.1
Particle size, micrometry and nano	3.2.1, 5.5.2, 5.7.1, 8.2.1, 8.2.2, 8.4.2, B8.1, P3.7
Chromatography	5.3.1, 5.3.2, 5.3.3, 7.4.1, 7.5.1, 7.5.3
Spectroscopy	5.4.1, 5.4.2, 5.4.3, 3.2.2, 3.3.2
Microscopy and raw drug identity	5.5.1, 5.5.2, 5.5.3, B2.2, P3.3
Calibration	5.6.1, 5.6.2, 5.6.3, 5.4.3, B5.1, P3.8
Formulation design and excipients	1.3.1, 1.3.2, 1.3.3, 1.3.4, 7.5.4
Biopharmaceutics and bioavailability	B1.5, 8.4.1, 3.3.2
Animal studies, ethics and disease	1.3.4, 8.4.2, P2.2, B6.4, P3.4
Databases	P2.1, B2.2, P3.6

Topic	Questions
QC of dosage forms	Churna 7.3.1 · Vati and Varti 7.3.2 · Rasaushadhi, Kshara, Lavana 7.3.3 · Avaleha P1, 7.4.3 · Rasakriya 7.4.1 · Anjana 7.4.2 · Asava-Arishta 7.5.1 · Netra and Karna Bindu 7.5.2 · Kashaya, Arka, Sharkara 7.5.3 · Sprays 7.5.4 · Liquid forms, overview P3.5

Sources

The model answer frameworks rest on the companion textbook and on the official documents below. Where a source appears in the NCISM reference list for AYPG-AB-RB, its curriculum number is given. No source has been cited that was not checked.

The textbook and the curriculum

- Chatrabhuji MP. *Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana.* Vadodara: Parul Institute of Ayurved and Research, Parul University; 2026. ISBN 978-81-955675-2-2.
- National Commission for Indian System of Medicine. *Curriculum for MD/MS Ayurveda, Semester II — Applied Basics of Rasashastra and Bhaishajya Kalpana (AYPG-AB-RB).* New Delhi: NCISM; applicable from the 2024-25 batch.
- Parul University, Faculty of Ayurved. MS/MD (Ayurved) Preliminary Examination, Semester 2, October 2026 — Applied Basics of Rasashastra and Bhaishajya Kalpana — the complete paper, Questions 1, 2.1 to 2.4 and 3.1 to 3.8 (reproduced for teaching).

Official compendia, law and guidelines

- Ayurvedic Pharmacopoeia of India, Part I and Part II. Pharmacopoeia Commission for Indian Medicine and Homoeopathy, Ministry of Ayush, Government of India. (NCISM references 156, 161 and 162)
- Lohar D. *Protocol for Testing of Ayurvedic, Siddha and Unani Medicines.* Department of AYUSH, Ministry of Health and Family Welfare, Government of India; 2008. (NCISM reference 163)
- Ayurvedic Formulary of India. Ministry of Ayush, Government of India.
- Drugs and Cosmetics Act, 1940 and Drugs Rules, 1945, as amended. Central Drugs Standard Control Organisation.
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- Department of Biotechnology, Indian Council of Medical Research and Central Drugs Standard Control Organisation. *Guidelines for Evaluation of Nanopharmaceuticals in India*. October 2019.
- International Council for Harmonisation. ICH Q1A(R2): Stability Testing of New Drug Substances and Products; ICH Q2(R2): Validation of Analytical Procedures.
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Research papers cited in the frameworks

- Badal R, Kaushik S, Yadav P, Prajapati PK. Critical appraisal of documented shelf life of Ayurvedic formulations: current status and future implications. *Indian Journal of Pharmaceutical Sciences.* 2024;86(4):1207–1217.
- Shoba G, Joy D, Joseph T, Majeed M, Rajendran R, Srinivas PS. Influence of piperine on the pharmacokinetics of curcumin in animals and human volunteers. *Planta Medica.* 1998;64(4):353–356.

Online databases and standards named in the answers

- Mindat (mindat.org); Webmineral (webmineral.com); RRUFF Project (rruff.info); Crystallography Open Database (crystallography.net); American Mineralogist Crystal Structure Database; U.S. Geological Survey mineral resources data; NIST Chemistry WebBook; Materials Project.
- World Flora Online. *WFO Plant List — Background* (wfoplantlist.org/background): The Plant List not updated since 2013; WFO Plant List launched in May 2021 as its replacement. (Accessed 13 October 2026.)
- PlantGDB website notice of 1 July 2015: the grant has ended and the site is no longer being updated. (Accessed 13 October 2026.)
- Bureau of Indian Standards. *Code for Breeding, Care, Management and Housing of Laboratory Animals, Part 1: Mice and Rats* (IS 5701 Part 1, second revision), draft for comments, Doc. FAD 32 (25869) WC, June 2024 — aligned with the CPCSEA Guidelines for Laboratory Animal Facility, 2015.

A note on the cases

Every case in this book, except the three questions from the October 2026 paper, was constructed for teaching. Names of firms are not given, and any resemblance to a real firm or batch is coincidental. Numbers in the cases are illustrative unless a source is named. They were chosen to be realistic, so that the reasoning they demand is the reasoning a real case would demand.

About the Author



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Alongside his academic work he has been in clinical practice for more than a decade at Suvarna Ayurveda Clinic, Vadodara.

His teaching and research interests lie in the pharmaceutical processing of metals and minerals, the standardisation of classical Ayurvedic preparations, and the application of modern analytical methods to the verification of classical pharmaceutical claims. This question bank is a companion to his *Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana* (2026).

Corrections and suggestions. This is a first edition. The author would welcome corrections, comments and suggestions from teachers and scholars, especially on outdated regulatory references, so that later editions may be improved.

Practise the paper as it is now set.

University papers in Applied Basics of Rasashastra and Bhaishajya Kalpana now set the Application-Based Question as a long case: a pharmacy, an inspection, a laboratory report — with numbered findings and sub-parts tagged by Bloom's domain and Miller's level.

This companion to the textbook gives scholars that practice. Every case is drawn only from units the curriculum allows for ABQ. Every answer framework links the findings to the marks, and marks the boundary between what the evidence shows and what is merely claimed.

WHAT THIS BOOK CONTAINS

- 53 case-based ABQs (20 marks) covering all 17 ABQ-eligible units of Table 6F
- 10 case-based analytical structured LAQs (10 marks), including Modules 2 and 6
- The complete October 2026 preliminary paper solved — all 14 questions — and a seven-step method for any case
- Model answer frameworks with mark splits, tables and "where marks are lost"
- Ready Reckoner of verified limits and rules, including the 2026 amendments, and a Topic Finder

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